Catalytic Gate-Set Distortion

Temporary Escape in Quantum and Reversible Circuits

A Symbolic, Algebraic, and Computational Monograph

Komorebi P.

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Abstract

This monograph develops a general theory of *catalytic gate-set distortion*: a target belongs to a restricted circuit model, yet a shorter ambient realization temporarily leaves that model, reuses the resulting resource, and returns with the resource fully removed. The framework covers metric defects, where both realizations exist but have different costs, and existence defects, where the restricted realization is impossible. It treats scalar and vector resource costs, subgroup and groupoid holonomy, first-arrival Green kernels, algebraic product loci, border effects, tensor regularization, representable matroids, quiver factorizations, and root-system commutators.

The established linear core is reproduced in full. A universal six-row transvection target has intrinsic length seven and ambient length six over every nonzero commutative ring with identity. Connected families show that two reverse-directed gates can save linearly many gates, while tensor products yield an additive separation of order $n \log n$. At fixed dimension, generic targets are rigid and defect-d loci have codimension at least d.

The monograph then extends the program to quantum gate logic. Modular SUM gates inherit the universal atom for every qudit dimension. New exact finite computations are supplied for a symplectic Clifford model: in the swap-free primitive-count metric the parabolic subgroup is rigid through two qubits and has twenty-four defective targets at three qubits, whereas an entangling-count metric already has four two-qubit defects and 1624 three-qubit defects. Adding SWAP as a unit-cost primitive removes all three-qubit total-count defects. By contrast, exhaustive three-bit enumeration finds no affine target shortened by adding Toffoli gates, under either total-count or entangling-count cost. These results illustrate that defects depend essentially on the target class, gate alphabet, ancilla convention, and cost vector.

The remaining chapters give rigorous symbolic skeletons for CSS and Clifford synthesis, affine-to-nonlinear reversible logic, Clifford+T and phase-polynomial circuits, Gaussian and fermionic systems, symmetry-preserving circuits, measurement and feedforward, architecture constraints, and finite packet or realization hierarchies. Proved statements, exact computations, formal skeletons, and conjectures are visibly separated throughout. The accompanying Python and C++ atlas reproduces the finite experiments and provides reusable search, certificate, and circuit-drawing components.

Reader's guide and status convention

The scope is intentionally broad, but the logical status is not blurred. Green boxes contain symbolic proofs or consequences of stated hypotheses. Blue boxes contain complete finite computations with an explicit state model and exhaustive search. Yellow boxes specify a mathematically complete research skeleton whose decisive lower bound or classification remains open. Red boxes state conjectures or candidate mechanisms. A statement in a yellow or red box is not promoted to a theorem elsewhere in the text.

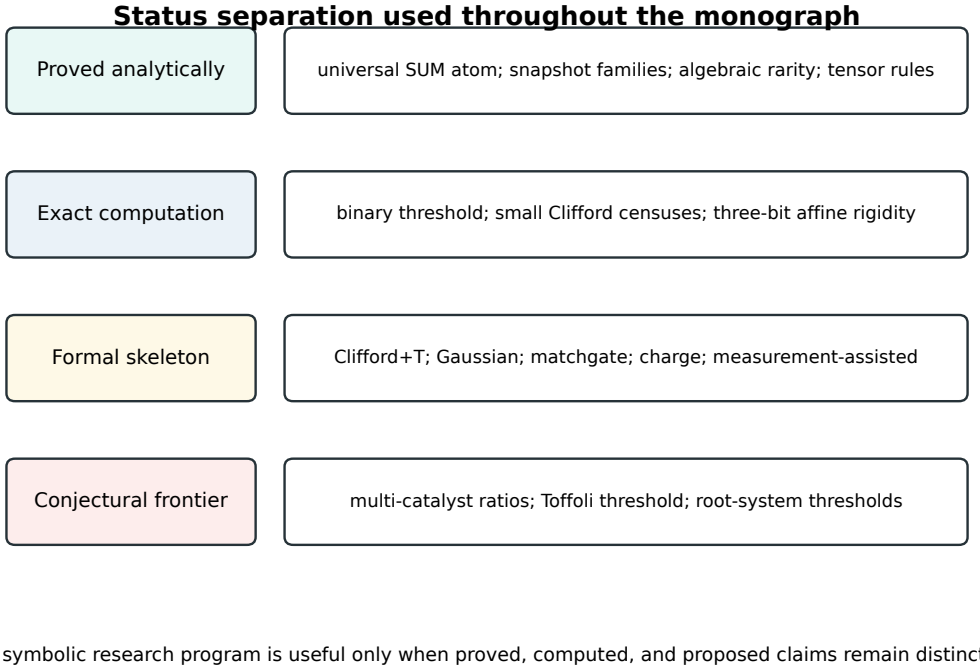


Figure 1: The status separation used throughout the monograph.

The established unitriangular paper is incorporated as a complete core chapter rather than paraphrased. Its computer-assisted claims retain their original finite-state completeness conditions. The new Clifford and affine-versus-Toffoli computations use exact binary symplectic matrices and exact permutations, respectively. Quantum Latin squares enter only as a contextual example of an existence defect; no circuit theorem is inferred from that example.

Notation and recurring objects

Symbol	Meaning
$H \leq G$	Restricted and ambient groups, when both models are group generated.
$\mathcal{S}_H \subseteq \mathcal{S}_G$	Restricted and ambient elementary gate alphabets.
$\ell_{\mathrm{res}}(U), \ell_{\mathrm{amb}}(U)$	Minimum restricted and ambient costs of a target U.
$\Delta(U)$	Gate-set defect $\ell_{\mathrm{res}}(U) - \ell_{\mathrm{amb}}(U)$, with $+\infty$ allowed for existence defects.
D_n	Worst additive defect at size n.
R_n	Worst multiplicative distortion at size n.
A_S	Convolution or adjacency operator associated with a finite generator set S.
$\mathcal{G}_{S,U}(z)$	Endpoint Green kernel $\langle \delta_U, (I - zA_S)^{-1}\delta_e \rangle$.
P_n	Binary symplectic parabolic subgroup preserving the Z Lagrangian.
$\tau_{\otimes}$	Stable tensor-normalized ambient cost.
$\kappa_{\otimes}$	Stable catalytic tensor capacity.

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Part I

General theory of temporary escape

Chapter 1

Nested gate models and distortion invariants

1.1 Why a separate theory is needed

A target transformation often belongs to a visibly structured class even when the machine used to realize it is capable of much more. A lower-triangular linear transformation may be synthesized by an unrestricted CNOT network. A diagonal unitary may be realized by a non-diagonal Clifford+T circuit. A charge-preserving unitary may be compiled by gates that temporarily visit other charge sectors. A Gaussian target may be realized by a circuit that briefly uses a non-Gaussian primitive. These situations share a question that is not answered by ordinary exact synthesis:

> How much does it cost to require the *entire path* of a computation to retain the structure possessed by its endpoint?

The distinction is path dependent. It is not determined only by membership of the final target in a subgroup, a subcategory, or a model class.

1.2 Exact gate models

Definition 1.1 (Gate model)**.** An *exact gate model* is a tuple

$$\mathfrak{M} = (\mathcal{X}, \mathcal{S}, c, \circ),$$

where $\mathcal{X}$ is a set of states or morphisms, $\mathcal{S}$ is a specified family of elementary generators, $c : \mathcal{S} \to \mathbb{R}_{\geq 0}$ is a cost function, and $\circ$ is the composition law. A word

$$w = s_L \cdots s_1$$

has cost

$$c(w) = \sum_{j=1}^{L} c(s_j).$$

For a target $U \in \mathcal{X}$, the exact synthesis cost is

$$\ell_{\mathfrak{M}}(U) = \inf\{c(w) : w = U\}.$$

The value is $+\infty$ when the target has no realization.

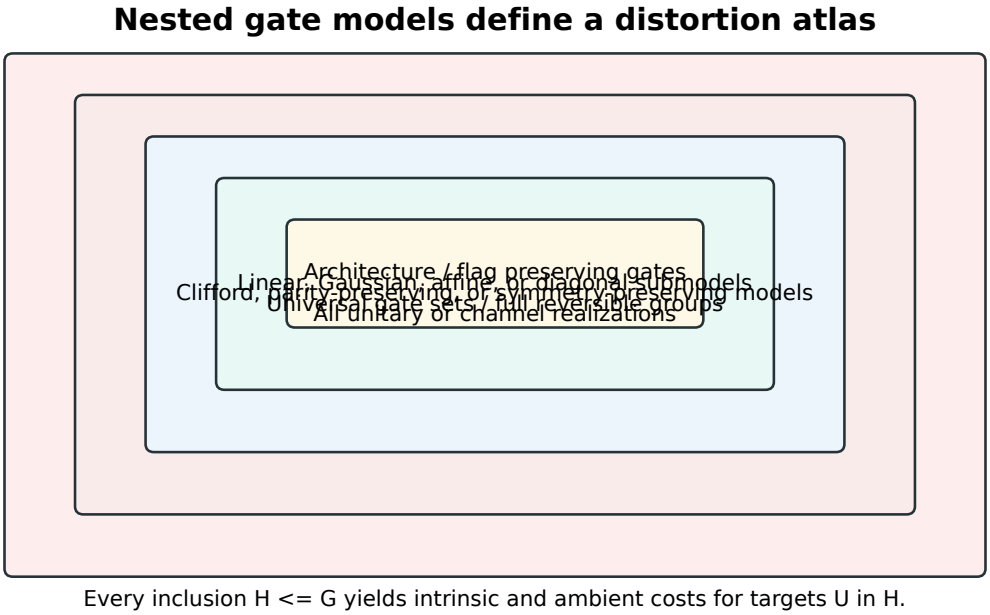


Figure 1.1: Representative nested gate models. The same target can be assigned different costs at several levels.

The composition law may be group multiplication, composition of permutations, composition of quantum channels, tensor-network contraction, or composition in a dagger symmetric monoidal category. The cost may count all gates, only entangling gates, only non-Clifford gates, depth, weighted error, or a vector of several resources.

Definition 1.2 (Nested realization pair)**.** A *nested realization pair* consists of restricted and ambient models

$$\mathfrak{M}_{\mathrm{res}} \subseteq \mathfrak{M}_{\mathrm{amb}}$$

with a common target class $\mathcal{T}$. For $U \in \mathcal{T}$, define

$$\ell_{\mathrm{res}}(U) = \ell_{\mathfrak{M}_{\mathrm{res}}}(U), \qquad \ell_{\mathrm{amb}}(U) = \ell_{\mathfrak{M}_{\mathrm{amb}}}(U).$$

The additive defect is

$$\Delta(U) = \ell_{\mathrm{res}}(U) - \ell_{\mathrm{amb}}(U).$$

When both costs are finite, $\Delta(U)$ is a metric or optimization defect. If $\ell_{\mathrm{res}}(U) = +\infty$ and $\ell_{\mathrm{amb}}(U) < +\infty$, it is an existence defect. This convention unifies finite gate savings with the failure of a restricted realization category.

Proposition 1.3 (Monotonicity)**.** *For every target in the common domain,*

$$\ell_{\mathrm{amb}}(U) \leq \ell_{\mathrm{res}}(U).$$

Consequently $\Delta(U) \geq 0$ *whenever both models use the same nonnegative cost on their common generators.*

Proof. Every restricted word is also an ambient word. □

1.3 Cost models are part of the theorem

A gate-set comparison is incomplete until the cost convention is fixed. The same pair of groups can display a defect under entangling count and no defect under total primitive count. A SWAP gate may be primitive, may cost three CNOTs, or may be implemented by physical motion. A local Clifford may be free in a fault-tolerant resource estimate and nonfree in a pulse-level model.

Definition 1.4 (Resource vector)**.** Let

$$\mathbf{c}(s) = (c_1(s), \dots, c_r(s)) \in \mathbb{R}^r_{\geq 0}$$

be the resource vector of a gate. The achievable resource region of a target is

$$\mathcal{P}_{\mathfrak{M}}(U) = \left\{ \sum_{j=1}^{L} \mathbf{c}(s_j) : s_L \cdots s_1 = U \right\}.$$

Its nondominated points form the Pareto boundary.

For a weight vector $\lambda \in \mathbb{R}^r_{\geq 0}$, scalarization gives

$$\ell^{\lambda}_{\mathfrak{M}}(U) = \inf_{x \in \mathcal{P}_{\mathfrak{M}}(U)} \lambda \cdot x.$$

The scalar defect $\Delta^{\lambda}(U)$ can vary with λ even when the target and gate alphabets are fixed.

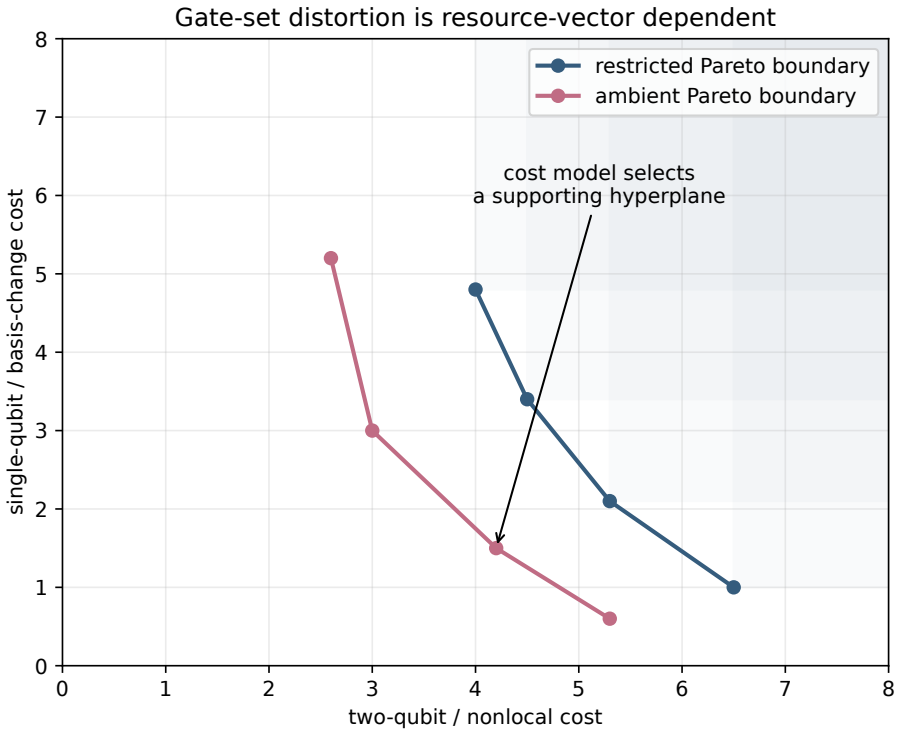


Figure 1.2: Restricted and ambient Pareto regions can differ without one model dominating by the same amount under every resource weighting.

Principle 1.5 (Model specificity)**.** *Every claimed shortcut must state the target class, elementary gate alphabet, ancilla convention, exact or approximate synthesis criterion, and cost vector. Changing any one of these may remove, create, or reverse a defect.*

1.4 Ancilla and restoration conventions

Temporary workspace is meaningful only after its restoration rule is specified.

Definition 1.6 (Ancilla conventions)**.** An ancilla is

1. *clean* if it begins in a specified state and must return to that state,
2. *borrowed* or *dirty* if its initial state is arbitrary but must be restored,
3. *consumable* if restoration is not required,
4. *catalytic* if it is restored while enabling a cheaper realization of the target.

The present theory focuses on restoration. If an auxiliary resource may be discarded, the comparison becomes a different channel or isometry synthesis problem.

1.5 Worst-case and threshold functions

For a size parameter n, let $\mathcal{T}_n$ denote the target family.

Definition 1.7 (Extremal distortion)**.** Define

$$D_n = \sup_{U \in \mathcal{T}_n} \Delta(U)$$

and, when $0 < \ell_{\mathrm{amb}}(U) < \infty$,

$$R_n = \sup_{U \in \mathcal{T}_n} \frac{\ell_{\mathrm{res}}(U)}{\ell_{\mathrm{amb}}(U)}.$$

The first defective size is

$$\tau = \min\{n : D_n > 0\}.$$

Several threshold notions should not be conflated:

$$\begin{aligned} \tau_{\mathrm{metric}} &= \min\{n : \exists U,\ 0 < \Delta(U) < \infty\}, \\ \tau_{\mathrm{exist}} &= \min\{n : \exists U,\ \ell_{\mathrm{res}}(U) = \infty,\ \ell_{\mathrm{amb}}(U) < \infty\}, \\ \tau_q &= \min\{n : \exists U,\ \ell_{\leq q}(U) < \ell_{\leq 0}(U)\}, \end{aligned}$$

where $\ell_{\leq q}$ permits at most q ambient-only gates. A model may have an early metric threshold and a later existence threshold, or no threshold at all.

1.6 Equivalence, orbits, and classification

A finite census should be quotiented by transformations preserving both gate models and the cost. Depending on the problem, these may include wire permutations, local Clifford conjugation, isotopy, simultaneous row and column relabeling, or graph automorphisms.

Definition 1.8 (Cost-preserving equivalence)**.** Targets U and V are equivalent if an automorphism φ of the ambient realization system satisfies

$$\varphi(U) = V, \qquad \varphi(\mathcal{S}_{\mathrm{res}}) = \mathcal{S}_{\mathrm{res}}, \qquad \varphi(\mathcal{S}_{\mathrm{amb}}) = \mathcal{S}_{\mathrm{amb}},$$

and preserves the gate costs.

Under this equivalence, both lengths and the defect are orbit invariants. A useful threshold theorem should classify minimal defective orbits, not merely list raw matrices.

1.7 Approximate synthesis and border defects

For continuous gate models, exact word length can differ from the minimum length achievable as a limit.

Definition 1.9 (Border cost)**.** Let $\overline{\mathcal{W}}_L$ be the topological or Zariski closure of targets realizable with cost at most L. The border cost is

$$\underline{\ell}(U) = \inf\{L : U \in \overline{\mathcal{W}}_L\}.$$

The border defect is

$$\underline{\Delta}(U) = \underline{\ell}_{\mathrm{res}}(U) - \underline{\ell}_{\mathrm{amb}}(U).$$

This is the circuit analogue of the distinction between tensor rank and border rank. An exact defect can disappear under degeneration, persist, or become larger after approximation changes the relevant cost geometry.

1.8 Metric and existence defects in one category

Let $\mathcal{C}_{\mathrm{res}} \subseteq \mathcal{C}_{\mathrm{amb}}$ be categories with weighted generating morphisms. For a morphism f in the restricted target class, the same definition

$$\Delta(f) = c_{\mathrm{res}}(f) - c_{\mathrm{amb}}(f)$$

includes group-word metrics, adaptive quantum channels, tensor-network realizations, and orthogonality designs. This categorical form is essential when the ambient model permits measurement, branching, postselection, or entangled objects that do not lie in a common group.

Scope boundary

The abstract definition does not prove that a defect exists. It provides a common language in which a finite search, a symbolic identity, and a lower-bound certificate can be compared without changing the meaning of the target or the cost.

1.9 A standard theorem package

A mature gate-set distortion result should contain as many of the following components as the model allows:

1. a precisely defined nested gate pair and cost vector,
2. a smallest defective target or a no-defect theorem below a threshold,
3. an explicit ambient word,
4. a restricted lower-bound certificate,
5. classification modulo cost-preserving equivalence,
6. a reusable catalytic normal form,
7. direct-sum, recursive, or tensor amplification,
8. generic-rigidity or sparsity results,

9. a certificate-producing implementation,
10. a clear account of which statements are symbolic, computed, or conjectural.

The remainder of the monograph follows this package.

Chapter 2

Catalytic holonomy, excursion geometry, and resource reuse

2.1 Subgroup metrics and Schreier loops

Assume first that the restricted and ambient models are groups

$$H = \langle \mathcal{S}_H \rangle \leq G = \langle \mathcal{S}_G \rangle.$$

For $h \in H$, the intrinsic and ambient word metrics are

$$\ell_H(h) = |h|_{\mathcal{S}_H}, \qquad \ell_G(h) = |h|_{\mathcal{S}_G}.$$

The quotient geometry is the right Schreier graph $H\backslash G$.

Theorem 2.1 (Holonomy-compression identity)**.** *Let every generator have unit cost. For $h \in H$, $\ell_G(h)$ is the minimum length of a based loop in $H\backslash G$ whose lifted word has holonomy h. The intrinsic length $\ell_H(h)$ is the minimum length among loops that remain in the base fiber.*

Proof. An ambient word $s_k \cdots s_1 = h$ projects to the loop

$$H, Hs_1, Hs_2s_1, \ldots, Hh = H.$$

Conversely, a based Schreier loop has product in H. Restricting the path to the base fiber forces every prefix to lie in H, hence every edge label can be replaced by a restricted generator under the chosen intrinsic Cayley structure. □

Thus

$$\Delta_{H\hookrightarrow G}(h) = \text{intrinsic fiber cost} - \text{shortest holonomy loop cost}.$$

This is the geometric content of a dirty-workspace shortcut.

2.2 The basic conjugation inequality

Let $g \in G$ and $k \in H$ satisfy $gkg^{-1} \in H$. Then

$$\ell_G(gkg^{-1}) \leq 2\ell_G(g) + \ell_H(k).$$

Define the intrinsic conjugation stretch

$$\mathrm{Stretch}_g(k) = \ell_H(gkg^{-1}) - \ell_H(k).$$

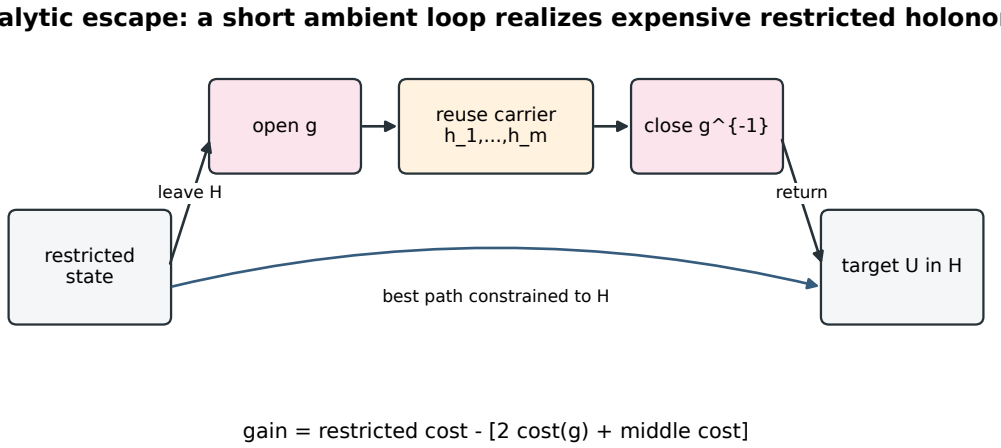


Figure 2.1: A catalytic excursion leaves a restricted fiber, reuses an ambient state, and returns with nontrivial holonomy.

Proposition 2.2 (Catalytic stretch criterion)**.** *If*

$$\mathrm{Stretch}_g(k) > 2\ell_G(g),$$

then gkg^{-1} *has positive defect.*

The opening and closing cost is paid once. Every useful consumer inside the excursion contributes to the stretch.

2.3 Many consumers and catalytic excess

Suppose the middle word is

$$k = h_m \cdots h_1, \qquad h_i \in H.$$

The ambient word

$$gh_m \cdots h_1 g^{-1}$$

has cost at most

$$2c(g) + \sum_i c(h_i).$$

If the restricted images gh_ig^{-1} require costs r_i and a lower-bound invariant proves that their obligations cannot be shared, then

$$\ell_H(gkg^{-1}) \geq \sum_i r_i.$$

A sufficient inequality is

$$\sum_i (r_i - c(h_i)) > 2c(g).$$

The left side is the gross catalytic benefit. The right side is the restoration cost.

Definition 2.3 (Catalytic excess)**.** For a normal form with r matched excursions and C useful cross interactions, define

$$\mathcal{E} = C - 2r$$

when each consumer contributes one unit and each opening or closing costs one unit.

Positive excess is necessary in the simplest transvection model, but it is not sufficient. An all-cross configuration may admit a mirror word of equal restricted length. Temporal asymmetry or phase holonomy is also required.

2.4 Excursion budgets

Counting ambient-only gates is only one measure of how far a circuit leaves its restricted model.

Definition 2.4 (Excursion profile)**.** For a circuit path $X_0, X_1, \ldots, X_L$, an excursion profile may include

1. the number q of ambient-only gates,
2. the maximum Schreier distance $d(HX_t, H)$,
3. the rank of a forbidden block,
4. the bandwidth of a forbidden dependency,
5. the number of simultaneously open catalytic intervals,
6. the intersection graph of those intervals,
7. the maximum algebraic degree created temporarily,
8. the maximum ancilla dimension or branch count.

The established linear family shows that a rank-one, bandwidth-one excursion using exactly two forbidden gates can save linearly many operations. Thus geometric excursion size need not control computational benefit.

2.5 Catalytic interval geometry

Associate to every matched opening and closing operation an interval

$$I_j = [t_j^{\text{open}}, t_j^{\text{close}}].$$

The family of intervals can be disjoint, nested, laminar, or crossing.

Principle 2.5 (First higher-order frontier)**.** *Independent or disjoint intervals often reduce to additive copies of a one-catalyst analysis. Crossing intervals are the first place where one dirty resource can alter another and where a multiplicative distortion larger than the clean one-carrier value may occur.*

Define the catalytic width

$$w_{\text{cat}} = \max_t \#\{j : t \in I_j\}.$$

One may then study

$$D_n^{(w)} = \max_U \left(\ell_{\text{res}}(U) - \ell_{\text{amb}}^{(w)}(U) \right),$$

where the ambient circuit has width at most w. The unitriangular snapshot family gives $D_n^{(1)} = \Omega(n)$ in its corresponding interval model.

2.6 Four recurrent catalytic mechanisms

2.6.1 Basis-change catalysis

An ambient gate g changes the frame in which several restricted operations are cheap:

$$h_i \mapsto g h_i g^{-1}.$$

Hadamard and Fourier transforms are the basic quantum examples. The frame is restored at the end.

2.6.2 Commutator catalysis

An ambient commutator creates a new direction:

$$[g, h] = ghg^{-1}h^{-1}.$$

For continuous controls,

$$e^{\varepsilon X} e^{\varepsilon Y} e^{-\varepsilon X} e^{-\varepsilon Y} = e^{\varepsilon^2 [X,Y] + O(\varepsilon^3)}.$$

The desired effective generator may lie in the restricted algebra even though the path does not.

2.6.3 Degree catalysis

A Toffoli, CCZ, or non-Gaussian gate can create higher algebraic degree or higher moments temporarily. If the nonlinear object is reused and canceled, the final target may remain affine, diagonal, or Gaussian.

2.6.4 Branch and workspace catalysis

Ancillas, measurements, reset, teleportation, and classical feedforward enlarge the realization category. The resource is a temporary branch, subsystem, or borrowed state rather than a group element.

2.7 When defects are impossible

A useful positive theorem is a criterion guaranteeing geodesic convexity.

Proposition 2.6 (One-Lipschitz retraction criterion)**.** *Suppose there is a retraction $\pi : G \to H$ with $\pi|_H = \mathrm{id}_H$ such that every ambient generator $s \in \mathcal{S}_G$ satisfies*

$$\ell_H(\pi(s)) \le c(s).$$

Then

$$\ell_H(h) \le \ell_G(h)$$

for every $h \in H$, hence the two metrics agree on H.

Proof. Apply π gate by gate to an ambient word for h. The projected factors form a restricted word of no greater cost. □

The unitriangular counterexample proves that no such generator-wise one-Lipschitz retraction exists for the lower transvection model inside the all-transvection model.

2.8 Weighted restoration and phase transitions

Let every restricted gate cost one and every ambient-only gate cost λ. For a catalyst used by v consumers with unit benefit, the displayed construction has net gain at least

$$v - 2\lambda.$$

The critical weight is therefore target and mechanism dependent.

Definition 2.7 (Critical catalyst penalty)**.** For a target family U_n, define

$$\lambda_c(U_n) = \sup\{\lambda : \ell_{\lambda,\mathrm{amb}}(U_n) < \ell_{\mathrm{res}}(U_n)\}.$$

Determining λ_c exactly is a refined synthesis problem. It can be more physically meaningful than unweighted gate count.

2.9 Restoration is the source of nontriviality

If the ambient resource need not be restored, the opening cost can dominate the comparison and many examples become trivial. Catalytic escape is strongest when the following all hold:

1. the final target lies entirely in the restricted class,
2. every auxiliary subsystem or violated constraint is restored,
3. the ambient-only resource is used more than once,
4. the restricted lower bound charges the same repeated obligations separately,
5. the gain survives a realistic weighted cost.

Conjectural frontier

In many finite gate hierarchies, minimal defects should admit a bounded-width catalytic normal form. Proving or refuting this would turn an unrestricted synthesis search into a search over a finite family of excursion motifs.

Chapter 3

First arrival, spectral packets, algebraic rarity, and tensor regularization

3.1 Word distance as first arrival

Let G be finite and let $S = S^{-1}$ be a symmetric generating multiset. In the right regular representation, define

$$A_S = \sum_{s \in S} \rho(s).$$

Then

$$\langle \delta_g, A_S^k \delta_e \rangle$$

counts length-k words ending at g.

Theorem 3.1 (Endpoint Green valuation)**.** *The word distance is*

$$d_S(e, g) = \operatorname{ord}_{z=0} \left\langle \delta_g, (I - zA_S)^{-1} \delta_e \right\rangle.$$

Proof. Expand the resolvent as a formal power series:

$$(I - zA_S)^{-1} = \sum_{k \geq 0} z^k A_S^k.$$

The first nonzero endpoint coefficient occurs at the shortest word length. □

For weighted nonnegative integer costs, replace each generator s by the monomial $z^{c(s)}\rho(s)$. The coefficient of z^k then counts words of total cost k. Multivariate variables can track several resources at once.

3.2 Why eigenvalues alone are insufficient

If

$$A_S = \sum_{\lambda} \lambda \Pi_\lambda,$$

then

$$\mathcal{G}_{S,g}(z) = \sum_{\lambda} \frac{\langle \delta_g, \Pi_\lambda \delta_e \rangle}{1 - \lambda z}.$$

The eigenvalues and multiplicities do not determine the endpoint coefficients

$$\langle \delta_g, \Pi_\lambda \delta_e \rangle.$$

Two Cayley or Schreier systems can therefore be cospectral and still have different first-arrival behavior at distinguished targets.

Principle 3.2 (Embedded packet principle)**.** *For threshold questions, a spectral label is usually weaker than an embedded spectral projector. The matrix coefficients of the projector relative to the initial and target states retain positional information that multiplicities erase.*

This distinction recurs in the Latin-square example discussed later. The natural commuting row, column, and symbol operators have universal eigenvalue packets at fixed order, while the embedded incidence and transition data still vary.

3.3 Subgroup-holonomy Fourier packets

Let $H \leq G$. Let $\mu_k(h)$ count ambient words of length k with product $h \in H$. For every irreducible representation $\tau \in \widehat{H}$, define

$$\widehat{\mu}_k(\tau) = \sum_{h \in H} \mu_k(h) \tau(h).$$

Fourier inversion gives

$$\mu_k(h) = \frac{1}{|H|} \sum_{\tau \in \widehat{H}} \dim(\tau) \operatorname{tr}\big(\tau(h^{-1}) \widehat{\mu}_k(\tau)\big).$$

The matrix-valued Green packet

$$\widehat{\mathcal{G}}_\tau(z) = \sum_{k \geq 0} \widehat{\mu}_k(\tau) z^k$$

recovers every endpoint series after Fourier inversion.

The trivial representation records only the total number of loops returning to the subgroup. Nontrivial H-types retain their holonomy. This yields a hierarchy:

$$\text{spherical return data} \prec \text{type-resolved holonomy packets} \prec \text{exact endpoint realization}.$$

3.4 Word maps and product loci

Let K be a field and let $X \subseteq G$ be an algebraic gate family. For a fixed length L, multiplication defines

$$\mu_L : X^L \to G, \qquad (x_L, \ldots, x_1) \mapsto x_L \cdots x_1.$$

The exact length-L locus is $\mu_L(X^L)$, and the length-at-most-L locus is a finite union of such images when the gate family has finitely many components.

Theorem 3.3 (Dimension per gate)**.** *Suppose every irreducible component of X has dimension at most r. Then the Zariski closure of the locus reachable in at most L gates has dimension at most rL.*

Proof. Every component of X^L has dimension at most rL. A morphism cannot increase dimension. A finite union has dimension equal to the maximum dimension of its components. □

For scaled transvections, $r = 1$. This immediately yields generic lower bounds in a target variety whose dimension exceeds L.

3.5 Generic rigidity

Suppose the restricted target variety V has dimension N, every target has a restricted realization of cost at most N, and each ambient gate contributes at most one algebraic parameter. Then ambient realizations of length below N occupy a proper closed subset.

Theorem 3.4 (Abstract generic rigidity)**.** *Under the preceding hypotheses, there is a nonempty Zariski-open subset $\Omega \subseteq V$ such that*

$$\ell_{\mathrm{res}}(U) = \ell_{\mathrm{amb}}(U) = N \qquad (U \in \Omega).$$

The unitriangular theorem in the core chapter is a concrete instance with

$$N = \binom{n}{2}.$$

3.6 Defect codimension

If every target has restricted cost at most N, then defect at least d forces ambient cost at most $N - d$. The dimension-per-gate theorem gives

$$\operatorname{codim}_V \overline{\{U : \Delta(U) \geq d\}} \geq d$$

when each ambient gate carries one parameter.

Principle 3.5 (Rare but amplifiable)**.** *At fixed dimension, large defects require increasingly special algebraic conditions. Under direct sums, recursive lifts, or tensor products, a point on a thin exceptional stratum can nevertheless seed a macroscopic family.*

This is one of the central tensions of the subject. Catalysis is neither generic nor negligible.

3.7 Finite-field sparsity

A word-counting analogue applies over $\mathbb{F}_q$. If the ambient alphabet has at most $M(q-1)$ nontrivial scaled gates, then the number of targets reached in L gates is at most

$$\sum_{j=0}^{L} [M(q-1)]^j.$$

When the target space has q^N points and defect d implies ambient length at most $N - d$, the crude bound is

$$\mathbb{P}[\Delta \geq d] = O_{N,M}(q^{-d}).$$

The estimate can be sharpened by orbit counting, stabilizers, and the geometry of word-map fibers.

3.8 Direct sums and monoidal amplification

Suppose targets admit a direct sum $U \oplus V$ and the restricted metric has an exact projection onto both factors. Then

$$\ell_{\mathrm{res}}(U \oplus V) = \ell_{\mathrm{res}}(U) + \ell_{\mathrm{res}}(V).$$

The ambient model always satisfies the constructive upper bound

$$\ell_{\mathrm{amb}}(U \oplus V) \le \ell_{\mathrm{amb}}(U) + \ell_{\mathrm{amb}}(V),$$

so

$$\Delta(U \oplus V) \ge \Delta(U) + \Delta(V).$$

One finite defect therefore gives an unbounded additive family.

3.9 Tensor products and normalized rates

Let U have dimension m and V dimension n. In linear transvection models with tree-tight certificates,

$$\ell_{\mathrm{res}}(U \otimes V) = n\ell_{\mathrm{res}}(U) + m\ell_{\mathrm{res}}(V).$$

The ambient model satisfies the corresponding inequality. Normalizing by dimension gives an entropy-like law:

$$\frac{\ell_{\mathrm{res}}(U \otimes V)}{mn} = \frac{\ell_{\mathrm{res}}(U)}{m} + \frac{\ell_{\mathrm{res}}(V)}{n}.$$

The normalized defect is superadditive under tensor products.

Definition 3.6 (Tensor synergy)**.** Define

$$\mathrm{Syn}(U, V) = n\ell_{\mathrm{amb}}(U) + m\ell_{\mathrm{amb}}(V) - \ell_{\mathrm{amb}}(U \otimes V).$$

Positive synergy means that the joint ambient realization improves on independent lifted circuits.

Restricted tree-tight metrics have zero tensor synergy. Ambient synergy is a possible source of collective activation.

Definition 3.7 (Stable tensor cost)**.** For a size-m target U, define

$$\tau_{\otimes}(U) = \lim_{r\to\infty} \frac{\ell_{\mathrm{amb}}(U^{\otimes r})}{rm^{r-1}},$$

when the subadditivity argument applies. Define

$$\kappa_{\otimes}(U) = \ell_{\mathrm{res}}(U) - \tau_{\otimes}(U).$$

Fekete's lemma gives the limit from the tensor subadditivity inequality.

Definition 3.8 (Tensor activation)**.** A target is tensor activated if

$$\Delta(U) = 0 \quad \text{but} \quad \kappa_{\otimes}(U) > 0.$$

This is analogous to regularized capacities and asymptotic ranks. It asks whether collective synthesis can unlock a catalyst invisible in one copy.

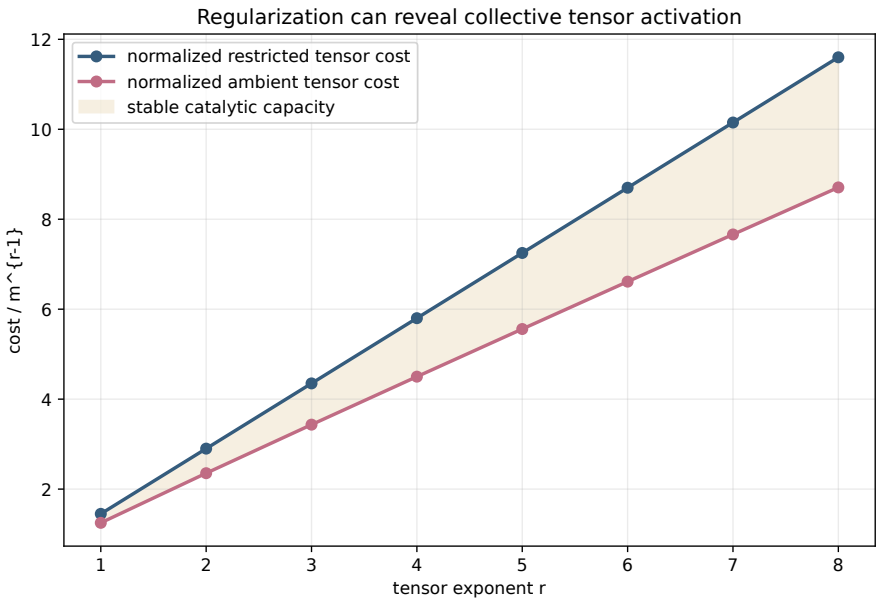


Figure 3.1: Schematic stable tensor capacity. The figure illustrates definitions, not measured data.

3.10 Border catalysis

Let X_{res} and X_{amb} be algebraic gate varieties. The exact product rank is

$$\mathrm{rk}_X(U) = \min\{L : U \in X^L\},$$

while border product rank is

$$\underline{\mathrm{rk}}_X(U) = \min\{L : U \in \overline{X^L}\}.$$

The four possible comparisons are:

1. exact and border defects both positive,
2. exact defect positive but border defect zero,
3. one-copy defect zero but tensor-regularized defect positive,
4. ambient and restricted border ranks both smaller than exact ranks but by different amounts.

Each requires a separate certificate. Numerical approximation alone does not distinguish them.

3.11 A packet-resolved threshold principle

Combining the preceding constructions gives the following architecture.

1. A coarse commutative algebra supplies spherical spectral packets.
2. Embedded projectors retain incidence and endpoint position.
3. Noncommutative transition data record holonomy.
4. A realization theorem decides whether the packet has a restricted representative and at what cost.

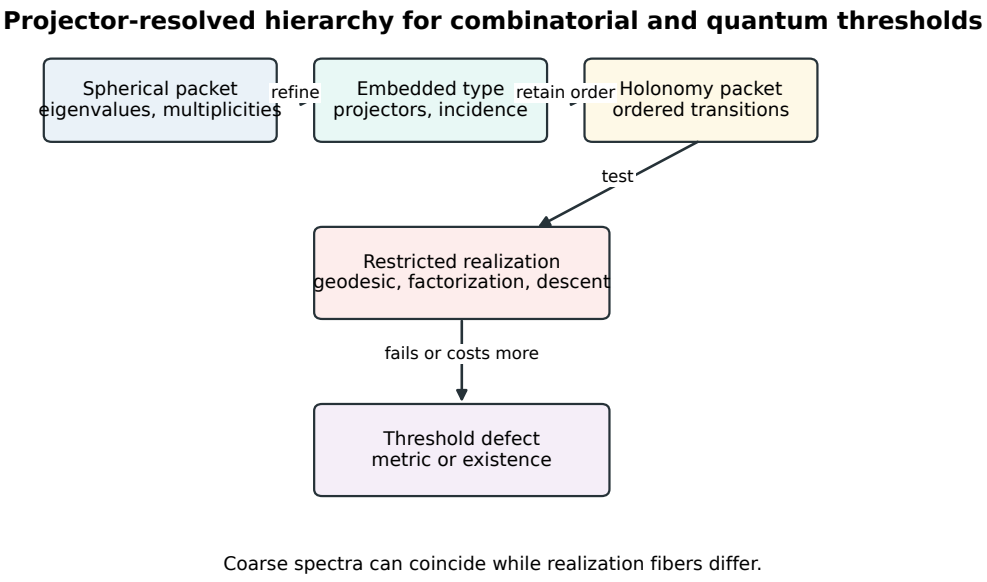


Figure 3.2: The spectral-to-realization hierarchy.

5. Direct sums and tensor products propagate local defects.

Scope boundary

A packet hierarchy organizes evidence. It does not replace the final lower bound, positivity argument, orthogonality construction, or descent theorem. Classification and realization remain distinct layers.

Chapter 4

Matroidal state geometry, quiver factorizations, and root-system catalysis

4.1 Intermediate states as a representable matroid

A linear circuit visits vectors that satisfy exact dependence relations. If $\mathcal{V}$ is the set of all row states appearing during a computation, then $\mathcal{V}$ determines a representable matroid. A transvection

$$u \longmapsto u + av$$

creates a three-point projective line

$$[u],\ [v],\ [u + av].$$

The ambient circuit chooses a tour through this projective configuration. A flag-preserving circuit must lift the same dependencies through prescribed pivot positions.

Definition 4.1 (Flagged matroid tour)**.** Let M be a representable matroid with designated initial basis, target basis, and consumer states. A *flagged tour* is a sequence of basis exchanges or transvections in which the ith carrier remains inside a specified filtration layer. An ambient tour drops this constraint.

The six-row atom realizes the graphic regular matroid $M(K_4)$. Its ambient state tour is shorter than every compatible flag lift. Regularity explains why the same dependence pattern survives every field and, through integer matrices, every nonzero commutative coefficient ring in the stated model.

4.2 Matroidal classification problem

The first general classification problem is:

> Which regular matroids admit designated carrier and consumer roles for which an ambient tour is shorter than every flag-compatible tour?

Natural subclasses are graphic, cographic, and network matrices. Potential invariants include branch width, circuit size, excluded minors, and the number of consumer incidences supported by one projective line.

Conjectural frontier

Minimal field-universal catalytic atoms should be governed by regular matroids, while characteristic-dependent atoms should arise from representable matroids with excluded-characteristic minors.

This conjecture is a programmatic guide. The core paper establishes only the $M(K_4)$ instance.

4.3 Dynamic rank-one factorization

Consider a height-two block shear

$$E(C) = \begin{pmatrix} I_s & 0 \\ C & I_t \end{pmatrix}.$$

A restricted circuit may change source and output bases while transmitting rank-one terms. Let $P_r \in U_s$ be the source basis before the rth transmission and let $Q_{>r} \in U_t$ be the future output basis. A cross gate contributes

$$Q_{>r} e_{i_r} e_{j_r}^T P_{<r}$$

to the final block. Hence

$$C = \sum_{r=1}^{c} Q_{>r} e_{i_r} e_{j_r}^T P_{<r},$$

with closed basis paths

$$P_{\text{final}} = I_s, \qquad Q_{\text{final}} = I_t.$$

Definition 4.2 (Dynamic rank-one cost)**.** The dynamic rank-one cost of C is the minimum of

$$\text{source basis motion} + \text{output basis motion} + \text{number of transmissions}$$

over such timed decompositions.

Ordinary rank retains only the number of transmissions after arbitrary bases are given for free. Circuit complexity charges the closed paths needed to expose and restore those bases.

4.4 Quiver gate models

Let $Q = (Q_0, Q_1)$ be a quiver. Associate a vector space V_v to every vertex. Internal gates change bases inside V_v, while edge gates transmit low-rank maps along arrows. A target is a representation of the path algebra or an automorphism of a filtered quiver module.

This generalization includes:

- block-triangular linear circuits,
- network coding with reversible local state,
- layered dataflow architectures,
- incidence-algebra transformations,
- tensor networks with reusable bond states.

A catalytic excursion can temporarily activate a forbidden arrow or enlarge a vertex space. The restored target still lies in the original quiver representation class.

Formal research skeleton

A quiver threshold theorem should specify a forbidden-arrow motif, an exact ambient word, a projection-based restricted lower bound, and an amplification operation under quiver substitution. The unitriangular snapshot family is the linearly ordered one-dimensional-vertex case.

4.5 Pattern groups and architecture constraints

For a poset P, define the pattern group

$$U_P(K) = \{I + A : A_{ij} = 0 \text{ unless } j <_P i\}.$$

When the allowed positions are transitively closed, U_P is a group. Compare the intrinsic metric generated by P-allowed transvections with the metric induced by all transvections.

The first threshold now depends on the forbidden-edge geometry of P. One may ask whether a six-role catalytic motif embeds as a generator-preserving retract. If it does, the larger architecture inherits a defect.

Proposition 4.3 (Retraction inheritance)**.** *Suppose a smaller nested pair $H \leq G$ embeds into $H' \leq G'$ and there is a generator-nonincreasing retraction $H' \to H$. Then every defect in $H \leq G$ gives a defect at least as large in $H' \leq G'$.*

The proof projects a restricted word in the larger system and embeds the shorter ambient word from the smaller system.

4.6 Root subgroups and Chevalley relations

Let G be a split reductive group [9, 16] with root system Φ. Let $\Psi \subseteq \Phi$ be a closed set of allowed roots and let U_Ψ be generated by the root groups $x_\beta(t)$ for $\beta \in \Psi$. Define

$$\ell_\Psi(u) = \min\{L : u = x_{\beta_L}(t_L) \cdots x_{\beta_1}(t_1),\ \beta_i \in \Psi\},$$

and let ℓ_Φ permit all roots.

The Chevalley conjugation formula has the schematic form

$$x_\alpha(a)x_\beta(b)x_\alpha(a)^{-1} = x_\beta(b) \prod_{k>0} x_{\beta+k\alpha}(c_k a^k b).$$

A forbidden root α can therefore create several allowed root effects before being removed.

4.7 Type A and beyond

Row transvections are root-group elements of type A. The established unitriangular atom occurs in A_5. In nonsimply laced systems, root strings can have length greater than two. This raises the possibility that one ambient root can create multiple restricted effects per consumer.

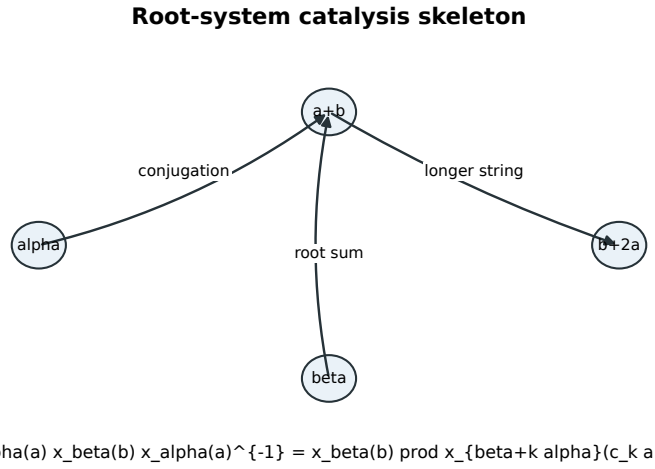


Figure 4.1: A root-string template for catalytic generation. Longer strings may support more gain per opening and closing pair.

Conjecture 4.4 (Long-root catalytic gain). There exist gate alphabets in types B, C, or G_2 for which a clean one-catalyst multiplicative distortion exceeds the type-A snapshot limit of $3/2$.

No lower-bound certificate is presently supplied. The conjecture identifies a concrete reason to test nonsimply laced root systems rather than merely repeating type A computations.

4.8 Symplectic root systems and Clifford logic

The binary Clifford group modulo phases acts through $\mathrm{Sp}(2n, 2)$. A parabolic subgroup stabilizing a Lagrangian is the symplectic analogue of a Borel or flag-preserving circuit class. Hadamard gates leave this parabolic, while S, CZ, and CX preserve it. The exact Clifford censuses in Chapter 7 are therefore a first root-system extension of the linear theory.

4.9 Categorical realization defects

The group framework does not cover measurement, feedforward, postselection, or entangled design objects. Let

$$\mathcal{C}_{\mathrm{res}} \subseteq \mathcal{C}_{\mathrm{amb}}$$

be monoidal categories with costed generators. A target morphism can have a finite metric defect or an infinite existence defect.

The categorical viewpoint supports:

- unitary versus measurement-assisted circuits,
- product-state versus entangled realizations,
- clean versus borrowed ancilla categories,
- symmetry-preserving versus symmetry-breaking channels,
- exact versus approximate completion.

The common invariant is still the cost of a restored target. What changes is the geometry of the path and the meaning of holonomy.

4.10 A general research dichotomy

The broad program separates into two tasks.

1. **Construction:** exhibit an ambient excursion whose repeated consumers produce a cheaper target.
2. **Certification:** prove that the restricted category cannot share the same obligations as efficiently.

Conjugation identities often solve the first task. The second is the difficult part and requires model-specific invariants such as interval-tree rank, symplectic flag length, algebraic degree, phase-polynomial rank, covariance constraints, or branch complexity.

Part II

Established unitriangular theory

Chapter 5

The complete linear core

Scope boundary

This chapter incorporates the publication-ready unitriangular manuscript. Its symbolic theorems, exact finite searches, and asymptotic constructions are the proved foundation for the broader gate-set program. The notation $\ell_{\downarrow}$ and ℓ_* specializes ℓ_{res} and ℓ_{amb}.

5.1 Introduction

A CNOT circuit on n labeled wires represents an element of $\mathrm{GL}_n(\mathbb{F}_2) = \mathrm{SL}_n(\mathbb{F}_2)$, and minimizing its gate count is the word-metric problem for transvection generators. This formulation supports asymptotically efficient synthesis, architecture-aware elimination, computable lower bounds, and exact small-width optimization [28, 25, 17, 14, 6, 8, 10, 33]. Recent work has also produced stronger explicit lower-bound families and established NP-completeness of exact all-to-all CNOT distance [18, 12, 2]. The present paper studies a different comparison: the target is lower unitriangular, but the circuit is either required to respect that ordered structure or allowed to leave it temporarily.

Let R be a nonzero commutative ring with identity. A row transvection is

$$\tau_{t,s}(a) = I + aE_{t,s}, \qquad t \neq s, \qquad a \in R \setminus \{0\},$$

acting by

$$R_t \longleftarrow R_t + aR_s.$$

The transvection is *lower directed* when $s < t$. The lower-directed gates generate $\mathrm{U}_n(R)$. We compare

$$\ell_{\downarrow}(A) = \min\{\text{number of lower-directed transvections producing } A\}$$

with

$$\ell_*(A) = \min\{\text{number of arbitrary row transvections producing } A\}, \qquad A \in \mathrm{U}_n(R).$$

For a field $\mathbb{F}$, the first metric is intrinsic to the unipotent radical and the second is induced from the elementary group $\mathrm{E}_n(\mathbb{F}) = \mathrm{SL}_n(\mathbb{F})$.

The equality $\ell_{\downarrow}(A) = \ell_*(A)$ is tempting because the target and the obvious Gaussian-elimination paths are lower unitriangular. It is false. The failure first occurs at six wires over $\mathbb{F}_2$ and $\mathbb{F}_3$, and it is caused by a reusable temporary carrier rather than by a better ordering of the same lower gates.

Main results

The first result is entirely analytic and does not require a field.

Theorem 5.1 (Universal catalytic atom over rings)**.** *For every nonzero commutative ring R with identity, let*

$$C = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{pmatrix} \in \mathrm{U}_6(R). \tag{5.1}$$

Then

$$\ell_{\downarrow}(C) = 7, \qquad \ell_*(C) = 6.$$

The six-gate ambient circuit uses row 1 as a dirty workspace. It first adds row 2 into row 1, exports the carrier $e_1 + e_2$, changes the carrier to $e_0 + e_1 + e_2$, exports it twice, and finally removes the temporary e_2 component. No extra wire is introduced. The workspace is one of the target rows and is restored to its prescribed value.

The next result is computer assisted. All state spaces are finite, every transition is exact, and the complete source is included in the ancillary archive.

Theorem 5.2 (Exact binary threshold and census)**.** *Over $\mathbb{F}_2$:*

(i) *$\ell_{\downarrow}(A) = \ell_*(A)$ for every $A \in \mathrm{U}_n(\mathbb{F}_2)$ when $n \le 5$.*

(ii) *Among the $2^{15} = 32768$ matrices in $\mathrm{U}_6(\mathbb{F}_2)$, exactly twenty satisfy $\ell_{\downarrow}(A) > \ell_*(A)$.*

(iii) *Every defective target satisfies*

$$\ell_{\downarrow}(A) = 7, \qquad \ell_*(A) = 6.$$

No other distance pair is defective.

The twenty targets have a uniform combinatorial explanation.

Theorem 5.3 (Minimal catalytic classification)**.** *The twenty defective targets in $\mathrm{U}_6(\mathbb{F}_2)$ are exactly the targets generated by a simple one-catalyst word having three cross interactions, one phase interaction, and a nontrivial before/after split across the phase. The numbers arising from the three possible left/right auxiliary splits are*

$$6, \qquad 8, \qquad 6,$$

for a total of twenty.

The field-independent atom and exact ternary search also give the same finite threshold over $\mathbb{F}_3$.

Theorem 5.4 (Exact ternary threshold)**.** *For $n \le 5$ and every $A \in \mathrm{U}_n(\mathbb{F}_3)$,*

$$\ell_{\downarrow}(A) = \ell_*(A).$$

The matrix $C \in \mathrm{U}_6(\mathbb{F}_3)$ of Equation (5.1) has distances $(7, 6)$. Hence six is the first defective dimension over $\mathbb{F}_3$.

To pass from the first threshold to arbitrarily large connected examples, we introduce a multiscale rank certificate and a family that exports successive carrier states.

Theorem 5.5 (Connected snapshot family)**.** *Let $u \geq 1$, let $1 \leq v \leq u+1$, and choose distinct phases*

$$0 \leq s_1 < \cdots < s_v \leq u.$$

There is a connected target

$$A(u; s_1, \ldots, s_v) \in \mathrm{U}_{u+v+2}(\mathbb{F})$$

with

$$\ell_{\downarrow}\big(A(u; s_1, \ldots, s_v)\big) = u + 2v$$

and

$$\ell_{*}\big(A(u; s_1, \ldots, s_v)\big) \leq u + v + 2.$$

Hence its certified defect is at least $v - 2$.

Corollary 5.6 (Connected defects in every dimension)**.** *For every field and every $n \geq 7$, there is a connected target $A_n \in \mathrm{U}_n(\mathbb{F})$ satisfying*

$$\ell_{\downarrow}(A_n) - \ell_{*}(A_n) \geq \left\lfloor \frac{n-5}{2} \right\rfloor.$$

The same construction remains effective under stringent restrictions on the ambient resource.

Theorem 5.7 (Two forbidden gates and weighted penalties)**.** *For every field and every $n \geq 7$, there is a connected target $A_n \in \mathrm{U}_n(\mathbb{F})$ such that*

$$\ell_{\leq 0}(A_n) - \ell_{\leq 2}(A_n) \geq \left\lfloor \frac{n-5}{2} \right\rfloor,$$

where $\ell_{\leq q}$ permits at most q upper-directed gates. More generally, if lower gates have cost 1 and upper gates have cost $\lambda > 0$, then

$$\ell_{\downarrow}(A_n) - \ell_{\lambda}(A_n) \geq \left\lfloor \frac{n-1}{2} \right\rfloor - 2\lambda.$$

Thus no fixed upper-gate budget of two, and no fixed finite upper-gate penalty, suppresses catalytic escape asymptotically.

The same paths remain extremely close to the restricted subgroup. For a matrix X over a field, let X^{+} be its strict upper part, let $r^{+}(X) = \operatorname{rank} X^{+}$, and let $b^{+}(X)$ be the largest upper displacement $j - i$ with $X_{ij} \neq 0$. If $\ell^{(r,b)}$ denotes the minimum length among circuits satisfying $r^{+}(X_t) \leq r$ and $b^{+}(X_t) \leq b$ at every prefix, then the connected family can be chosen so that

$$\ell^{(0,0)}(A_n) - \ell^{(1,1)}(A_n) \geq \left\lfloor \frac{n-5}{2} \right\rfloor.$$

Thus a rank-one excursion across one adjacent flag layer can create a linear saving.

At fixed dimension, however, defects occupy thin algebraic strata.

Theorem 5.8 (Generic rigidity and defect codimension)**.** *Let K be an infinite field and put $N = \binom{n}{2}$. There is a nonempty Zariski-open subset $\Omega_n \subseteq \mathrm{U}_n$ with $\Omega_n(K) \neq \varnothing$ such that*

$$\ell_{\downarrow}(A) = \ell_*(A) = N \qquad (A \in \Omega_n(K)).$$

Moreover, after base change to an algebraic closure $\overline{K}$, for

$$\mathcal{D}_{n,\geq d}(\overline{K}) = \{A \in \mathrm{U}_n(\overline{K}) : \Delta(A) \geq d\},$$

one has

$$\operatorname{codim}_{(\mathrm{U}_n)_{\overline{K}}} \overline{\mathcal{D}_{n,\geq d}(\overline{K})} \geq d.$$

Corollary 5.9 (Finite-field sparsity)**.** *Let $N = \binom{n}{2}$ and $M = n(n-1)$. For $1 \leq d \leq N$,*

$$\mathbb{P}_{A \in \mathrm{U}_n(\mathbb{F}_q)}[\Delta(A) \geq d] \leq \min\{1, M^{N-d} q^{-d}\}.$$

For fixed n, a uniformly random target therefore has defect zero with probability $1 - O_n(q^{-1})$ as $q \to \infty$.

Tensor products produce a stronger asymptotic separation.

Theorem 5.10 (Tensor-amplified snapshot family)**.** *Let $A = A(u; s_1, \ldots, s_v)$ have dimension $m = u + v + 2$. For $N = m^r$,*

$$\ell_{\downarrow}(A^{\otimes r}) = r m^{r-1}(u + 2v),$$

while

$$\ell_*(A^{\otimes r}) \leq r m^r.$$

Consequently,

$$\ell_{\downarrow}(A^{\otimes r}) - \ell_*(A^{\otimes r}) \geq \frac{v-2}{m \log_2 m} N \log_2 N.$$

Corollary 5.11 (Explicit $n \log n$ distortion)**.** *For*

$$D_n(\mathbb{F}) = \sup_{A \in \mathrm{U}_n(\mathbb{F})} \big(\ell_{\downarrow}(A) - \ell_*(A)\big),$$

one has, for every field and every n,

$$D_n(\mathbb{F}) \geq \frac{8}{21 \log_2 21} n \log_2 n - \frac{82}{105} n.$$

The coefficient is attained within the snapshot family by the twenty-one-dimensional seed

$$A(9; 0, 1, \ldots, 9).$$

Conceptual interpretation

The ambient circuit is shorter because one forbidden-direction gate creates a carrier that is read by several later gates before it is cleaned. The saving is therefore a reuse effect. The relevant object is not an ancillary wire in the usual sense. It is a temporary excursion from a flag-preserving frame space into the full frame graph.

Over $\mathbb{F}_2$, the carrier states used by the canonical word form six of the seven points of $\mathrm{PG}(2,2)$. The four projective lines supported on those six points form a Pasch configuration. The same

six vectors represent the graphic matroid $M(K_4)$ over every field; this regular-matroid structure isolates the characteristic-independent core of the atom. At the spectral level, the distance to a target is the first nonzero coefficient of an endpoint Green kernel. For C, the ambient kernel begins in degree six and the restricted kernel begins in degree seven.

The higher-dimensional families reveal three additional principles. First, an interval-tree certificate can make the restricted metric exactly additive across scales even when the target is connected. Second, two forbidden gates can support a linear saving, even if each forbidden gate carries any fixed finite penalty. Third, tensor products add normalized restricted complexity while ambient complexity remains subadditive. This converts one local carrier advantage into a superlinear global defect. Finally, the algebraic-geometric bounds show that these amplifiable atoms are exceptional at fixed dimension: a defect of size d imposes at least d algebraic codimensions. The resulting picture is one of algebraic rarity coupled with asymptotic amplification.

Scope

The results concern in-place reversible linear circuits on labeled rows. They do not by themselves imply a faster general CNOT synthesis algorithm, a lower bound for unrestricted XOR directed acyclic graphs, or a result about integer multiplication. Their content is a precise comparison between a flag-preserving gate model and the ambient transvection model.

The paper is organized as follows. Section 5.2 fixes the circuit and metric conventions and gives the Schreier-holonomy interpretation. Section 5.3 proves the universal ring-valued atom. Section 5.4 develops the catalytic normal form and counts the twenty minimal motifs. Section 5.5 gives the exact binary and ternary enumerations and proves the search procedure complete. Section 5.6 gives the projective, regular-matroid, and Green-kernel formulations. Section 5.7 develops interval-tree certificates, exact cases through dimension nine, and connected families in every larger dimension. Section 5.8 proves bounded, weighted, and microscopic-excursion theorems. Section 5.9 proves generic rigidity, defect-codimension bounds, and finite-field sparsity. Section 5.10 proves block, recursive, and tensor amplification, introduces stable tensor capacity, and derives the $n \log n$ extremal bound. The appendices provide the exact target list, algorithms, and reproducibility data.

Proof and computation status

The universal atom, catalytic identities, interval-tree certificates, connected families, resource bounds, algebraic-rarity theorems, amplification results, and stable tensor limit are proved symbolically. Complete search is used only for the first-threshold censuses, the equality between the twenty computed defects and the twenty analytic motifs, and the exact ambient distances of the selected seven-, eight-, and nine-dimensional targets. Every computer-assisted statement is paired with a finite-state completeness argument and a reproducible implementation in the supplement.

5.2 Circuit model and metric comparison

5.2.1 Transvections and row conventions

Fix a nonzero commutative ring R with identity and an integer $n \geq 2$. Rows and columns are indexed by

$$0, 1, \ldots, n-1.$$

For $s \neq t$ and $a \in R \setminus \{0\}$, set

$$\tau_{t,s}(a) = I + aE_{t,s}.$$

Left multiplication by $\tau_{t,s}(a)$ performs

$$R_t \longleftarrow R_t + aR_s.$$

We call s the source and t the target. A circuit word

$$w = (\tau_{t_m,s_m}(a_m), \ldots, \tau_{t_1,s_1}(a_1))$$

produces

$$[w] = \tau_{t_m,s_m}(a_m) \cdots \tau_{t_1,s_1}(a_1).$$

Let

$$S_n^{\downarrow}(R) = \{\tau_{t,s}(a) : 0 \le s < t < n,\ a \in R \setminus \{0\}\}$$

and

$$S_n^{*}(R) = \{\tau_{t,s}(a) : 0 \le s, t < n,\ s \neq t,\ a \in R \setminus \{0\}\}.$$

Both sets are symmetric because

$$\tau_{t,s}(a)^{-1} = \tau_{t,s}(-a).$$

Let

$$\mathrm{E}_n(R) = \langle S_n^{*}(R) \rangle.$$

When $R = \mathbb{F}$ is a field and $n \ge 2$, elementary elimination gives

$$\mathrm{E}_n(\mathbb{F}) = \mathrm{SL}_n(\mathbb{F}).$$

The ambient metric below is therefore the transvection word metric in $\mathrm{SL}_n(\mathbb{F})$, viewed inside $\mathrm{GL}_n(\mathbb{F})$. When $\mathbb{F} = \mathbb{F}_2$, one has $\mathrm{SL}_n(\mathbb{F}_2) = \mathrm{GL}_n(\mathbb{F}_2)$, every generator is an involution, and it is exactly a CNOT gate under the usual row-action convention.

Lemma 5.12. *The set $S_n^{\downarrow}(R)$ generates $\mathrm{U}_n(R)$. Every prefix of a lower-directed circuit belongs to $\mathrm{U}_n(R)$.*

Proof. A lower-directed transvection is lower unitriangular, and products of lower-unitriangular matrices remain lower unitriangular. Conversely, ordinary elimination of the strict lower entries, processed in a compatible order, expresses every element of $\mathrm{U}_n(R)$ as a product of lower-directed transvections. □

Definition 5.13 (Intrinsic and ambient lengths)**.** For $A \in \mathrm{U}_n(R)$, define

$$\ell_{\downarrow}(A) = |A|_{S_n^{\downarrow}(R)}, \qquad \ell_{*}(A) = |A|_{S_n^{*}(R)}.$$

The *catalytic defect* is

$$\Delta(A) = \ell_{\downarrow}(A) - \ell_{*}(A) \ge 0.$$

The inequality follows because $S_n^{\downarrow}(R) \subseteq S_n^{*}(R)$. The central question is whether the inclusion can be strict on a lower-unitriangular target.

5.2.2 Frame and flag interpretation

For the geometric interpretation in this subsection, assume $R = \mathbb{F}$ is a field. An invertible matrix is an ordered basis

$$(R_0, \dots, R_{n-1})$$

of $\mathbb{F}^n$. A transvection replaces one basis vector by its sum with a nonzero multiple of another. Let

$$V_i = \operatorname{span}(e_0, \dots, e_i).$$

A matrix lies in $\mathrm{U}_n(\mathbb{F})$ exactly when

$$R_i \in e_i + V_{i-1}$$

for every i. Thus lower circuits move among frames compatible with the standard complete flag

$$V_0 \subset V_1 \subset \cdots \subset V_{n-1}.$$

An upper-directed gate leaves this fiber of the flag map. A strict defect means that the flag-compatible frame subgraph is not geodesically convex in the full transvection graph.

5.2.3 Schreier holonomy

Set

$$G = \mathrm{E}_n(R), \qquad H = \mathrm{U}_n(R).$$

Consider the right Schreier graph on the left cosets $H\backslash G$: a generator $s \in S_n^*(R)$ gives an edge

$$Hg \longrightarrow Hgs.$$

A word representing $h \in H$ traces a based loop at the coset H. Its label product is the holonomy h.

Proposition 5.14 (Holonomy-compression interpretation)**.** *For $h \in H$, the ambient length $\ell_*(h)$ is the least length of a labeled based loop in $H\backslash G$ with holonomy h. The restricted length $\ell_\downarrow(h)$ is the least length among such words whose every prefix remains in H. Consequently,*

$$\Delta(h) = \textit{intrinsic flag-preserving cost} - \textit{shortest ambient holonomy-loop cost.}$$

Proof. A word $w = s_1 \cdots s_k$ ends at the coset Hw. It is a based loop exactly when $w \in H$, and its label product is then the represented element of H. Minimizing over all transvection words representing h gives $\ell_*(h)$. Requiring every generator to be lower directed keeps every prefix in H and gives $\ell_\downarrow(h)$. □

A catalytic shortcut is therefore a nonconstant quotient excursion whose accumulated holonomy is more expensive to produce inside the base fiber.

5.2.4 Two elementary counting lemmas

The proofs of the universal atom use only target counts and pivot support.

Lemma 5.15 (Target count)**.** *Suppose a circuit starts at the identity and finishes at A.*

(i) *If row i of A differs from e_i, then row i is targeted at least once.*

(ii) *If row i of A equals e_i and is targeted, then it is targeted at least twice.*

Proof. A row changes only when it is targeted. Every row of an invertible matrix over R is unimodular. If $aR_s = 0$ for such a row, then multiplying a unimodular relation for R_s by a gives $a = 0$. Hence a gate with nonzero coefficient changes its target by a nonzero vector, so one targeting cannot leave that row equal to its initial value. □

Lemma 5.16 (One-target source condition)**.** *Let $W \leq R^n$. Suppose row t starts at e_t, is targeted exactly once, and finishes at*

$$e_t + w, \qquad w \in W,$$

where at least one coordinate of w is a unit of R. At the moment of its unique targeting, the gate coefficient is a unit and the source row is a unit multiple of w.

Proof. If the unique gate is $R_t \leftarrow R_t + aR_s$, then $aR_s = w$. Choose a coordinate j for which w_j is a unit. Since $a(R_s)_j = w_j$, the factor a is a unit in the commutative ring R. Hence $R_s = a^{-1}w$. □

The second lemma is especially restrictive when W is a low-dimensional source space and output rows carry unique pivot coordinates outside W.

5.3 The universal six-row atom

Let R be a nonzero commutative ring with identity, and let

$$v = e_1 + e_2, \qquad w = e_0 + e_1 + e_2, \qquad W = \operatorname{span}(e_0, e_1, e_2).$$

The target C of Equation (5.1) has rows

$$\begin{aligned} R_0 &= e_0, & R_1 &= e_1 + e_0, & R_2 &= e_2, \\ R_3 &= e_3 + v, & R_4 &= e_4 + w, & R_5 &= e_5 + w. \end{aligned}$$

5.3.1 A six-gate ambient word

Proposition 5.17. *For every nonzero commutative ring R with identity,*

$$\ell_*(C) \leq 6.$$

Proof. Apply the following six gates:

$$R_1 \leftarrow R_1 + R_2, \tag{5.2}$$
$$R_3 \leftarrow R_3 + R_1, \tag{5.3}$$
$$R_1 \leftarrow R_1 + R_0, \tag{5.4}$$
$$R_4 \leftarrow R_4 + R_1, \tag{5.5}$$
$$R_5 \leftarrow R_5 + R_1, \tag{5.6}$$
$$R_1 \leftarrow R_1 - R_2. \tag{5.7}$$

After Equation (5.2), row 1 contains v. Gate Equation (5.3) copies v to row 3. Gate Equation (5.4) changes the carrier to w, which is copied to rows 4 and 5 by Equations (5.5) and (5.6). Gate Equation (5.7) removes the temporary e_2 component, leaving $e_1 + e_0$. Every row now agrees with C. □

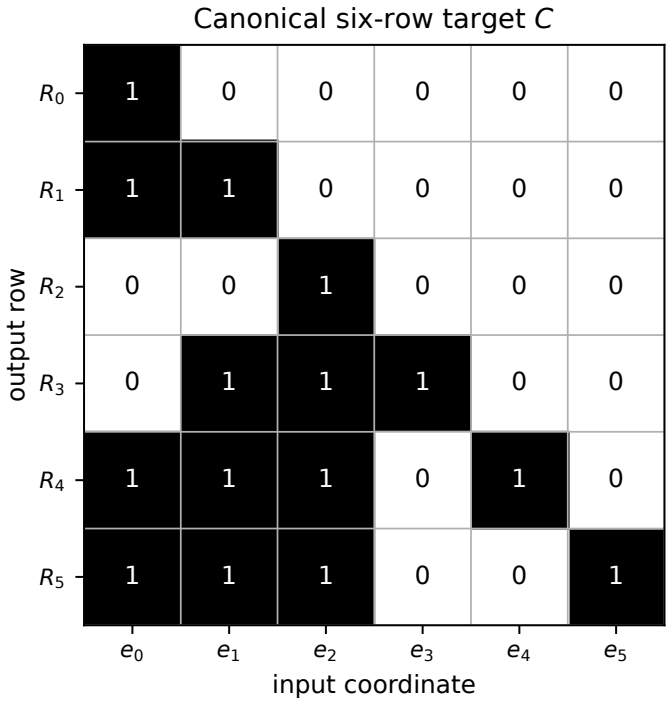


Figure 5.1: The canonical target C. Black cells are nonzero. Rows 0 and 2 are restored exactly, while rows $3, 4, 5$ require two distinct additions v and w from the source space W.

5.3.2 A seven-gate lower word

Proposition 5.18. *For every nonzero commutative ring R with identity,*

$$\ell_{\downarrow}(C) \leq 7.$$

Proof. The lower-directed word

$$\begin{aligned} R_1 &\leftarrow R_1 + R_0, \\ R_2 &\leftarrow R_2 + R_1, \\ R_3 &\leftarrow R_3 - R_0, \\ R_3 &\leftarrow R_3 + R_2, \\ R_4 &\leftarrow R_4 + R_2, \\ R_5 &\leftarrow R_5 + R_2, \\ R_2 &\leftarrow R_2 - R_1 \end{aligned}$$

produces C. After the first two gates, row 2 contains w. Rows 4 and 5 copy that state. The two gates on row 3 cancel the unwanted e_0 and leave v. The final gate restores row 2. □

5.3.3 Restricted lower bound

Proposition 5.19. *For every nonzero commutative ring R with identity,*

$$\ell_{\downarrow}(C) \geq 7.$$

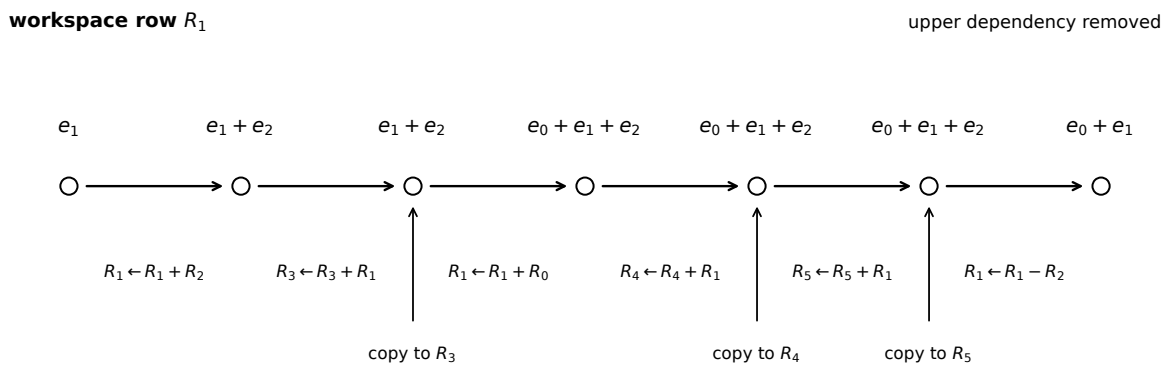


Figure 5.2: Carrier timeline for the six-gate ambient word. The two upper-directed operations are the opening and closing gates on row 1. The state $e_1 + e_2$ is exported once and $e_0 + e_1 + e_2$ is exported twice.

Proof. Rows $1, 3, 4, 5$ differ from their initial values, so they must each be targeted. Assume for contradiction that a lower-directed word of length at most six produces C.

First suppose row 2 is never targeted. Row 0 remains e_0, row 2 remains e_2, and every state ever assumed by row 1 lies in $\mathrm{span}(e_0, e_1)$ because the only possible source below row 1 is row 0. Every output row retains coefficient 1 on its own pivot coordinate in a lower-directed circuit, so no output row can lie in W. Hence no available source row is a unit multiple of either

$$v = e_1 + e_2 \quad \text{or} \quad w = e_0 + e_1 + e_2.$$

By Theorem 5.16, each of rows $3, 4, 5$ must be targeted at least twice. Together with the required targeting of row 1, this gives at least seven gates.

Now suppose row 2 is targeted. Since its final value is e_2, Theorem 5.15 forces at least two targetings of row 2. A six-gate word must therefore target row 2 exactly twice and each of rows $1, 3, 4, 5$ exactly once. Each output row $3, 4, 5$ must receive its complete addition in its unique targeting. Row 0 cannot supply v or w, and row 1 never contains an e_2 component. Therefore row 2 must supply all three outputs, up to unit scaling. But two targetings of row 2 expose at most one noninitial intermediate state before row 2 returns to e_2, while the outputs require both distinct states v and w. This is impossible. □

5.3.4 Ambient lower bound

Proposition 5.20. *For every nonzero commutative ring R with identity,*

$$\ell_*(C) \geq 6.$$

Proof. Assume that a five-gate ambient word produces C. Rows $1, 3, 4, 5$ must each be targeted. Rows 0 and 2 are unchanged. If either were targeted, it would require at least two targetings, giving at least six gates in total. Thus rows 0 and 2 are never targeted. Among rows $1, 3, 4, 5$, exactly one is targeted twice and the other three are targeted once.

Project the rows to the quotient module R^6/W. Rows $0, 1, 2$ have zero image, while rows $3, 4, 5$ begin and end at the three distinct quotient basis vectors of the free quotient module. An output row targeted exactly once must therefore use a source lying in W, by Theorem 5.16.

Suppose first that row 1 is the twice-targeted row. The three output rows are each targeted once. Rows 0 and 2 supply only scalar multiples of e_0 and e_2. Row 1 begins at e_1, has one intermediate state, and ends at $e_1 + e_0$. Thus at most one of the two required unit-multiple classes of v and w can appear as an available source. The three outputs require both, a contradiction.

Suppose instead that an output row, say row $r \in \{3, 4, 5\}$, is targeted twice. The other two outputs and row 1 are targeted once. We claim that row r cannot lie in W between its two targetings. To cancel its unique quotient pivot e_r at the first targeting, the source would already have to contain an e_r component. Before row r is first targeted, the only way another row can acquire such a component is to be targeted using row r as source. That other row is targeted only once and would then retain an unwanted e_r component in its final value. Therefore row r never becomes a W-valued carrier. The only W-valued source states are rows 0 and 2 and the two states of singly targeted row 1, namely its initial and final states. None supplies both v and w. The two singly targeted outputs cannot be completed, again a contradiction. □

Proof of Theorem 5.1. Combine Theorems 5.17 to 5.20. □

Remark 5.21. The proof uses only unimodular rows, unit coordinates, and pivot support. It does not use division or a characteristic assumption. In characteristic two the closing subtraction equals addition.

Corollary 5.22 (Failure of geodesic convexity)**.** *For every field $\mathbb{F}$ and every $n \geq 6$, the lower-unitriangular frame subgraph is not geodesically convex in the transvection Cayley graph of* $\mathrm{E}_n(\mathbb{F}) = \mathrm{SL}_n(\mathbb{F})$.

Proof. Embed C as $C \oplus I_{n-6}$. A six-gate ambient word acts on the first six rows. In a lower-directed circuit, rows with index at least six cannot be used as sources for targets among the first six rows, so the restricted lower bound of seven persists. □

For the remainder of the paper, unless explicitly stated otherwise, the coefficient domain is a field $\mathbb{F}$.

5.4 Catalytic normal forms and the six-role threshold

The universal atom is one instance of a root-conjugation mechanism. This section isolates that mechanism and explains the count twenty.

5.4.1 Root conjugation identities

Let $p < q < y$ and $x < p < q$. Direct multiplication gives

$$\tau_{p,q}(-a)\,\tau_{y,p}(b)\,\tau_{p,q}(a) = \tau_{y,p}(b)\,\tau_{y,q}(ab), \tag{5.8}$$

and

$$\tau_{p,q}(-a)\,\tau_{q,x}(b)\,\tau_{p,q}(a) = \tau_{q,x}(b)\,\tau_{p,x}(-ab). \tag{5.9}$$

A single forbidden upper transvection can therefore turn one lower interaction into two lower effects. Interactions of the forms $x \to p$ and $q \to y$ commute through the excursion without duplication. We call the duplicated interactions *cross interactions* and the unduplicated interactions *phase interactions*.

Definition 5.23 (Simple one-catalyst word)**.** A simple one-catalyst word has the form

$$\tau_{p,q}(-a)\, h_m \cdots h_1\, \tau_{p,q}(a), \qquad p < q,$$

where every h_i is lower directed and is one of

$$\tau_{q,x}(b), \quad \tau_{p,x}(b), \quad \tau_{y,p}(b), \quad \tau_{y,q}(b)$$

with $x < p < q < y$. Auxiliary rows are distinct unless stated otherwise.

The circuit is read from right to left in the matrix product and from left to right as a gate list. The opening and closing operations temporarily place the q component in row p.

Proposition 5.24 (Cross-interaction accounting)**.** *Let a simple one-catalyst word have m middle interactions, of which c are cross interactions. Its target admits a lower-directed realization of length at most*

$$m + c.$$

Consequently, if the ambient word of length $m + 2$ is strictly shorter than every lower word, then

$$c \geq 3.$$

Proof. Conjugate each middle generator using Equations (5.8) and (5.9). A phase interaction remains one lower transvection, while a cross interaction becomes a product of two lower transvections. This gives length at most $m + c$. If $c \leq 2$, then $m + c \leq m + 2$, so the ambient excursion cannot be strictly better. □

Three consumers are therefore the first possible point at which the cost two of opening and closing can be amortized.

5.4.2 Why an all-cross word is insufficient

Within the distinct-leaf normal form, cross count alone is not enough.

Lemma 5.25 (All-cross mirror)**.** *Suppose a simple one-catalyst word contains only cross interactions, with left inputs X and right outputs Y. Then its target has a lower-directed realization with the same number of gates.*

Proof. Write the opening coefficient as $a \in \mathbb{F}^\times$, the left-update coefficients as b_x, and the right-read coefficients as c_y. Since the all-cross middle gates commute past one another, the ambient gate list may be written

$$R_p \leftarrow R_p + aR_q; \quad [R_q \leftarrow R_q + b_x R_x]_{x\in X}; \quad [R_y \leftarrow R_y + c_y R_p]_{y\in Y}; \quad R_p \leftarrow R_p - aR_q.$$

Put $B = \sum_{x\in X} b_x e_x$. Its final nontrivial rows are

$$R_p = e_p - aB, \qquad R_q = e_q + B, \qquad R_y = e_y + c_y e_p + ac_y e_q \quad (y \in Y).$$

The following lower-directed gate list produces the same rows:

$$R_q \leftarrow R_q + a^{-1}R_p; \quad [R_y \leftarrow R_y + ac_y R_q]_{y\in Y}; \quad [R_p \leftarrow R_p - ab_x R_x]_{x\in X}; \quad R_q \leftarrow R_q - a^{-1}R_p.$$

Indeed, after the first gate $R_q = e_q + a^{-1}e_p$, so the output gates create exactly $c_y e_p + ac_y e_q$. The source updates then leave $R_p = e_p - aB$, and the final gate changes R_q to $e_q + B$. Both words have $|X| + |Y| + 2$ gates. □

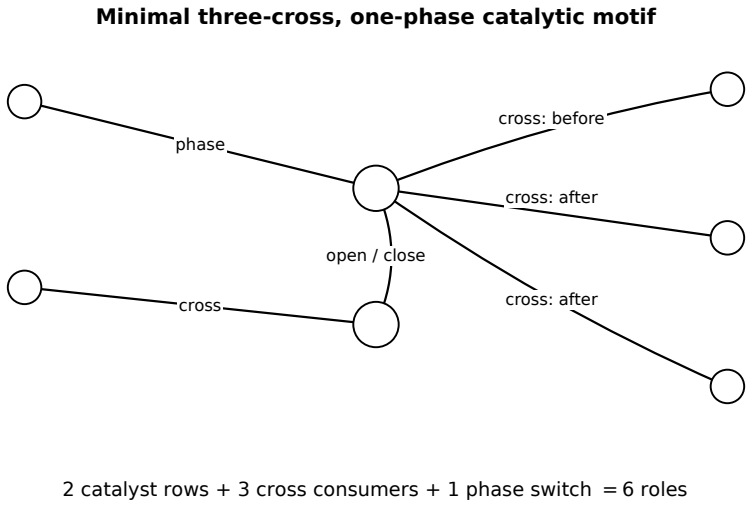


Figure 5.3: The minimal one-catalyst motif. Cross interactions reuse the opened carrier. The phase gate separates consumers that observe two different carrier states and prevents the all-cross mirror.

Thus a minimal strict shortcut needs both positive catalytic excess and a phase switch that makes temporal order matter.

Proposition 5.26 (Six-role threshold in the simple model)**.** *A strict shortcut in the distinct-leaf simple one-catalyst model requires at least six row roles:*

$$2 \text{ catalyst rows} + 3 \text{ cross interactions} + 1 \text{ phase interaction}.$$

The bound is attained by the matrix C.

Proof. By Theorem 5.24, at least three cross interactions are required. By Theorem 5.25, at least one phase interaction is required. With distinct leaves these use four auxiliary rows in addition to the two catalyst rows. The word of Theorem 5.17 has exactly this form. □

5.4.3 Enumeration of the minimal motifs

Fix six ordered rows. The catalyst pair must be adjacent in a minimal six-role word. Write

$$p < q = p + 1,$$

let L be the set of rows left of p, and let R be the set of rows right of q. Then

$$|L| + |R| = 4, \qquad |L|, |R| > 0.$$

There are two orientations.

Left phase. Choose a phase leaf $x \in L$. The remaining $|L| - 1$ left leaves interact by $u \to q$. Every right leaf is a cross read $p \to y$. To make the phase effective, the right leaves are divided into a nonempty set read before $x \to p$ and a nonempty set read after it. The number of targets is

$$|L|\big(2^{|R|} - 2\big).$$

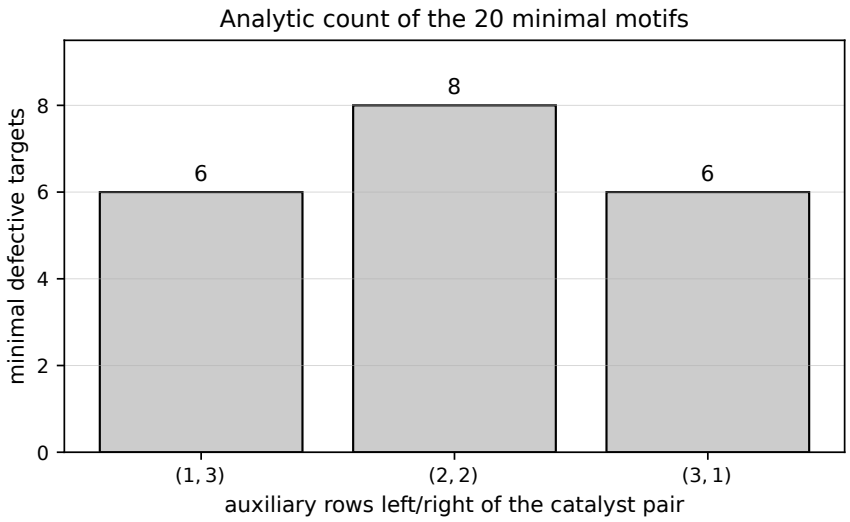


Figure 5.4: The analytic count of the twenty minimal catalytic targets.

Right phase. Choose a phase leaf $y \in R$. Every other right leaf is a cross read. The left updates into q are divided nontrivially before and after the phase gate $q \to y$. The number is

$$|R|\big(2^{|L|} - 2\big).$$

Proposition 5.27 (Motif count)**.** *For a split $(L, R) = (|L|, |R|)$ with $L + R = 4$, the number of minimal phase-separated targets is*

$$N_{L,R} = L(2^R - 2) + R(2^L - 2).$$

For the three possible splits,

$$N_{1,3} = 6, \qquad N_{2,2} = 8, \qquad N_{3,1} = 6.$$

Hence there are twenty targets.

Proof. The preceding left-phase and right-phase constructions are disjoint because the phase row lies on opposite sides of the catalyst. Within each orientation, the final support records the phase leaf and the before/after subset, so distinct choices give distinct targets. Summing gives the formula and the stated values. □

Every target counted by Theorem 5.27 has an explicit six-gate ambient word and a seven-gate lower word obtained by eliminating the three cross interactions. The nontrivial statement is that there are no other defects in dimension six and that none of these targets has a lower word of length six. That is the exact classification established in Section 5.5.

5.5 Exact finite enumeration

This section describes the computer-assisted part of the paper. The computations use integer encodings only. No floating-point decisions occur.

5.5.1 Restricted breadth-first search

Over $\mathbb{F}_2$, an element of $\mathrm{U}_n(\mathbb{F}_2)$ is determined by its $n(n-1)/2$ strict lower entries. For $n = 6$ there are $2^{15} = 32768$ states. Breadth-first search from the identity using the fifteen lower transvections gives every restricted distance and the restricted diameter.

For $n = 6$, the restricted diameter is eight. Therefore every lower-unitriangular target has ambient distance at most eight. This makes a radius-four meet-in-the-middle search complete.

5.5.2 Ambient radius-four ball and right-coset pairing

Let

$$B_4 = \{g \in \mathrm{GL}_6(\mathbb{F}_2) : \ell_*(g) \le 4\}.$$

Exact breadth-first search gives

$$|B_4| = 110626.$$

Group B_4 by right cosets of $\mathrm{U}_6(\mathbb{F}_2)$. If a, b lie in the same right coset, then

$$b^{-1}a \in \mathrm{U}_6(\mathbb{F}_2),$$

and

$$\ell_*(b^{-1}a) \le \ell_*(b) + \ell_*(a) \le 8.$$

The implementation uses a canonical right-coset key obtained by reducing matrix columns from right to left. Right multiplication by a lower-unitriangular matrix changes column j only by a combination of columns with larger index, so this reduction is constant on right cosets and separates them.

Lemma 5.28 (Meet-in-the-middle completeness)**.** *For every $u \in \mathrm{U}_6(\mathbb{F}_2)$,*

$$\ell_*(u) = \min\{\ell_*(a) + \ell_*(b) : a, b \in B_4,\ b^{-1}a = u\}.$$

Proof. Every pair on the right gives a word for u, so the right-hand side is at least $\ell_*(u)$. Conversely, $\ell_*(u) \le 8$. Take a shortest path from the identity to u and split it at a midpoint. The prefix endpoint a has distance at most four. The inverse of the remaining suffix is an element b of distance at most four, and $u = b^{-1}a$. Hence the minimum is at most $\ell_*(u)$. □

The radius-four ball splits into 6948 right-coset buckets. The largest bucket has 3775 elements. The computation checks 45178326 ordered pairs and recovers an ambient distance for every one of the 32768 targets.

5.5.3 The exact binary census

Proof of Theorem 5.2. For $n = 2, 3, 4, 5$, complete restricted breadth-first search and complete ambient meet-in-the-middle search return zero defects. The state counts are respectively $2, 8, 64, 1024$. For $n = 6$, Theorem 5.28 proves that the ambient calculation is exhaustive. The resulting distribution is Table 5.1; its counts sum to 32768, no target is missing, and the only strict pair is $(7, 6)$ with multiplicity twenty. □

Proof of Theorem 5.3. The analytic generator of Theorem 5.27 produces twenty distinct targets without reference to the distance census. Exact set comparison shows that these twenty targets are precisely the twenty off-diagonal targets in Table 5.1. Each therefore has distances $(7, 6)$, and no other dimension-six target is defective. □

Table 5.1: Complete distance distribution in $\mathrm{U}_6(\mathbb{F}_2)$.

$\ell_\downarrow$	ℓ_*	number of targets
0	0	1
1	1	15
2	2	125
3	3	720
4	4	2914
5	5	7851
6	6	12656
7	6	20
7	7	7926
8	8	540

Table 5.2: Complete no-defect computations below dimension six.

field	n	$\lvert \mathrm{U}_n(\mathbb{F})\rvert$	restricted diameter
$\mathbb{F}_2$	2	2	1
$\mathbb{F}_2$	3	8	2
$\mathbb{F}_2$	4	64	4
$\mathbb{F}_2$	5	1024	6
$\mathbb{F}_3$	2	3	1
$\mathbb{F}_3$	3	27	3
$\mathbb{F}_3$	4	729	5
$\mathbb{F}_3$	5	59049	7

5.5.4 Ternary and additional field checks

For $\mathbb{F}_3$, every nonzero scaled transvection is included. Complete searches give the data in Table 5.2.

The $n = 5$ ternary computation uses radius-three and radius-four balls, checks 38415205 right-coset pairs, and recovers all 59049 targets. It finds no defect. The field-independent Theorem 5.1 supplies a defect at $n = 6$, proving Theorem 5.4. Separate exact searches over $\mathbb{F}_3$ and $\mathbb{F}_5$ verify the distances of the canonical atom directly. The $\mathbb{F}_5$ calculation is an audit of the atom, not a complete threshold computation below dimension six.

5.5.5 Verification status

The analytic proofs of Theorems 5.1 and 5.52 do not depend on computation. The claims in Theorems 5.2 to 5.4 depend on exact finite enumeration. The source archive contains:

- the C++ programs for every breadth-first and meet-in-the-middle search;
- the Python program that independently generates the twenty motifs and compares sets;
- symbolic checks for the bounded-resource formulas and the totally unimodular $M(K_4)$ representation;
- exact stdout files and JSON packet data;
- SHA-256 checksums;

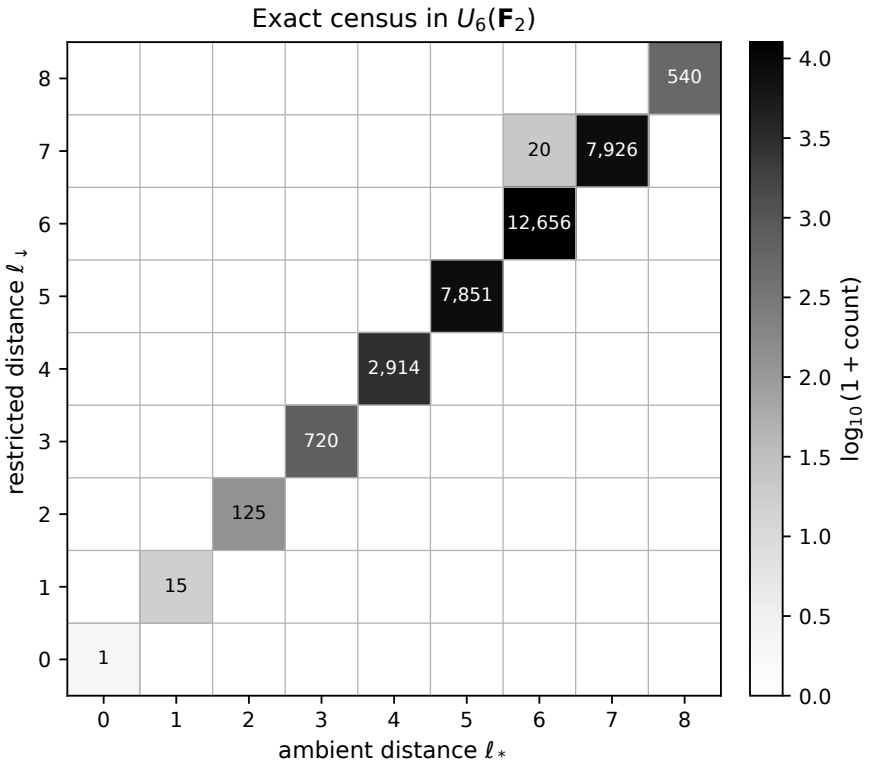


Figure 5.5: Complete binary dimension-six census. The only off-diagonal cell contains the twenty targets with distance pair $(7, 6)$.

- build and reproduction commands.

The programs were rebuilt from source and rerun independently when this manuscript was assembled; the outputs matched byte for byte.

5.6 Projective, matroidal, and spectral structure

5.6.1 The flagged Pasch packet

Over $\mathbb{F}_2$, write

$$x = e_0, \qquad p = e_1, \qquad q = e_2.$$

The canonical ambient word uses the six nonzero vectors

$$x, \quad p, \quad q, \quad p + q, \quad p + q + x, \quad p + x$$

inside $\operatorname{span}(x, p, q) \cong \mathbb{F}_2^3$. The only missing nonzero vector is $x + q$.

In $\mathrm{PG}(2, 2)$, three distinct points form a line exactly when their vector representatives sum to zero. The six points support the four lines

$$\{x, p, x + p\}, \quad \{x, p + q, x + p + q\}, \quad \{p, q, p + q\}, \quad \{x + p, q, x + p + q\}.$$

Every point lies on exactly two of these lines. This is the Pasch configuration, equivalently the Fano plane with one point and its three incident lines removed from the full incidence structure.

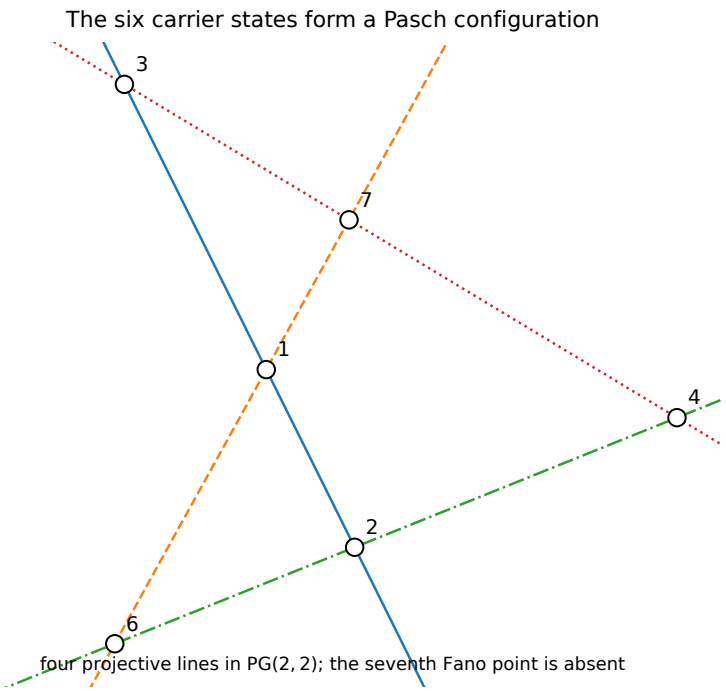


Figure 5.6: A real drawing of the four-line Pasch incidence structure. The labels are the binary point encodings $1, 2, 3, 4, 6, 7$; point 5 is the missing Fano point.

The configuration is flagged because the rows have fixed roles. Points x and q are stationary sources, p is the carrier, and the remaining points occur as carrier states. Forgetting those roles loses the circuit ordering information.

5.6.2 The regular matroid $M(K_4)$

The six carrier states admit a field-independent representation. In the ordered basis (x, p, q), place them as the columns of

$$B = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}, \tag{5.10}$$

corresponding respectively to

$$x, \quad p, \quad q, \quad x+p, \quad p+q, \quad x+p+q.$$

Proposition 5.29 (Regular-matroid core)**.** *Every square minor of B belongs to $\{0, \pm 1\}$, so B is totally unimodular. Hence the column-dependence matroid is the same over every field. Under the edge labeling*

state	p	x	$x+p$	$p+q$	q	$x+p+q$
K_4 *edge*	12	13	23	14	24	34,

it is the graphic matroid $M(K_4)$.

Proof. A direct calculation of all square minors of B gives only 0, 1, and -1. Thus every subdeterminant has characteristic-independent vanishing, and a column set is independent over one field

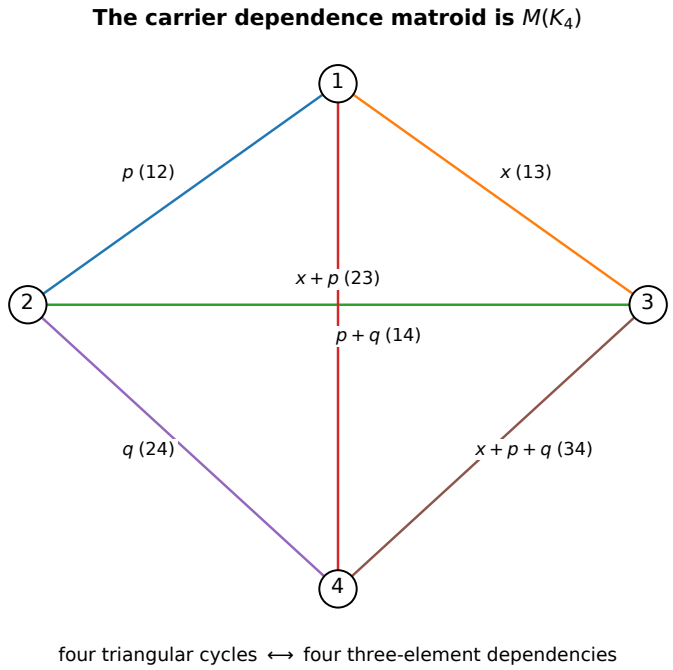


Figure 5.7: The six carrier states as the edges of K_4. Each triangular cycle is one of the four three-element dependencies in the Pasch configuration. The resulting regular matroid is representable over every field.

exactly when it is independent over every field. The four minimal dependent triples are

$$\{x, p, x+p\}, \quad \{p, q, p+q\}, \quad \{x, p+q, x+p+q\}, \quad \{q, x+p, x+p+q\}.$$

Under the displayed edge labeling, these are precisely the four triangular cycles of K_4. They determine the rank-three cycle matroid $M(K_4)$, which is regular because it has a totally unimodular representation; see, for example, [24]. □

The field-independent circuit theorem is stronger than matroid representability alone, because it also depends on the ordering and the designated carrier role. Nevertheless, Theorem 5.29 explains why the local dependence pattern survives unchanged in every characteristic.

5.6.3 Distance as first arrival

Let G be a finite group and $S = S^{-1}$ a symmetric generating set. On $\ell^2(G)$, let λ be the left regular representation,

$$\lambda(s)\delta_h = \delta_{sh},$$

and define

$$A_S = \sum_{s \in S} \lambda(s).$$

For $g \in G$, define the endpoint Green series

$$\mathcal{G}_{S,g}(z) = \left\langle \delta_g, (I - zA_S)^{-1}\delta_e \right\rangle.$$

Table 5.3: The five endpoint Green-signature packets. Each row occurs for four defective targets.

	restricted coefficients				ambient coefficients		
packet	$[z^7]$	$[z^8]$	$[z^9]$	$[z^{10}]$	$[z^6]$	$[z^7]$	$[z^8]$
1	326	15430	316118	7576819	4	570	23318
2	326	15536	321870	7727183	4	570	23318
3	543	25080	510703	10824940	2	776	39454
4	543	25639	511524	10984988	2	776	39454
5	543	26091	511819	11121486	2	776	39454

Proposition 5.30 (First-arrival identity)**.** *The coefficient of z^k in $\mathcal{G}_{S,g}(z)$ is the number of length-k words in S with product g. Hence*

$$d_S(e,g) = \operatorname{ord}_{z=0} \mathcal{G}_{S,g}(z).$$

Proof. Formally,

$$(I - zA_S)^{-1} = \sum_{k\geq 0} z^k A_S^k.$$

The matrix coefficient $\langle \delta_g, A_S^k \delta_e \rangle$ counts k-step walks from e to g, which are exactly length-k words with product g. The first nonzero coefficient occurs at the word distance. □

If

$$A_S = \sum_\lambda \lambda \Pi_\lambda$$

is the spectral resolution, then

$$\mathcal{G}_{S,g}(z) = \sum_\lambda \frac{\langle \delta_g, \Pi_\lambda \delta_e \rangle}{1 - \lambda z}.$$

Thus eigenvalues alone do not determine first arrival. The endpoint matrix coefficients of the spectral projectors are essential.

For the canonical target C, exact word counts begin

$$\mathcal{G}_{\downarrow,C}(z) = 543z^7 + 25080z^8 + 510703z^9 + 10824940z^{10} + \cdots,$$
$$\mathcal{G}_{*,C}(z) = 2z^6 + 776z^7 + 39454z^8 + \cdots.$$

The metric defect is the valuation drop

$$\operatorname{ord}_0 \mathcal{G}_{\downarrow,C} - \operatorname{ord}_0 \mathcal{G}_{*,C} = 1.$$

Across the twenty defects, the truncated endpoint data split into five Green-signature classes, each of multiplicity four. The signatures are listed in Table 5.3.

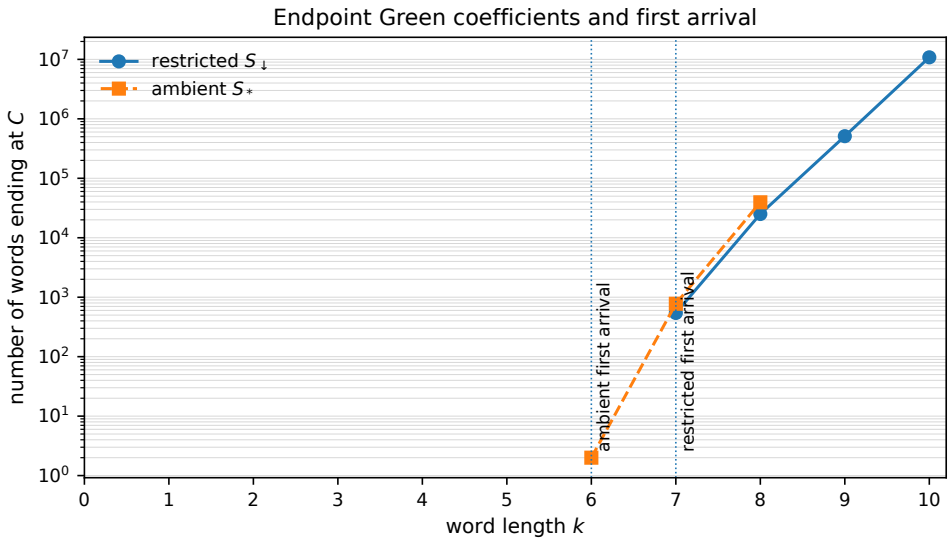


Figure 5.8: Exact endpoint word counts for the canonical target. The ambient walk first arrives at length six. The restricted walk first arrives at length seven.

5.7 Interval-tree certificates and connected higher-dimensional families

From this point through the rank and tensor arguments, the coefficient domain is a field $\mathbb{F}$. The six-row atom is the first defect, but it is not an isolated low-dimensional accident. This section gives a field-independent family of connected targets in every dimension at least seven. The lower bound is certified by a multiscale rank potential adapted to the ordered flag.

5.7.1 Interval-tree rank potential

Definition 5.31 (Interval tree)**.** An *interval tree* on the ordered coordinates $0, 1, \dots, n-1$ is a rooted binary tree whose leaves occur in that order and whose every node is a contiguous interval. If an internal node v splits its interval into consecutive left and right children L_v and R_v, define

$$\Phi_T(A) = \sum_{v \in \mathrm{Int}(T)} \operatorname{rank} A[R_v, L_v], \qquad A \in \mathrm{U}_n(\mathbb{F}). \tag{5.11}$$

Lemma 5.32 (One-cut locality)**.** *Let $X \in \mathrm{U}_n(\mathbb{F})$ and let X' be obtained from X by one lower-directed transvection. Then*

$$\left|\Phi_T(X') - \Phi_T(X)\right| \le 1.$$

Proof. Suppose the gate is $s \to t$ with $s < t$. Let v_0 be the lowest node of T whose two children separate s and t. The order of the leaves forces $s \in L_{v_0}$ and $t \in R_{v_0}$. At this one node, a single row of the cross block $X[R_{v_0}, L_{v_0}]$ changes, so its rank changes by at most one.

At every ancestor of v_0, the two rows lie in the same child. If that child is the right child, the gate is an internal row operation on the relevant cross block and preserves rank; if it is the left child, the cross block is unchanged. At a descendant containing t but not s, every column of the descendant interval lies strictly to the right of s. Since X remains lower triangular, row s vanishes

on those columns, so the descendant block is unchanged. All other descendants are also unchanged. Hence only the term at v_0 can vary. □

Corollary 5.33 (Interval-tree lower bound)**.** *For every $A \in \mathrm{U}_n(\mathbb{F})$ and every interval tree T,*

$$\ell_{\downarrow}(A) \geq \Phi_T(A).$$

Proof. The identity has potential zero. Apply Theorem 5.32 along any lower-directed word from the identity to A. □

Definition 5.34 (Tree-tight target)**.** A lower-unitriangular target A is *tree tight* if some interval tree T satisfies

$$\Phi_T(A) = \ell_{\downarrow}(A).$$

Tree tightness is stronger than possessing a rank lower bound. It says that one ordered hierarchy accounts for the entire restricted word length. The family below is tree tight by construction.

5.7.2 The snapshot family

Fix integers

$$u \geq 1, \qquad 1 \leq v \leq u+1,$$

and choose distinct phases

$$0 \leq s_1 < s_2 < \cdots < s_v \leq u.$$

Order the coordinates as

$$x_1, \ldots, x_u,\ p,\ q,\ t_1, \ldots, t_v.$$

For $0 \leq s \leq u$, put

$$w_s = e_p + \sum_{i=1}^{s} e_{x_i}. \tag{5.12}$$

Define

$$A(u; s_1, \ldots, s_v) \in \mathrm{U}_{u+v+2}(\mathbb{F}) \tag{5.13}$$

by

$$R_{x_i} = e_{x_i}, \qquad R_p = w_u, \qquad R_q = e_q,$$

and

$$R_{t_j} = e_{t_j} + e_q + w_{s_j} \qquad (1 \leq j \leq v). \tag{5.14}$$

The output rows record selected intermediate versions of the carrier p.

Proof of Theorem 5.5. For the ambient construction, first apply

$$q \to p.$$

Sweep through the updates

$$x_1 \to p, \ldots, x_u \to p.$$

Whenever phase s_j is reached, apply

$$p \to t_j.$$

Finally apply $q \to p$ with coefficient -1. At phase s, the carrier is $e_q + w_s$, so every output has the required value, and the final gate removes e_q from row p. The word has length

$$1 + u + v + 1 = u + v + 2.$$

Thus

$$\ell_*\big(A(u; s_1, \dots, s_v)\big) \le u + v + 2.$$

For a lower word, perform the same u updates without opening the catalyst. At each selected phase, use the two gates

$$p \to t_j, \qquad q \to t_j.$$

This gives length $u + 2v$.

It remains to prove optimality in the lower model. At the root, split

$$\{x_1, \dots, x_u, p\} \quad | \quad \{q, t_1, \dots, t_v\}.$$

The nonzero rows of the root cross block are $w_{s_1}, \dots, w_{s_v}$. Distinct prefix vectors are linearly independent: in a linear relation, inspect successively a coordinate occurring in w_{s_j} but in none of the earlier prefixes, starting from the largest phase. Hence the root contributes v.

Inside the left child, use the comb that splits off x_1, then x_2, and so on while retaining p in the remaining interval. Every one of the u cuts has rank one because the final row p contains every x_i. Inside the right child, repeatedly split off the last output while retaining q in the left interval. Each of the v cuts has rank one because every output contains q. Therefore

$$\Phi_T\big(A(u; s_1, \dots, s_v)\big) = v + u + v = u + 2v.$$

The interval-tree lower bound matches the displayed lower circuit. □

Remark 5.35 (Support connectivity). For $u, v \ge 1$, the undirected support graph of $A(u; s_1, \dots, s_v) - I$ is connected. Row p joins all update coordinates x_i, while every output joins both p and q. The separation below therefore does not arise from a direct sum of independent six-row gadgets.

5.7.3 Exact higher binary examples

The interval tree gives the exact restricted distance in all dimensions. For the first three higher targets, a separate bidirectional search also proves that the displayed ambient construction is optimal.

Theorem 5.36 (Exact binary cases in dimensions seven through nine)**.** *Over* $\mathbb{F}_2$*, let*

$$H_7 = A(2; 0, 1, 2), \qquad H_8 = A(3; 0, 1, 2), \qquad H_9 = A(3; 0, 1, 2, 3).$$

Then

$$(\ell_\downarrow(H_7), \ell_*(H_7)) = (8, 7),$$
$$(\ell_\downarrow(H_8), \ell_*(H_8)) = (9, 8),$$

and

$$(\ell_\downarrow(H_9), \ell_*(H_9)) = (11, 9).$$

Proof. The restricted values follow from Theorem 5.5. The ambient words have lengths 7, 8, and 9. To exclude shorter words, the supplement computes exact ambient balls and checks that the two relevant balls do not intersect. The certificate sizes are

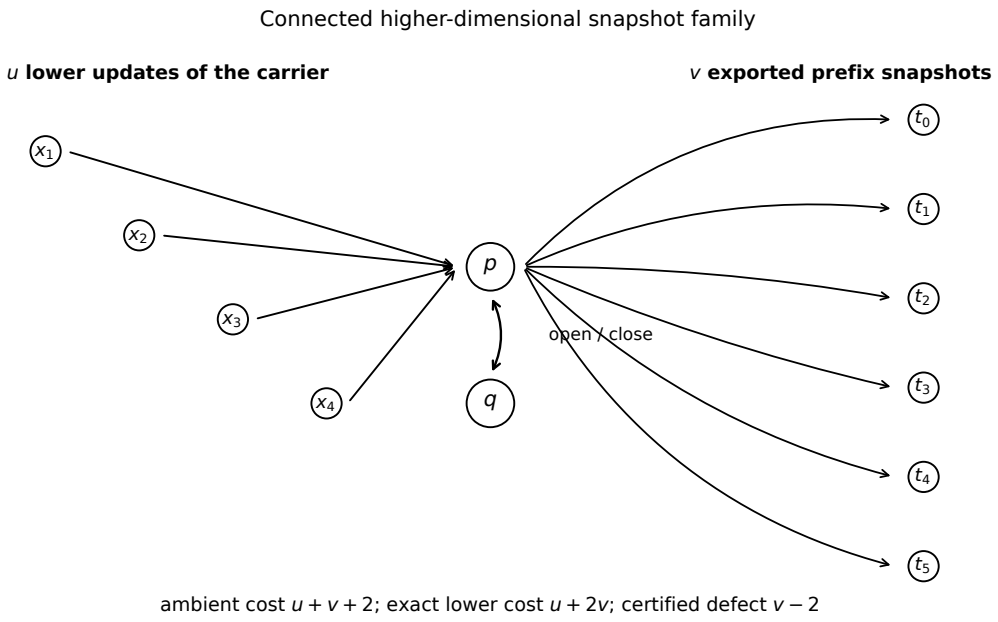


Figure 5.9: The connected snapshot family. One temporary upper dependency lets the ambient circuit export v selected prefix states. A lower-only circuit must insert the q component separately into every output.

n	tested split	left ball	right ball
7	$3+3$	23276	23276
8	$3+4$	52277	1174097
9	$4+4$	2994949	2994949

Every intersection is empty. Since the generating sets are symmetric, an empty radius-r versus radius-s intersection rules out every word of length at most $r+s$. □

These are exact higher-dimensional examples, not merely padded copies of C.

5.7.4 A connected defect in every dimension

Proof of Theorem 5.6. For $n \geq 7$, put

$$v_n = \left\lfloor \frac{n-1}{2} \right\rfloor, \qquad u_n = n - v_n - 2,$$

and take the phases

$$0, 1, \ldots, v_n - 1.$$

The inequality $v_n \leq u_n + 1$ holds, so the snapshot target is defined. By Theorem 5.5,

$$\ell_{\downarrow}(A_n) = u_n + 2v_n = n + v_n - 2,$$

while

$$\ell_*(A_n) \leq u_n + v_n + 2 = n.$$

Table 5.4: Successive connected cases from the optimized snapshot family. The restricted value is exact; the ambient column is a certified construction and is exact for $n = 7, 8, 9$ over $\mathbb{F}_2$.

n	u	v	$\ell_\downarrow$	ambient upper bound / certified gap
7	2	3	8	7 / 1
8	3	3	9	8 / 1
9	3	4	11	9 / 2
10	4	4	12	10 / 2
11	4	5	14	11 / 3
12	5	5	15	12 / 3
13	5	6	17	13 / 4
15	6	7	20	15 / 5
17	7	8	23	17 / 6
19	8	9	26	19 / 7
21	9	10	29	21 / 8

Hence

$$\ell_\downarrow(A_n) - \ell_*(A_n) \geq v_n - 2 = \left\lfloor \frac{n-5}{2} \right\rfloor.$$

Connectivity follows from the preceding remark. □

The ratio also increases beyond the block-sum value $7/6$.

Corollary 5.37 (Asymptotic multiplicative distortion)**.** *For every field,*

$$\limsup_{n\to\infty} \sup_{A\in \mathrm{U}_n(\mathbb{F}),\, A\neq I} \frac{\ell_\downarrow(A)}{\ell_*(A)} \geq \frac{3}{2}.$$

Proof. Along the optimized connected family,

$$\frac{\ell_\downarrow(A_n)}{\ell_*(A_n)} \geq \frac{n + \lfloor (n-1)/2 \rfloor - 2}{n},$$

which tends to $3/2$. □

5.8 Bounded and weighted catalytic resources

The connected snapshot family uses a single matched excursion: one upper-directed gate opens the dirty carrier and one upper-directed gate closes it. This permits a sharper statement than the unrestricted inequality $\ell_* < \ell_\downarrow$.

5.8.1 A bounded number of upper-directed gates

For a circuit word w, let $u(w)$ denote its number of upper-directed gates.

Definition 5.38 (Upper-gate budget)**.** For $q \geq 0$ and $A \in \mathrm{U}_n(\mathbb{F})$, define

$$\ell_{\leq q}(A) = \min\{|w| : [w] = A,\ u(w) \leq q\}.$$

Thus

$$\ell_{\leq 0}(A) = \ell_\downarrow(A), \qquad \ell_{\leq q+1}(A) \leq \ell_{\leq q}(A), \qquad \ell_*(A) \leq \ell_{\leq q}(A).$$

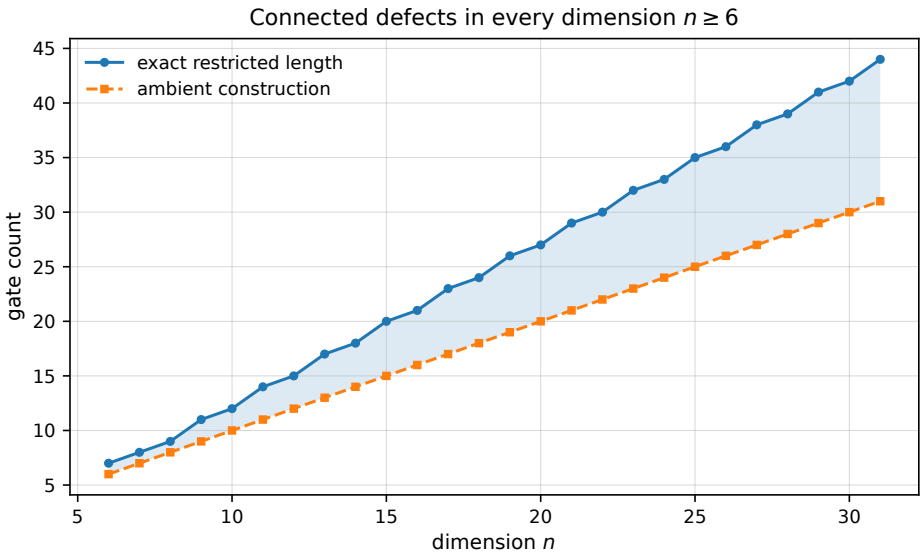


Figure 5.10: The six-row atom is the first exact defect. From dimension seven onward, the connected snapshot family gives an exact restricted length and a shorter ambient construction in every dimension.

Proposition 5.39 (Two upper gates give a linear advantage)**.** *For every snapshot target* $A = A(u; s_1, \dots, s_v)$*,*

$$\ell_{\leq 0}(A) = u + 2v, \qquad \ell_{\leq 2}(A) \leq u + v + 2.$$

Consequently,

$$\ell_{\leq 0}(A) - \ell_{\leq 2}(A) \geq v - 2.$$

For the optimized connected target A_n *of Theorem 5.6,*

$$\ell_{\leq 0}(A_n) - \ell_{\leq 2}(A_n) \geq \left\lfloor \frac{n-5}{2} \right\rfloor.$$

Proof. The exact restricted value is Theorem 5.5. Its ambient construction has precisely two upper-directed gates, namely the opening and closing operations on the carrier row; all u updates and all v reads are lower directed. The optimized choice has $v = \lfloor (n-1)/2 \rfloor$. □

Corollary 5.40 (No budget-only additive simulation bound)**.** *There is no finite constant* K *such that*

$$\ell_{\downarrow}(A) \leq \ell_{\leq 2}(A) + K$$

for all dimensions, fields, and lower-unitriangular targets A*.*

Proof. The left-minus-right difference in Theorem 5.39 tends to infinity with n. □

Thus the power of the excursion is not controlled by the number of forbidden gates. It is controlled by how many useful carrier states are exported between opening and closing.

5.8.2 Rank-one, bandwidth-one flag escape

The number of upper gates does not measure how far a circuit travels from the lower subgroup. We now control every intermediate state.

For $X \in \mathrm{SL}_n(\mathbb{F})$, let X^+ be the matrix obtained by retaining only the strict upper entries of X. Define

$$r^+(X) = \operatorname{rank} X^+$$

and

$$b^+(X) = \max\{j - i : X_{ij} \neq 0,\ i < j\},$$

with $b^+(X) = 0$ when $X^+ = 0$. Let $\ell^{(r,b)}(A)$ be the minimum length of a transvection circuit from I to A such that every prefix matrix X satisfies

$$r^+(X) \leq r, \qquad b^+(X) \leq b.$$

Lemma 5.41. *For $A \in \mathrm{U}_n(\mathbb{F})$,*

$$\ell^{(0,0)}(A) = \ell_{\downarrow}(A).$$

Proof. A lower-directed gate preserves lower triangularity. Conversely, if X is lower unitriangular and an upper-directed gate $R_t \leftarrow R_t + aR_s$ with $t < s$ is applied, then the new (t, s) entry equals

$$X_{t,s} + aX_{s,s} = a \neq 0.$$

The next state is therefore not lower triangular. A circuit whose every prefix has zero strict upper part uses only lower-directed gates. □

Theorem 5.42 (Microscopic escape, macroscopic saving)**.** *For every field $\mathbb{F}$ and every $n \geq 7$, there is a connected target $A_n \in \mathrm{U}_n(\mathbb{F})$ such that*

$$\ell^{(0,0)}(A_n) - \ell^{(1,1)}(A_n) \geq \left\lfloor \frac{n-5}{2} \right\rfloor.$$

The length-n circuit witnessing the second term uses exactly two upper-directed gates.

Proof. Use the connected family from Theorem 5.6, with the carrier coordinates $p < q$ adjacent. Between the opening and closing gates, the only strict upper entry is the coefficient at position (p, q). Hence every prefix has upper rank at most one and upper bandwidth at most one. The displayed ambient circuit has length n, whereas

$$\ell^{(0,0)}(A_n) = \ell_{\downarrow}(A_n) = n + \left\lfloor \frac{n-1}{2} \right\rfloor - 2.$$

Subtracting proves the claim. □

5.8.3 A weighted upper-gate metric

A hardware or fault model may penalize an upper-directed interaction more heavily than a lower-directed one. Fix $\lambda > 0$. If w has $d(w)$ lower-directed gates and $u(w)$ upper-directed gates, put

$$\mathrm{cost}_\lambda(w) = d(w) + \lambda u(w),$$

and define

$$\ell_\lambda(A) = \inf\{\mathrm{cost}_\lambda(w) : [w] = A\}.$$

At $\lambda = 1$ this is the ordinary ambient length, while large λ increasingly penalizes flag excursions.

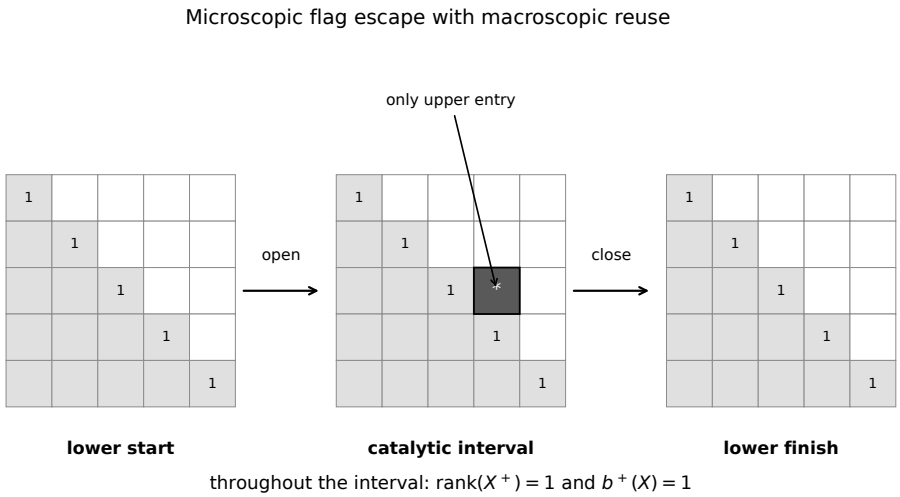


Figure 5.11: The snapshot circuit leaves the lower subgroup through one adjacent upper entry. Throughout the excursion the strict upper part has rank one and bandwidth one, yet the accumulated saving grows linearly.

Proposition 5.43 (Fixed penalties do not suppress catalysis)**.** *For every snapshot target* $A = A(u; s_1, \dots, s_v)$ *and every* $\lambda > 0$*,*

$$\ell_\lambda(A) \le u + v + 2\lambda.$$

Consequently,

$$\ell_\downarrow(A) - \ell_\lambda(A) \ge v - 2\lambda.$$

For the optimized connected family,

$$\ell_\downarrow(A_n) - \ell_\lambda(A_n) \ge \left\lfloor \frac{n-1}{2} \right\rfloor - 2\lambda.$$

In particular, for every fixed finite λ*, the weighted advantage grows linearly with* n*.*

Proof. The displayed catalytic word contains $u + v$ lower-directed gates and two upper-directed gates, so its weighted cost is $u + v + 2\lambda$. Subtract this from the exact restricted value $u + 2v$. The optimized family again has $v = \lfloor (n-1)/2 \rfloor$. □

Remark 5.44 (Penalty threshold)*.* For the displayed realization of a snapshot target, catalysis beats the optimal lower-only cost whenever

$$\lambda < \frac{v}{2}.$$

The critical penalty therefore scales with the number of consumers, not with the dimension of the dirty workspace.

Proof of Theorem 5.7. Combine Theorems 5.39 and 5.43. □

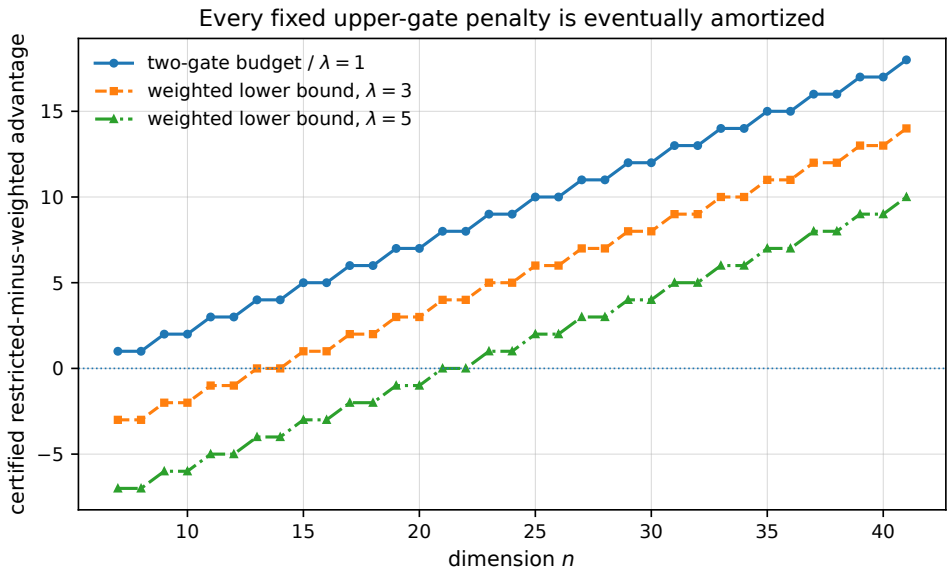


Figure 5.12: Certified lower-minus-weighted advantage for the optimized connected family. Every fixed upper-gate penalty is eventually amortized because the same matched excursion serves a growing number of output snapshots.

5.9 Algebraic rarity of catalytic defects

The preceding sections construct highly structured targets with large defects. This section proves the complementary statement that catalytic defects are nongeneric at every fixed dimension. Fix a field K. Algebraic dimensions and closures are taken after base change to an algebraic closure of K. Put

$$N = \dim \mathrm{U}_n = \binom{n}{2}, \qquad M = n(n-1).$$

5.9.1 Transvection product loci

For a direction skeleton

$$\sigma = ((t_1, s_1), \dots, (t_L, s_L)), \qquad t_i \neq s_i,$$

define the polynomial word map

$$\Phi_\sigma : \mathbb{A}^L \longrightarrow \mathrm{SL}_n, \qquad (a_1, \dots, a_L) \longmapsto (I + a_L E_{t_L, s_L}) \cdots (I + a_1 E_{t_1, s_1}).$$

The coefficients are allowed to vanish, so a word of length at most L can be padded to length L. Let

$$\mathcal{R}^*_{n,\le L} = \mathrm{U}_n \cap \bigcup_{\sigma \in \{(t,s) : t \neq s\}^L} \operatorname{im} \Phi_\sigma$$

be the ambient reachability locus, and define $\mathcal{R}^{\downarrow}_{n,\le L}$ analogously using only lower directions.

Equivalently, put

$$X_* = \bigcup_{t \neq s} \{I + aE_{t,s} : a \in K\}, \qquad X_{\downarrow} = \bigcup_{s < t} \{I + aE_{t,s} : a \in K\}.$$

Then $\ell_*(A)$ and $\ell_\downarrow(A)$ are the corresponding transvection product lengths: the least L for which $A \in X_*^L$ or $A \in X_\downarrow^L$.

Proposition 5.45 (Dimension of bounded-length loci)**.** *For every $L \geq 0$, the sets $\mathcal{R}^*_{n,\leq L}$ and $\mathcal{R}^\downarrow_{n,\leq L}$ are constructible and*

$$\dim \overline{\mathcal{R}^*_{n,\leq L}} \leq L, \qquad \dim \overline{\mathcal{R}^\downarrow_{n,\leq L}} \leq L.$$

Proof. There are finitely many direction skeletons of length L. For each skeleton, Φ_σ is a morphism from an L-dimensional affine space. Chevalley's theorem makes its image constructible, and the closure of the image has dimension at most L [15]. Finite unions preserve constructibility and the maximum dimension bound. Intersecting with the closed subvariety U_n cannot increase dimension. □

The argument is an instance of a general parameter-counting principle.

Proposition 5.46 (Dimension per gate)**.** *Let G be an algebraic group, let $V \subseteq G$ be an irreducible subvariety of dimension D, and let $X \subseteq G$ be a finite union of locally closed subvarieties, each of dimension at most r, with the identity in X. Write*

$$X^{(L)} = \{x_L \cdots x_1 : x_1, \ldots, x_L \in X\}.$$

Then

$$\dim \overline{V \cap X^{(L)}} \leq rL.$$

In particular, if $rL < D$, a Zariski-generic point of V has X-product length greater than L.

Proof. Every irreducible component of the Cartesian power $X^{\times L}$ has dimension at most rL. Multiplication $G^L \to G$ is a morphism, so the closure of the image of each component has dimension at most rL. There are finitely many components, and intersecting the resulting finite union with V cannot increase dimension. Since the identity belongs to X, products of length at most L are already contained in $X^{(L)}$. □

For transvections, the discrete choice of direction contributes finitely many components and each component carries one scalar parameter. Thus $r = 1$. Every element of $\mathrm{U}_n(K)$ has lower length at most N: process the strict lower entries in any root order compatible with elimination. The dimension-per-gate bound shows that this worst-case value is generic.

Theorem 5.47 (Generic rigidity)**.** *Let K be an infinite field. There is a nonempty Zariski-open subvariety $\Omega_n \subseteq \mathrm{U}_n$, defined over K, such that $\Omega_n(K)$ is nonempty and*

$$\ell_*(A) = \ell_\downarrow(A) = N \qquad (A \in \Omega_n(K)).$$

The same statement holds with exact product length replaced by border product length.

Proof. After base change to an algebraic closure $\overline{K}$, Theorem 5.45 gives

$$\dim \overline{\mathcal{R}^*_{n,\leq N-1}} \leq N - 1 < N = \dim \mathrm{U}_n\,.$$

Because the word maps are defined over K, this proper closed set descends to a proper closed subset of U_n over K. Its complement is a nonempty Zariski-open set Ω_n. Since K is infinite and $\mathrm{U}_n \cong \mathbb{A}^N$, the set $\Omega_n(K)$ is nonempty and Zariski dense. For any $A \in \Omega_n(K)$, one has $\ell_*(A) \geq N$. The inequalities

$$\ell_*(A) \leq \ell_\downarrow(A) \leq N$$

force equality. Replacing each image by its Zariski closure gives the border-length statement by the same argument. □

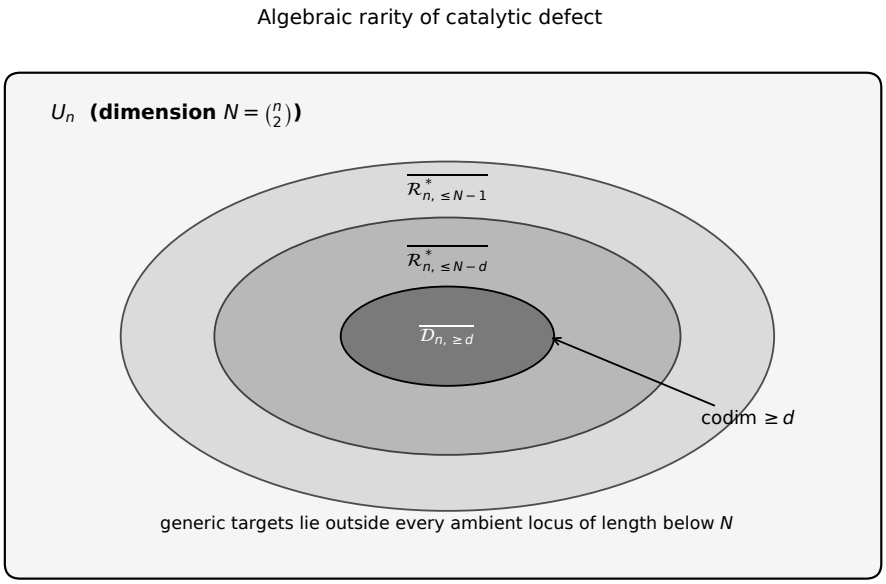


Figure 5.13: Bounded ambient product loci form nested algebraic strata inside U_n. A defect of size at least d lies inside the ambient reachability locus of length $N - d$, whose closure has codimension at least d.

5.9.2 Defect codimension

For an algebraically closed field K and $d \geq 1$, define

$$\mathcal{D}_{n,\geq d} = \{A \in \mathrm{U}_n(K) : \Delta(A) \geq d\}.$$

Theorem 5.48 (Defect–codimension bound)**.** *Let K be algebraically closed. For every $1 \leq d \leq N$,*

$$\dim \overline{\mathcal{D}_{n,\geq d}} \leq N - d.$$

Equivalently,

$$\operatorname{codim}_{\mathrm{U}_n} \overline{\mathcal{D}_{n,\geq d}} \geq d.$$

More finely, on the stratum where $\ell_\downarrow(A) = L$ and $\Delta(A) \geq d$, the closure has dimension at most $L - d$.

Proof. If $\Delta(A) \geq d$, then

$$\ell_*(A) \leq \ell_\downarrow(A) - d \leq N - d.$$

Thus $\mathcal{D}_{n,\geq d} \subseteq \mathcal{R}^*_{n,\leq N-d}$, and Theorem 5.45 gives the first assertion. On the intrinsic-length-L stratum one instead has $\ell_*(A) \leq L - d$, proving the refinement. □

5.9.3 Finite-field sparsity

The dimension argument has an elementary counting counterpart. Allowing zero coefficients, a padded word of length L is specified by L choices among M directed positions and L coefficients in $\mathbb{F}_q$. Consequently

$$|\mathcal{R}^*_{n,\leq L}(\mathbb{F}_q)| \leq (Mq)^L.$$

Corollary 5.49 (Finite-field defect density)**.** *For* $1 \le d \le N$,

$$\mathbb{P}_{A \in \mathrm{U}_n(\mathbb{F}_q)}[\Delta(A) \ge d] \le \min\{1, M^{N-d} q^{-d}\}.$$

In particular, for fixed n,

$$\mathbb{P}[\Delta(A) > 0] = O_n(q^{-1}), \qquad \mathbb{E}[\Delta(A)] = O_n(q^{-1})$$

as $q \to \infty$.

Proof. The defect condition implies ambient length at most $N - d$, so the number of such targets is at most $(Mq)^{N-d}$. Divide by

$$|\,\mathrm{U}_n(\mathbb{F}_q)| = q^N.$$

The expectation bound follows from

$$\mathbb{E}[\Delta] = \sum_{d=1}^{N} \mathbb{P}[\Delta \ge d].$$

□

The same counting argument also gives

$$\mathbb{P}_{A \in \mathrm{U}_n(\mathbb{F}_q)}[\ell_*(A) = \ell_\downarrow(A) = N] \ge 1 - M^{N-1} q^{-1},$$

whenever the right side is positive. Thus maximal ambient and intrinsic length is not only Zariski generic over infinite fields but asymptotically typical over growing finite fields at fixed dimension.

5.9.4 Border catalysis

For an algebraic generating set X, define the border product length

$$\underline{\ell}_X(A) = \min\{L : A \in \overline{X^L}\}.$$

This is analogous to the passage from tensor rank to border rank [21]. The generic value is still N by Theorem 5.47, but exact and border defects can differ on exceptional strata. It is presently open whether the canonical atom C belongs to the closure of the six-factor lower transvection locus. A negative answer would show that its one-gate defect is stable under algebraic degeneration; a positive answer would exhibit a rank-versus-border-rank phenomenon inside reversible linear synthesis.

Remark 5.50 (Rare but amplifiable). The codimension theorem and the tensor construction point in opposite directions without contradiction. At each fixed n, a defect of size d lies on a codimension-at-least-d subset. Carefully selected points on these thin strata can nevertheless be tensorized into families with $\Omega(n \log n)$ additive defect. Catalytic compression is therefore algebraically rare but asymptotically amplifiable.

5.10 Recursive, block, and tensor amplification

5.10.1 Restricted block additivity

The lower metric is exactly additive on contiguous block-diagonal targets.

Proposition 5.51 (Block additivity)**.** *Let $A_i \in \mathrm{U}_{n_i}(\mathbb{F})$ for $1 \le i \le k$. Then*

$$\ell_{\downarrow}\big(\mathrm{diag}(A_1, \dots, A_k)\big) = \sum_{i=1}^{k} \ell_{\downarrow}(A_i).$$

Proof. The upper bound follows by running optimal lower circuits independently inside the blocks.

For the lower bound, view an element of $\mathrm{U}_{n_1+\cdots+n_k}(\mathbb{F})$ as a lower block-triangular matrix and project to its diagonal blocks:

$$\pi(X) = (X_{11}, \dots, X_{kk}).$$

Because diagonal blocks multiply independently in a lower block-triangular product, π is a homomorphism. A lower transvection internal to one block projects to one generator in that factor, while a crossing transvection projects to the identity in every factor. If m_i gates survive in factor i, then the total word length is at least

$$\sum_i m_i \ge \sum_i \ell_{\downarrow}(A_i).$$

□

The ambient metric is subadditive:

$$\ell_*\big(\mathrm{diag}(A_1, \dots, A_k)\big) \le \sum_i \ell_*(A_i).$$

Cross-block ambient workspace can only improve this upper bound.

Corollary 5.52 (Block-sum amplification)**.** *For every field and every $n \ge 6$, there is an $A_n \in \mathrm{U}_n(\mathbb{F})$ with*

$$\ell_{\downarrow}(A_n) - \ell_*(A_n) \ge \left\lfloor \frac{n}{6} \right\rfloor.$$

For $n = 6k$, the target $C^{\oplus k}$ satisfies

$$\ell_{\downarrow}(C^{\oplus k}) = 7k, \qquad \ell_*(C^{\oplus k}) \le 6k.$$

Proof. For $n = 6k + r$, take $C^{\oplus k} \oplus I_r$ and apply block additivity together with k copies of the ambient six-gate word. □

5.10.2 A recursive zeta-shaped lift

For $A \in \mathrm{U}_m(\mathbb{F})$, define

$$\mathcal{Z}(A) = \begin{pmatrix} A & 0 \\ A & A \end{pmatrix} \in \mathrm{U}_{2m}(\mathbb{F}). \tag{5.15}$$

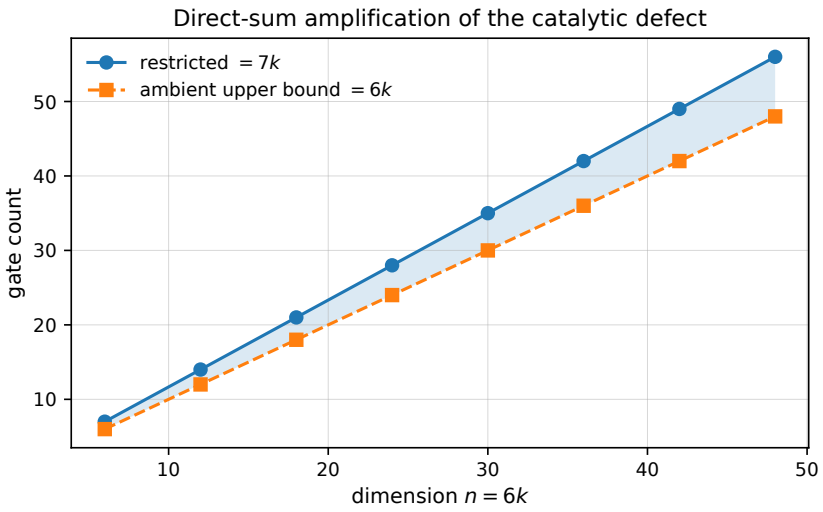


Figure 5.14: Block-sum amplification. The restricted length is exactly $7k$, while the displayed ambient construction has length $6k$.

Proposition 5.53 (Exact restricted lift)**.** *For every* $A \in \mathrm{U}_m(\mathbb{F})$,

$$\ell_{\downarrow}(\mathcal{Z}(A)) = 2\ell_{\downarrow}(A) + m,$$

and

$$\ell_{*}(\mathcal{Z}(A)) \le 2\ell_{*}(A) + m.$$

Consequently,

$$\Delta(\mathcal{Z}(A)) \ge 2\Delta(A).$$

Proof. The construction first realizes $\mathrm{diag}(A, A)$ and then applies the m root-cross gates that add top row i into bottom row i.

For the lower bound, write a circuit state in $m \times m$ blocks as

$$X = \begin{pmatrix} P & 0 \\ Q & R \end{pmatrix}.$$

Projection to the two diagonal blocks forces at least $2\ell_{\downarrow}(A)$ internal gates. A top-internal gate leaves Q unchanged. A bottom-internal gate performs an invertible row operation on Q and preserves its rank. A root-cross gate changes Q by one rank-one row contribution, so it can increase $\operatorname{rank} Q$ by at most one. The final cross block is A, of rank m, hence at least m root-cross gates are necessary. □

Let $A^{[0]} = A$ and $A^{[r+1]} = \mathcal{Z}(A^{[r]})$. If A has size m, restricted length a, and ambient length b, then

$$\ell_{\downarrow}(A^{[r]}) = 2^r a + rm2^{r-1}, \tag{5.16}$$

and

$$\ell_{*}(A^{[r]}) \le 2^r b + rm2^{r-1}. \tag{5.17}$$

Thus

$$\Delta(A^{[r]}) \ge 2^r \Delta(A).$$

For the canonical atom C, the first values are

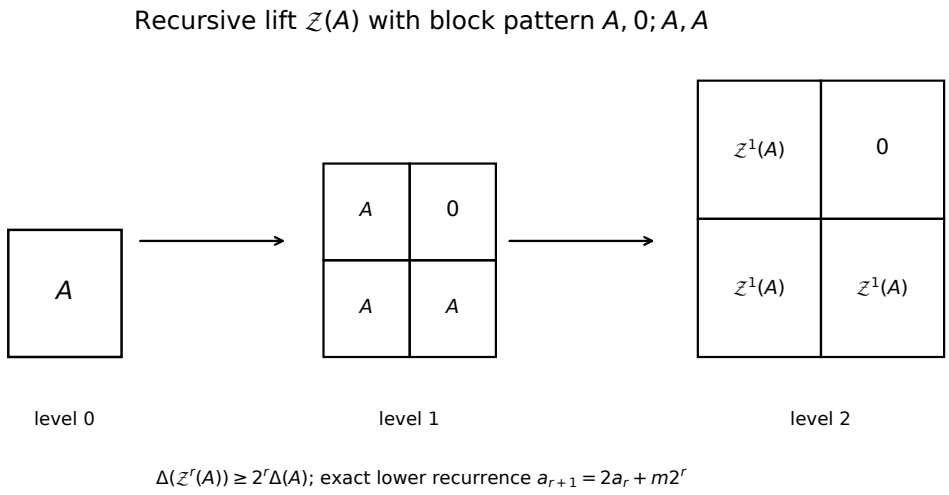


Figure 5.15: The recursive lift has the same outer shape as a Boolean zeta recursion. A finite defect at the leaf is doubled at every lift.

r	dimension	restricted exact	ambient upper bound	certified gap
0	6	7	6	1
1	12	20	18	2
2	24	52	48	4
3	48	128	120	8
4	96	304	288	16

This gives arbitrarily deep exact recursion with identical diagonal children and a full-rank cross block while preserving a positive defect.

5.10.3 Tensor products of tree-tight targets

The tensor operation gives the stronger asymptotic amplification.

Theorem 5.54 (Exact restricted tensor rule)**.** *Let $A \in \mathrm{U}_m(\mathbb{F})$ and $B \in \mathrm{U}_n(\mathbb{F})$ be tree tight. Then $A \otimes B$ is tree tight and*

$$\ell_\downarrow(A \otimes B) = n\ell_\downarrow(A) + m\ell_\downarrow(B). \tag{5.18}$$

Moreover,

$$\ell_*(A \otimes B) \leq n\ell_*(A) + m\ell_*(B), \tag{5.19}$$

so

$$\Delta(A \otimes B) \geq n\Delta(A) + m\Delta(B). \tag{5.20}$$

Proof. Use lexicographic order on the mn coordinates. Substitute a copy of a certificate tree for B at every leaf of a certificate tree for A. At an outer node, the cross block is

$$A[R, L] \otimes B,$$

whose rank is $n \operatorname{rank} A[R, L]$ because B is invertible. The outer nodes therefore contribute $n\ell_{\downarrow}(A)$. The m diagonal copies of the inner tree contribute $m\ell_{\downarrow}(B)$. This proves the lower bound.

For the matching lower construction, first realize $I_m \otimes B$ using m copies of a circuit for B. Then realize $A \otimes I_n$. Each gate of a circuit for A lifts to n coordinatewise gates. Since

$$(A \otimes I_n)(I_m \otimes B) = A \otimes B,$$

the cost is the right side of Equation (5.18). The same construction with ambient circuits proves Equation (5.19). □

If A is tree tight of dimension m, with restricted length a and an ambient construction of length b, then

$$\ell_{\downarrow}(A^{\otimes r}) = rm^{r-1}a, \tag{5.21}$$

and

$$\ell_{*}(A^{\otimes r}) \leq rm^{r-1}b. \tag{5.22}$$

A positive one-copy defect therefore becomes an order-$N \log N$ defect at dimension $N = m^r$.

Proof of Theorem 5.10. Apply Theorem 5.54 repeatedly to the tree-tight snapshot target $A = A(u; s_1, \ldots, s_v)$, for which

$$a = u + 2v, \qquad b = u + v + 2 = m.$$

Then

$$\ell_{\downarrow}(A^{\otimes r}) - \ell_{*}(A^{\otimes r}) \geq rm^{r-1}(v - 2).$$

Since $N = m^r$ and $r = \log_2 N / \log_2 m$, this is

$$\frac{v - 2}{m \log_2 m} N \log_2 N.$$

□

Within the snapshot family, the best coefficient is obtained by the twenty-one-dimensional seed

$$A_{21} = A(9; 0, 1, \ldots, 9),$$

for which

$$\ell_{\downarrow}(A_{21}) = 29, \qquad \ell_{*}(A_{21}) \leq 21, \qquad \Delta(A_{21}) \geq 8.$$

Proposition 5.55 (Optimal snapshot seed for tensor rate)**.** *Among all snapshot targets, the coefficient*

$$\frac{v - 2}{m \log_2 m}, \qquad m = u + v + 2,$$

is maximized at

$$(m, u, v) = (21, 9, 10).$$

Proof. For fixed m, maximize v subject to $v \leq u + 1$ and $u + v + 2 = m$. Thus

$$v = \left\lfloor \frac{m - 1}{2} \right\rfloor.$$

If m is even and at least eight, its coefficient is strictly smaller than that of $m - 1$, because the numerator is the same while

$$(m - 1) \log(m - 1) < m \log m.$$

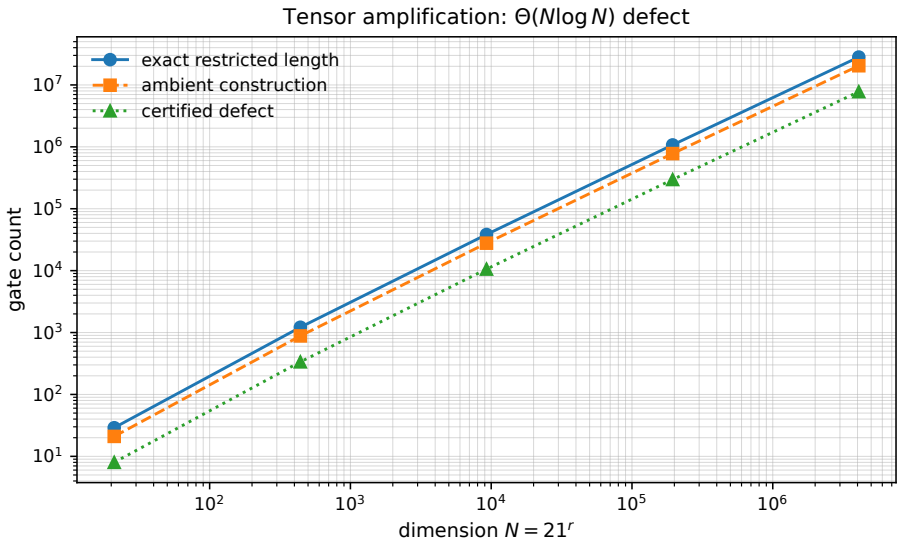


Figure 5.16: Tensor powers of the twenty-one-dimensional snapshot seed. Both displayed circuit sizes are of order $N \log N$, and their difference has a positive $N \log N$ coefficient.

It therefore suffices to consider odd m. Apart from the constant change of logarithm base, the odd-dimensional envelope is

$$f(x) = \frac{x-5}{2x \log x}.$$

Its derivative has the sign of

$$5(1 + \log x) - x.$$

For $x > 5$ this expression crosses zero exactly once, between 19 and 21. Direct comparison gives $f(21) > f(19)$, after which the envelope decreases. Hence the unique integer maximizer is $m = 21$. □

The first tensor powers are

r	dimension 21^r	restricted exact	ambient upper bound	certified gap
1	21	29	21	8
2	441	1218	882	336
3	9261	38367	27783	10584
4	194481	1074276	777924	296352

5.10.4 Stable tensor capacity and activation

The tensor inequality also defines a regularized ambient invariant. Let $A \in \mathrm{U}_m(\mathbb{F})$ and put

$$b_r(A) = \frac{\ell_*(A^{\otimes r})}{m^{r-1}}.$$

Applying Equation (5.19) to $A^{\otimes r} \otimes A^{\otimes s}$ gives

$$b_{r+s}(A) \le b_r(A) + b_s(A).$$

Proposition 5.56 (Stable tensor length)**.** *The limit*

$$\tau_*(A) = \lim_{r\to\infty} \frac{\ell_*(A^{\otimes r})}{rm^{r-1}} = \inf_{r\geq 1} \frac{\ell_*(A^{\otimes r})}{rm^{r-1}}$$

exists. If A is tree tight, then the stable catalytic capacity

$$\kappa_\otimes(A) = \ell_\downarrow(A) - \tau_*(A)$$

satisfies

$$\kappa_\otimes(A) \geq \Delta(A).$$

Proof. The sequence $b_r(A)$ is subadditive, so Fekete's lemma gives the displayed limit. Since the one-copy lifted construction implies $\tau_*(A) \leq \ell_*(A)$, one has

$$\kappa_\otimes(A) \geq \ell_\downarrow(A) - \ell_*(A) = \Delta(A).$$

For a tree-tight target, Equation (5.21) identifies $\ell_\downarrow(A)$ with its exact normalized restricted tensor cost. □

For the twenty-one-dimensional seed A_{21},

$$\kappa_\otimes(A_{21}) \geq 8.$$

This regularization introduces a new possibility: a target may have $\Delta(A) = 0$ but $\kappa_\otimes(A) > 0$. In that case one copy is rigid while many tensor copies admit a collective shortcut. We call this *tensor activation*.

5.10.5 Extremal consequences

Define

$$D_n(\mathbb{F}) = \sup_{A\in \mathrm{U}_n(\mathbb{F})} \big(\ell_\downarrow(A) - \ell_*(A)\big).$$

For finite fields the supremum is a maximum. The tensor family gives the following explicit all-dimension bound.

Proof of Theorem 5.11. Let $m = 21$ and write

$$n = \sum_{j=0}^{r} d_j m^j, \qquad 0 \leq d_j < m.$$

Take the block sum of d_j copies of $A_{21}^{\otimes j}$ for every j, with the convention $A_{21}^{\otimes 0} = I_1$. Block additivity and the tensor estimate give

$$D_n(\mathbb{F}) \geq \frac{8}{21} \sum_{j=0}^{r} j d_j 21^j.$$

Put $S = \sum_j j d_j 21^j$. Since $r = \lfloor \log_{21} n \rfloor$,

$$rn - S = \sum_{j<r} (r-j) d_j 21^j \leq 20 \sum_{h\geq 1} h 21^{r-h} \leq \frac{21^{r+1}}{20} \leq \frac{21}{20} n.$$

Also $r \geq \log_{21} n - 1$. Therefore

$$S \geq n \log_{21} n - \frac{41}{20} n.$$

Multiplying by 8/21 yields

$$D_n(\mathbb{F}) \geq \frac{8}{21 \log_2 21} n \log_2 n - \frac{82}{105} n.$$

□

Numerically,

$$\frac{8}{21 \log_2 21} = 0.0867315\ldots.$$

The elementary bottom-to-top construction gives

$$D_n(\mathbb{F}) \leq \frac{n(n-1)}{2}.$$

Thus the current general range is

$$0.0867315\, n \log_2 n - O(n) \leq D_n(\mathbb{F}) \leq \frac{n(n-1)}{2}. \tag{5.23}$$

5.11 Discussion and open problems

5.11.1 What the threshold and higher families establish

The six-row atom proves that an optimal circuit for a lower-unitriangular target need not remain in the lower-unitriangular subgroup. The exact binary classification shows that this first failure is rigid: it occurs for exactly twenty targets, every defect is $(7, 6)$, and every target is generated by the same three-cross, one-phase mechanism.

The higher-dimensional results change the scale of the phenomenon. The connected snapshot family shows that exactly two upper-directed gates can support a defect linear in dimension, and that the advantage persists under every fixed finite penalty on those gates. The recursive lift shows that a defective leaf survives arbitrarily many identical full-rank dyadic levels. Tensor products then turn any tree-tight positive-defect atom into an order-$N \log N$ additive separation. Thus the defect is not a finite anomaly and not merely linear in the worst case. At the same time, generic rigidity and the defect–codimension theorem show that large defects require correspondingly special algebraic conditions. The central synthesis is therefore not ubiquity but amplification: thin exceptional strata can control large structured families.

5.11.2 Relation to ordinary CNOT synthesis

Exact CNOT synthesis usually minimizes ℓ_* on all of $\mathrm{GL}_n(\mathbb{F}_2)$. The present problem fixes a structured target class and compares the ambient optimum with a structure-preserving optimum. Recent exact methods explore the full Cayley graph using symmetries, bidirectional search, A* methods, SAT, and related techniques [10, 33, 31]. The right-coset computation in this paper is tailored to a subgroup comparison, while the higher-dimensional exact audits use direct bidirectional balls for selected targets.

The 2026 NP-completeness result for all-to-all CNOT distance shows that cancellation and unrestricted parity reuse must be handled in full generality [2]. The present constructions isolate explicit reusable-carrier mechanisms under an ordered restriction. They do not resolve ambient synthesis complexity.

5.11.3 Open directions

Question 5.57 (Universal six-dimensional threshold). Is six the first defective dimension over every field, or even over every nonzero commutative ring? The atom gives a universal defective target in dimension six. Exact no-defect searches below six are proved here for $\mathbb{F}_2$ and $\mathbb{F}_3$.

Question 5.58 (Geometry of the defect locus). For fixed n and d, determine the irreducible components and exact dimension of $\overline{\mathcal{D}_{n,\geq d}}$. When is the lower bound codim $\geq d$ sharp?

Question 5.59 (Border catalysis). Does the canonical atom have lower border product length six or seven? More generally, when can exact catalytic defects disappear under algebraic degeneration?

Question 5.60 (Correct growth of the maximum defect). Is $D_n(\mathbb{F})$ of order $n \log n$, or can interacting catalysts produce a larger power of n? The current bounds are

$$\Omega(n \log n) \leq D_n(\mathbb{F}) \leq O(n^2).$$

Question 5.61 (Multiplicative distortion). Is

$$\sup_{n, A \in \mathrm{U}_n(\mathbb{F})} \frac{\ell_{\downarrow}(A)}{\ell_*(A)}$$

bounded? The connected snapshot family gives an asymptotic lower bound of $3/2$. Exceeding $3/2$ appears to require a mechanism beyond one clean carrier exporting independent snapshots.

Question 5.62 (Exact ambient complexity of snapshot targets). For which phase sets is the ambient construction of length $u + v + 2$ optimal? It is exact for the displayed binary targets in dimensions seven, eight, and nine, but the general ambient optimum remains open.

Question 5.63 (Tensor synergy and activation). Can $A \otimes B$ admit an ambient circuit shorter than the lifted bound

$$n\ell_*(A) + m\ell_*(B)?$$

Such synergy would only increase the defect, but understanding it is necessary for exact tensor complexity. Does there exist a tree-tight target with $\Delta(A) = 0$ but $\kappa_{\otimes}(A) > 0$?

Question 5.64 (Higher catalytic normal forms). Can interacting or crossing catalytic excursions produce asymptotic behavior impossible for a single clean carrier? A useful classification would separate laminar, crossing, and singular flag excursions.

Question 5.65 (Two-gate multiplicative ceiling). Among targets realizable with at most two upper-directed gates, is the asymptotic multiplicative distortion at most $3/2$? The snapshot family reaches this value from below, but the present paper proves no matching upper bound.

Question 5.66 (Matroidal classification). Which regular matroids admit a flagged realization whose shortest ambient state tour is strictly shorter than every flag-compatible lift? The first atom is the graphic matroid $M(K_4)$ with designated source, carrier, and output roles.

Question 5.67 (Weighted phase transition). For a fixed target family, determine the exact critical penalty λ_c at which a flag excursion ceases to improve the optimal weighted cost. The snapshot construction proves only the lower bound $\lambda_c \geq v/2$ for its displayed carrier word.

Question 5.68 (Other root systems). Let G be a split reductive group with unipotent radical U. Compare word length in positive root subgroups with ambient word length using all root subgroups. Which root systems admit a catalytic defect, and what is the first rank?

Question 5.69 (Certificate extraction). Can exact ambient lower bounds for large structured targets be accompanied by short human-readable certificates rather than exhaustive balls? The interval-tree method completely handles the restricted side, while the ambient side still lacks an equally general certificate.

5.12 The twenty binary dimension-six defects

A lower-unitriangular binary matrix is encoded by reading the strict lower entries row by row:

$$(1,0),(2,0),(2,1),\dots,(5,0),\dots,(5,4).$$

The resulting fifteen-bit integer is denoted by $\mathrm{code}(A)$. A row is displayed as the integer whose binary expansion is its support vector. Table 5.5 lists all defects. Every row has $\ell_{\downarrow} = 7$ and $\ell_{*} = 6$.

Table 5.5: The twenty defective matrices in $\mathrm{U}_6(\mathbb{F}_2)$.

code	R_0	R_1	R_2	R_3	R_4	R_5
6585	1	3	4	15	22	38
6641	1	3	4	14	23	38
6649	1	3	4	15	23	38
7601	1	3	4	14	22	39
7609	1	3	4	15	22	39
7665	1	3	4	14	23	39
10014	1	2	7	11	28	41
11038	1	2	7	11	28	42
12894	1	2	7	11	25	44
12958	1	2	7	11	26	44
13142	1	2	7	10	29	44
13198	1	2	7	9	30	44
14102	1	2	7	10	28	45
15118	1	2	7	9	28	46
17912	1	2	4	15	23	49
18936	1	2	4	15	23	50
19960	1	2	4	15	23	51
20984	1	2	4	15	23	52
22008	1	2	4	15	23	53
23032	1	2	4	15	23	54

For example, the canonical target has row encoding

$$(1,3,4,14,23,39)$$

and code 7601.

5.13 Search algorithms and correctness

5.13.1 Restricted search

For $\mathbb{F}_2$, each lower-unitriangular matrix is stored as a bit string. Standard breadth-first search gives exact distance because every edge has unit weight. The algorithm terminates after visiting all states.

Algorithm 5.70 (Restricted breadth-first search)**.** Initialize $d[I] = 0$ and a queue containing I. Repeatedly remove X from the queue. For every lower transvection $s \to t$, form $Y = \tau_{t,s}(1)X$. If Y has not been visited, set $d[Y] = d[X] + 1$ and append Y. Continue until the queue is empty.

5.13.2 Right-coset key

For an invertible binary matrix A, let c_j be column j. Starting at $j = n-1$ and moving left, reduce c_j by the already canonicalized columns $c_{j+1}, \dots, c_{n-1}$, using their pivot coordinates. The resulting ordered reduced columns form the key $\kappa(A)$.

Lemma 5.71. *For $A, B \in \mathrm{GL}_n(\mathbb{F}_2)$,*

$$\kappa(A) = \kappa(B) \quad \Longleftrightarrow \quad AU_n = BU_n.$$

Proof. Right multiplication by $U \in \mathrm{U}_n$ replaces column j by itself plus a combination of columns with index greater than j. Right-to-left reduction is therefore unchanged on a right coset. Conversely, equal reduced columns imply that the original column frames differ by a unique lower-unitriangular column transformation, so $A = BU$ for some $U \in \mathrm{U}_n$. □

5.13.3 Ambient search

Algorithm 5.72 (Radius-four right-coset meet in the middle)**.** Compute the exact radius-four ball B_4 in $\mathrm{GL}_6(\mathbb{F}_2)$. Bucket its elements by κ. For each bucket and every ordered pair (b, a) in that bucket, compute $u = b^{-1}a \in \mathrm{U}_6$ and update

$$d_*(u) \leftarrow \min\big(d_*(u), d(b) + d(a)\big).$$

Correctness is Theorem 5.28. The computation also checks that every generated pair lies in U_6 and that every one of the 32768 targets receives a finite distance.

5.13.4 Higher-dimensional bidirectional audits

For the targets H_7, H_8, H_9 of Theorem 5.36, the program `small_snapshot_exact.cpp` constructs exact ambient balls from the identity and from the target. If radius-r and radius-s balls are disjoint, symmetry of the generating set rules out every word of length at most $r + s$. The tested splits are $3 + 3$, $3 + 4$, and $4 + 4$, respectively. The source stores matrices in one 128-bit word and uses exact row-addition transitions; no probabilistic or floating-point step occurs.

5.13.5 Symbolic higher-family checks

The program `higher_family_tests.py` generates the snapshot targets over $\mathbb{F}_2$, $\mathbb{F}_3$, and $\mathbb{F}_5$, evaluates both displayed circuits, computes the interval-tree potentials, audits a tensor certificate for the seven-dimensional seed, verifies the recursive cross rank, and reproduces the twenty-one-dimensional rate optimization.

5.13.6 Further generalization audits

The program `further_generalization_tests.py` verifies the universal circuits over $\mathbb{Z}/4\mathbb{Z}$, $\mathbb{Z}/6\mathbb{Z}$, and $\mathbb{Z}/8\mathbb{Z}$, audits the rank-one bandwidth-one excursion, checks the finite-field counting bounds symbolically, and verifies the stable tensor subadditivity identities.

5.13.7 Motif comparison

The motif program independently loops over the three adjacent catalyst positions, the phase orientation, the phase leaf, and every nonempty proper before/after subset. It evaluates each six-gate word from the identity and stores its target. The resulting set has size twenty. Exact comparison with the twenty census defects returns equality.

5.14 Reproducibility ledger

The ancillary directory `supplement/` contains portable source, frozen expected outputs, and a one-command driver:

```
cd supplement
./run_all.sh
```

The driver compiles every C++ search, executes the Python motif, higher-family, matroid, and weighted-resource checks, and compares all generated outputs byte for byte with the frozen records. The supplement README gives the individual commands and the role of every program. A SHA-256 manifest covers every included source and expected-output file.

Acknowledgement of computational status

All theorem statements are labeled by proof type in the body. The universal ring-valued atom, Schreier-holonomy interpretation, regular-matroid identification, interval-tree family, bounded, weighted, and microscopic resource theorems, algebraic-rarity bounds, block additivity, recursive lift, tensor rule, stable tensor limit, and asymptotic amplification are proved symbolically. The first-threshold classification, finite censuses, and exact ambient values in dimensions seven through nine are computer assisted, with complete search-space coverage justified in Section 5.13.

Data and code availability

The source archive accompanying this paper contains the complete LaTeX source, all vector figures, every C++ and Python verification program, frozen expected outputs, and SHA-256 manifests. The command

```
cd supplement
./run_all.sh
```

rebuilds the finite searches and symbolic audits and compares every output byte for byte with its frozen record. No randomized or floating-point decision enters a theorem certificate.

Part III

Quantum and reversible gate atlas

Chapter 6

Qudit SUM gates and ring-valued linear logic

6.1 From transvections to quantum gates

Let $d \geq 2$. On n qudits with computational basis labeled by $\mathbb{Z}_d^n$, define the scaled SUM gate

$$\mathrm{SUM}_{s\to t}^{(a)}|x_0,\ldots,x_{n-1}\rangle = |x_0,\ldots,x_t + ax_s,\ldots,x_{n-1}\rangle, \qquad a \in \mathbb{Z}_d.$$

It is a permutation unitary. Its action on basis labels is the row transvection

$$I + aE_{t,s}.$$

For $d = 2$ and $a = 1$, it is CNOT. For prime d, these gates lie in the standard qudit Clifford group. For composite d, they remain exact reversible modular-addition gates.

Proved statement

The universal six-row atom in the core chapter is valid over every nonzero commutative ring with identity. Taking $R = \mathbb{Z}_d$ gives a six-qudit target with restricted directed-SUM length seven and ambient all-direction SUM length six for every $d \geq 2$.

6.2 The universal six-qudit circuit

Use the coordinate names

$$x, p, q, t_0, t_1, t_2.$$

The target acts on basis labels by

$$\begin{aligned} p &\mapsto p + x, \\ t_0 &\mapsto t_0 + p + q, \\ t_1 &\mapsto t_1 + x + p + q, \\ t_2 &\mapsto t_2 + x + p + q, \end{aligned}$$

with x and q unchanged. An ambient word is

$$\mathrm{SUM}_{q\to p},\quad \mathrm{SUM}_{p\to t_0},\quad \mathrm{SUM}_{x\to p},\quad \mathrm{SUM}_{p\to t_1},\quad \mathrm{SUM}_{p\to t_2},\quad \mathrm{SUM}_{q\to p}^{(-1)}.$$

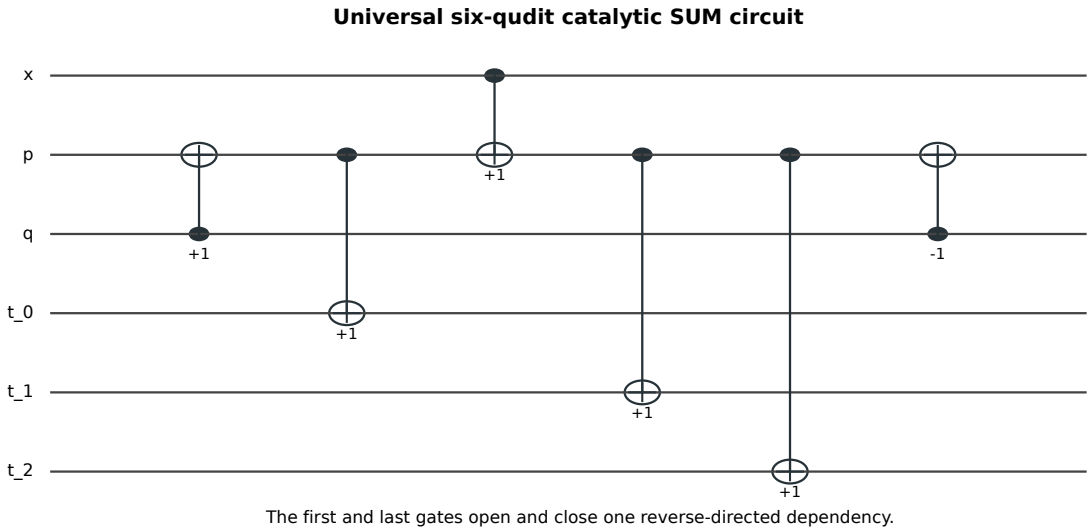


Figure 6.1: The universal six-qudit catalytic circuit. Coefficients are interpreted modulo d.

The first gate adds the dirty q component to p. The middle gates export two different carrier states. The final inverse gate restores p.

A lower-only word needs seven gates. The exact proof is coefficient-domain independent because it uses pivot support and the number of distinct carrier states rather than division.

6.3 Connected higher-dimensional qudit families

The snapshot target $A(u; s_1, \ldots, s_v)$ from the core chapter is defined over any coefficient ring for which the displayed additions make sense. Over $\mathbb{Z}_d$, it becomes a modular SUM circuit. Its explicit ambient word uses exactly two reverse-directed SUM gates and has length

$$u + v + 2.$$

The interval-tree proof of exact restricted length is most transparent over fields, but the displayed upper-bound and resource-amortization mechanism remain valid over every $\mathbb{Z}_d$.

For a field $\mathbb{F}_q$, the exact formula is

$$\ell_{\downarrow}(A) = u + 2v, \qquad \ell_*(A) \leq u + v + 2,$$

so

$$\Delta(A) \geq v - 2.$$

The connected family has asymptotic ratio approaching $3/2$.

6.4 Weighted physical costs

Suppose forward SUM gates cost one and reverse-directed SUM gates cost λ. The displayed word has weighted cost

$$u + v + 2\lambda,$$

while the restricted construction costs $u + 2v$. Hence the gain is at least

$$v - 2\lambda.$$

This formula can be combined with hardware calibration. If reverse orientation requires echo pulses or routing, the catalyst remains useful when the number of exported snapshots exceeds twice the effective penalty.

Scope boundary

A gate-count saving does not imply a lower physical error or depth. Device-specific conclusions require a cost vector containing duration, infidelity, crosstalk, and routing. The algebraic theorem identifies candidate rewrites and the exact break-even inequality.

6.5 Pauli action for prime qudits

For odd prime d, let X and Z denote generalized Pauli operators. The SUM gate acts by

$$X_s \mapsto X_s X_t^a, \qquad X_t \mapsto X_t,$$
$$Z_s \mapsto Z_s, \qquad Z_t \mapsto Z_s^{-a} Z_t.$$

Thus the modular linear action embeds into a symplectic transformation. The catalytic row excursion has a dual Pauli interpretation: a temporary reverse dependency changes both X propagation and the contragredient Z propagation, then restores them.

This makes the SUM family a bridge between the transvection paper and the Clifford chapter. It is the only quantum extension in this monograph that follows directly from the ring-universal theorem without a new finite search.

6.6 Continuous-variable shear analogue

On continuous variables, a linear coordinate map A embeds symplectically as

$$A \longmapsto \begin{pmatrix} A & 0 \\ 0 & A^{-T} \end{pmatrix}.$$

A row shear therefore gives a Gaussian SUM transformation. The six-row word identity remains a matrix identity in the corresponding shear subgroup.

Formal research skeleton

To claim an advantage in a full Gaussian gate library, one must compare against rotations, squeezers, beam splitters, and general symplectic decompositions. The current theorem proves a defect only inside the directed-shear versus all-shear comparison.

6.7 Compiler primitive

A compiler can search for a carrier row p, a temporary source q, and output rows requiring several successive states of p. The symbolic pattern is:

1. add q into p in the disallowed direction,
2. update p through a sequence of allowed additions,
3. copy selected intermediate states of p into outputs,
4. subtract q from p,
5. verify that the outputs contain the intended modular combinations.

The Python function `snapshot_target` constructs the target and `snapshot_ambient_word` emits the word.

Listing 6.1: Constructing a modular snapshot family.

```
from catalytic_gate_atlas.linear import (
    snapshot_target,
    snapshot_ambient_word,
    apply_row_word,
)

u = 4
phases = [0, 2, 4]
modulus = 5
A = snapshot_target(u, phases, modulus)
word = snapshot_ambient_word(u, phases, modulus)
assert apply_row_word(len(A), word, modulus) == A
```

6.8 Open qudit questions

Question 6.1. Does the first defective dimension remain six for every finite field and every modular ring $\mathbb{Z}_d$ under the directed-SUM metric?

Question 6.2. Can qudit Clifford gates outside the SUM subgroup produce a larger defect for a SUM target under total, entangling, or depth cost?

Question 6.3. Do composite dimensions admit characteristic-dependent catalytic motifs unavailable over fields?

Chapter 7

Catalytic basis escape in symplectic Clifford synthesis

7.1 Binary symplectic representation

Modulo global phase and Pauli translations, an n-qubit Clifford operation acts on Pauli exponent vectors

$$v = (x_0, \ldots, x_{n-1} \mid z_0, \ldots, z_{n-1})^T \in \mathbb{F}_2^{2n}$$

through a symplectic matrix

$$M \in \mathrm{Sp}(2n, 2).$$

The elementary gates used here act as follows:

$$\begin{aligned} H_i &: (x_i, z_i) \mapsto (z_i, x_i), \\ S_i &: z_i \mapsto z_i + x_i, \\ \mathrm{CX}_{i\to j} &: x_j \mapsto x_j + x_i, \quad z_i \mapsto z_i + z_j, \\ \mathrm{CZ}_{ij} &: z_i \mapsto z_i + x_j, \quad z_j \mapsto z_j + x_i. \end{aligned}$$

Let $P_n \leq \mathrm{Sp}(2n, 2)$ be the parabolic subgroup stabilizing the Z Lagrangian

$$L_Z = \{(0 \mid z) : z \in \mathbb{F}_2^n\}.$$

The gates S, CZ, and CX preserve L_Z. Hadamard gates do not.

Definition 7.1 (Parabolic and ambient Clifford metrics)**.** The restricted alphabet is

$$\mathcal{S}_P = \{S_i, \mathrm{CZ}_{ij}, \mathrm{CX}_{i\to j}\},$$

and the ambient alphabet is

$$\mathcal{S}_C = \mathcal{S}_P \cup \{H_i\}.$$

Targets lie in P_n. We study both total primitive count and entangling count, where H_i and S_i have cost zero while CX and CZ have cost one.

This is a precise basis-escape model. It is not the unique meaning of "CSS preserving" or "Clifford cost." It is chosen because both groups are finite and the restricted subgroup has a clean Lagrangian interpretation.

7.2 Exact group sizes

The finite state spaces are

$$|P_n| = 2^{n(n+1)/2} |\operatorname{GL}(n,2)|,$$

and

$$|\operatorname{Sp}(2n,2)| = 2^{n^2} \prod_{j=1}^{n} (2^{2j}-1).$$

For $n = 1, 2, 3$ this gives:

n	$\lvert P_n\rvert$	$\lvert\operatorname{Sp}(2n,2)\rvert$
1	2	6
2	48	720
3	10,752	1,451,520

These sizes permit exhaustive breadth-first or zero-one search through three qubits.

7.3 Total primitive count

Exact finite computation

With every gate in $\mathcal{S}_P$ and $\mathcal{S}_C$ assigned unit cost and with SWAP excluded as a primitive, exhaustive search gives:

n	restricted diameter	ambient diameter	defective targets	maximum defect
1	1	3	0	0
2	5	8	0	0
3	8	12	24	1

At $n = 3$, six targets have distance pair $(7, 6)$ and eighteen have pair $(8, 7)$.

Thus three qubits are the first total-count threshold in this exact swap-free model.

7.3.1 A three-qubit witness

One defective target has symplectic matrix

$$M_3 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \end{pmatrix}.$$

A restricted word of length seven is

$$\mathrm{CX}_{0\to2}, \mathrm{CZ}_{12}, \mathrm{CX}_{2\to0}, \mathrm{CX}_{0\to2}, \mathrm{CX}_{0\to1}, \mathrm{CX}_{1\to0}, \mathrm{CX}_{0\to1}\,.$$

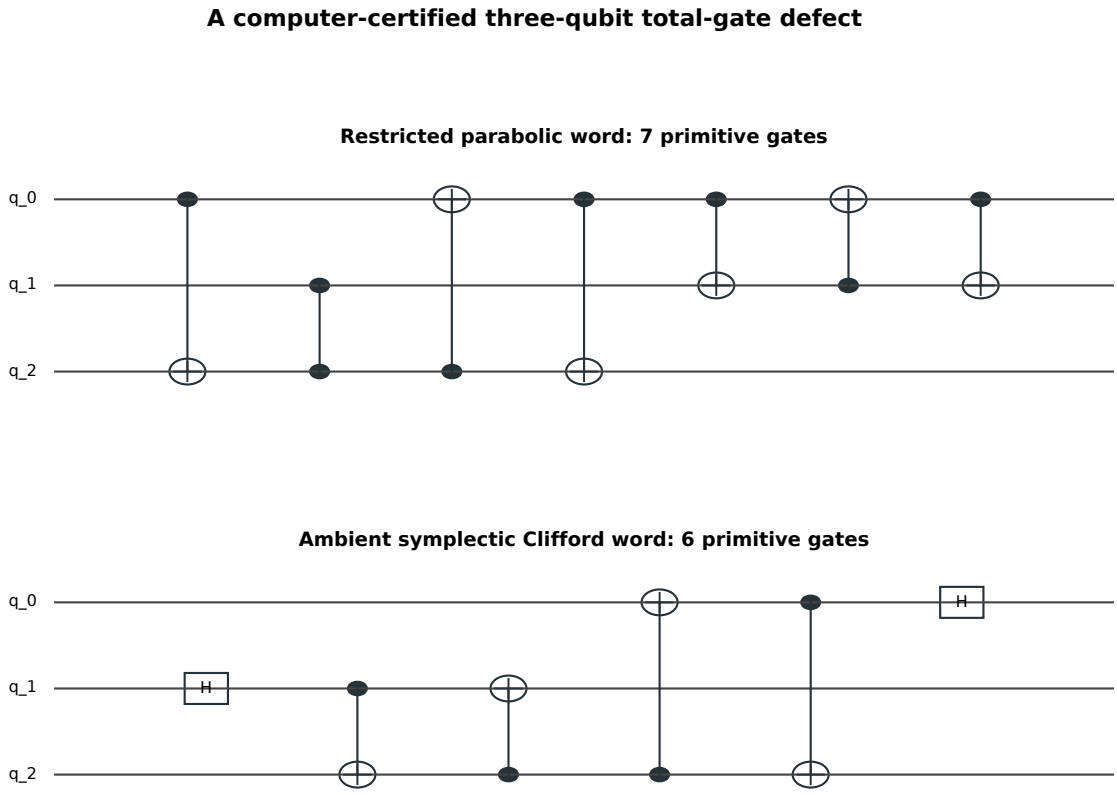


Figure 7.1: A swap-free three-qubit parabolic target with total primitive-count defect one.

An ambient word of length six is

$$H_1, \mathrm{CX}_{1\to2}, \mathrm{CX}_{2\to1}, \mathrm{CX}_{2\to0}, \mathrm{CX}_{0\to2}, H_0.$$

The Hadamards do not form a matched pair on the same wire. They open and close the Lagrangian excursion in different coordinates. This is a useful warning against assuming that every catalyst has the literal form $g \cdots g^{-1}$.

7.4 SWAP sensitivity

Add SWAP_{ij} to both alphabets as a unit-cost primitive. The target groups remain unchanged, but the metric changes.

Exact finite computation

For $n = 3$ with unit-cost SWAP gates, the restricted diameter drops from eight to seven, the ambient diameter drops from twelve to eleven, and every one of the $10,752$ parabolic targets has zero total-count defect.

The preceding twenty-four defects therefore measure a real gate-alphabet distinction, not an invariant of the subgroup inclusion alone. In many architectures SWAP is not elementary and costs

three CNOTs, while in others qubit movement or relabeling changes the accounting. The theorem must retain the convention.

7.5 Entangling-count defects

Now assign cost zero to H_i and S_i and cost one to CX and CZ.

Exact finite computation

The exact censuses are:

n	defective targets	distance pairs	maximum defect
1	0	none	0
2	4	$(3,2)$ with multiplicity 4	1
3	1,624	$(3,2):24,\ (4,3):432,\ (5,4):1056,$ $(6,4):48,\ (6,5):64$	2

The entangling metric has an earlier threshold and a larger exceptional set.

7.5.1 A two-qubit witness

A canonical target is

$$M_2 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}.$$

A restricted realization is

$$\mathrm{CX}_{0\to1}, S_1, \mathrm{CX}_{1\to0}, \mathrm{CX}_{0\to1},$$

with three entangling gates. An ambient realization is

$$H_1, \mathrm{CX}_{1\to0}, \mathrm{CX}_{0\to1}, H_0, S_1, S_0,$$

with two entangling gates.

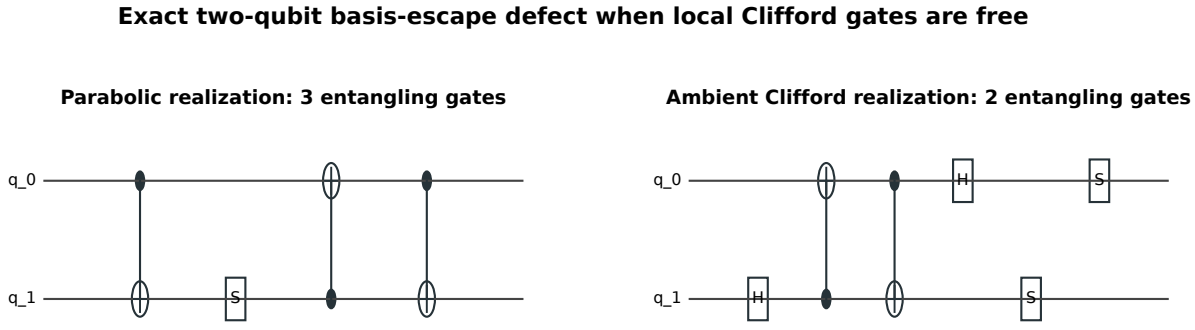


Figure 7.2: A two-qubit entangling-count defect. Local Clifford gates are free in this cost model.

7.6 Census summary

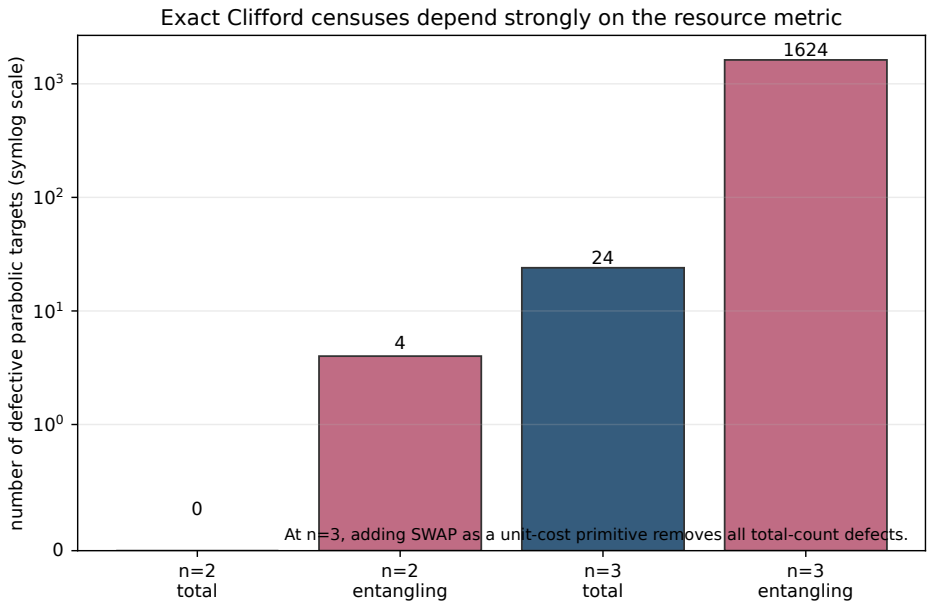


Figure 7.3: Exact parabolic-versus-ambient Clifford census. The vertical scale is symlogarithmic.

The results establish three distinct lessons.

1. Basis escape produces genuine Clifford defects, not only linear CNOT defects.
2. The first threshold depends on whether local gates are free.
3. Adding a SWAP primitive can eliminate a finite total-count threshold.

7.7 Search completeness

The search state is the exact $2n \times 2n$ binary symplectic matrix. Every generator acts by a fixed row update. Ordinary breadth-first search gives exact unit-cost distances. Zero-one breadth-first search gives exact entangling costs. Since every state of P_n and $\mathrm{Sp}(2n, 2)$ is reached, comparing the two distance maps is a complete census.

The optimized C++ verifier represents a 6×6 matrix by a 36-bit integer. It enumerates the full three-qubit group in a few seconds in the present environment. The Python implementation is slower but independently expresses the same state model.

7.8 Interpretation as symplectic holonomy

The parabolic subgroup is a fiber over the Lagrangian Grassmannian. A Hadamard moves one coordinate out of the chosen Lagrangian. The ambient path traverses other Lagrangian frames and returns to a target stabilizing L_Z.

This is the symplectic analogue of the unitriangular Schreier loop. The catalyst is a temporary exchange of X and Z roles. The consumers are entangling gates that become cheaper in the changed frame.

7.9 CSS-preserving extensions

The parabolic model fixes one Lagrangian, but CSS circuit classes can be defined more narrowly or more broadly. Possible restricted alphabets include:

1. CNOT and Pauli gates only,
2. CNOT, S, and diagonal controlled phases,
3. block-diagonal symplectic transformations preserving separate X and Z codes,
4. logical Clifford gates preserving a CSS stabilizer decomposition,
5. architecture-constrained versions of the preceding models.

Each produces a different subgroup and metric.

Formal research skeleton

A full CSS threshold paper should classify cost-preserving orbits of minimal defects, identify a symplectic-flag lower-bound invariant, and determine which defects survive when SWAP, qubit permutation, or local Cliffords are assigned alternative costs.

7.10 Depth and parallelism

Gate-count defects do not automatically imply depth defects. Given a gate sequence, define its dependency graph and schedule commuting or disjoint gates in parallel. The appropriate quantities are

$$\Delta_{\mathrm{depth}}(U), \qquad \Delta_{\mathrm{2q-depth}}(U), \qquad \Delta_{\mathrm{space-time}}(U).$$

A basis change applied transversally to many qubits may have small depth even when its total count is large. Conversely, a two-gate count saving can increase depth under locality constraints.

7.11 Open Clifford problems

Question 7.2. What are the defect orbits of the twenty-four three-qubit total-count targets under qubit permutations and parabolic conjugation?

Question 7.3. Does the maximum parabolic-versus-Clifford defect grow linearly, as $n \log n$, or faster under total primitive count?

Question 7.4. Can one prove a symbolic family with entangling-count defect greater than two, rather than relying on finite census data?

Question 7.5. Which finite symplectic parabolics are geodesically convex after SWAP is included as a primitive?

Conjectural frontier

The observed three-qubit defects should admit a small collection of symplectic root motifs analogous to the three-cross, one-phase classification in the unitriangular case. The present monograph records the exact census and witnesses but does not claim that classification.

Chapter 8

Temporary nonlinearity: affine logic versus Toffoli-assisted synthesis

8.1 The question

Let

$$\mathrm{AGL}(n,2) = \{x \mapsto Ax + b : A \in \mathrm{GL}(n,2),\ b \in \mathbb{F}_2^n\}$$

be the affine reversible group. It is generated by NOT and CNOT gates. The full classical reversible group on n bits is generated by NOT, CNOT, and Toffoli.

The catalytic question is:

> Can a circuit temporarily create nonlinear Boolean terms with Toffoli gates, reuse them, cancel them completely, and implement a final affine permutation with fewer gates than every affine circuit?

This is the nonlinear analogue of dirty linear memory.

8.2 Algebraic normal form

Every Boolean coordinate function has a unique algebraic normal form

$$f_i(x) = \bigoplus_{S \subseteq [n]} a_{i,S} \prod_{j \in S} x_j.$$

Affine maps have degree at most one. A Toffoli gate targeting t with controls a, b performs

$$x_t \longmapsto x_t + x_a x_b,$$

which can raise the degree of later coordinates. The final affine target requires every higher-degree coefficient to cancel.

A candidate lower-bound invariant must do more than track maximum degree. One Toffoli creates degree two and a second can remove it. Useful refinements include:

- the dimension of the quadratic monomial span,
- directional derivatives $D_u f(x) = f(x+u) + f(x)$,
- the rank of the bilinear polarization,

- the number of outputs consuming a nonlinear workspace term,
- restoration obligations for every nonlinear carrier.

8.3 Exact three-bit census

On three bits,

$$|\operatorname{AGL}(3,2)| = 2^3 |\operatorname{GL}(3,2)| = 1344,$$

while the group generated by NOT, CNOT, and Toffoli is the full permutation group

$$S_8, \qquad |S_8| = 40320.$$

Both spaces can be enumerated exactly.

Exact finite computation

For three bits, every affine target has the same minimum cost in the affine gate set and in the Toffoli-assisted gate set under each of the following metrics:

1. total primitive gate count, where NOT, CNOT, and Toffoli cost one,
2. entangling count, where NOT is free and CNOT and Toffoli cost one.

There are no defective affine targets among all 1344 targets.

The restricted total-count diameter is seven and the ambient diameter on S_8 is eight. Under entangling count, both relevant diameters are six. Equality on affine targets is stronger than a diameter comparison because every target is checked individually.

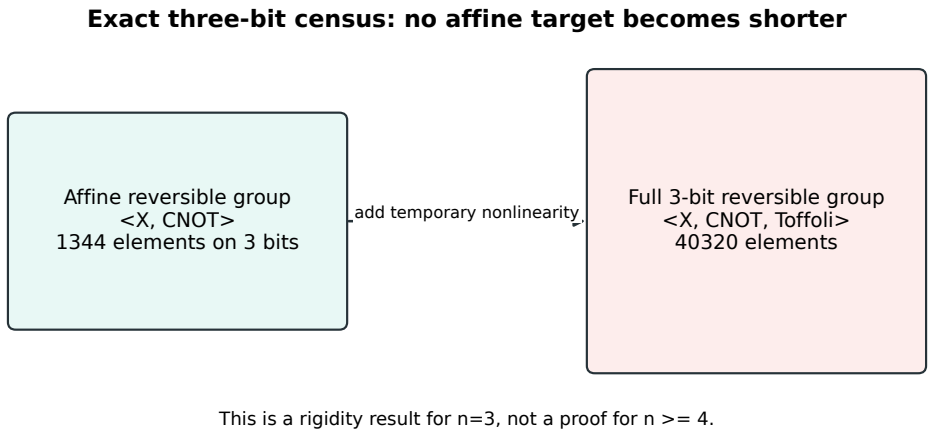


Figure 8.1: The complete three-bit affine-versus-Toffoli census is rigid.

8.4 Interpretation of the negative result

The census does not show that temporary nonlinearity is useless in general. It shows that three bits do not provide enough state space for a nonlinear carrier to repay its creation and cancellation

cost under the stated alphabets.

The negative result is valuable for three reasons.

1. It gives a certified lower threshold: any total- or entangling-count defect requires at least four bits.
2. It prevents a forced analogy with the six-wire linear catalyst.
3. It supplies a complete regression test for future synthesis code.

8.5 A formal nonlinear catalyst

A clean one-workspace template is

$$T\, h_m \cdots h_1\, T^{-1},$$

where T is Toffoli and the middle gates use the quadratic term created in the workspace. Since Toffoli is self-inverse, the displayed circuit uses two nonlinear gates.

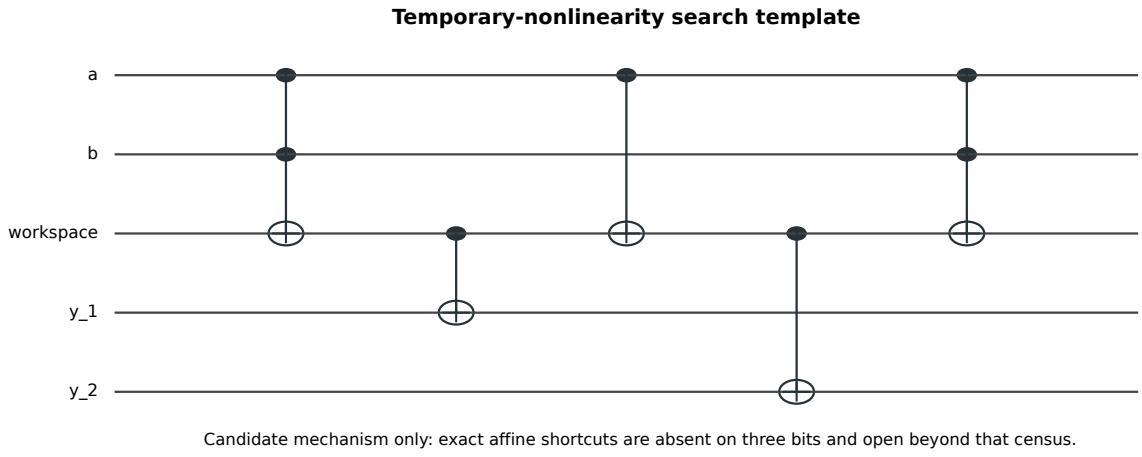


Figure 8.2: A temporary-nonlinearity template. It is a search skeleton, not a proved shortcut.

The key difficulty is that a final affine map cannot retain any product term. Every output that consumed the nonlinear carrier must be repaired, or the nonlinear contributions must cancel collectively.

8.6 Conjugation formulas

Let $T_{ab\to t}$ denote Toffoli. Conjugating an affine gate by T can produce nonlinear permutations. For example, if X_t flips the target,

$$T_{ab\to t} X_t T_{ab\to t} = X_t,$$

so no gain occurs. If a CNOT changes a control or exports the target, the conjugated map can contain several quadratic terms. The exact formula is easiest in algebraic normal form: substitute the Toffoli-updated coordinate into the affine map, then apply the inverse substitution.

A systematic search should classify pairs (T, h) for which

$$ThT^{-1} \in \mathrm{AGL}(n, 2).$$

For one Toffoli and an affine middle word, this is a normalizer-intersection problem. The exploratory four-bit search accompanying the monograph found no defect among simple two-Toffoli sandwiches with a middle affine word of restricted length at most five. This is a partial negative check, not an exhaustive four-bit theorem.

8.7 Potential lower-bound invariants

8.7.1 Quadratic support debt

Let $Q(f)$ be the span of quadratic monomials appearing across the coordinate functions. A Toffoli changes $Q(f)$ by at most a controlled rank-one contribution before subsequent substitutions. If several outputs consume the same quadratic term, a dynamic debt invariant may charge its creation, distribution, and cancellation.

8.7.2 Derivative rank

For an affine map, every derivative $D_u f$ is constant. During a nonlinear excursion, the derivative map

$$u \mapsto D_u f(x)$$

can depend on x. One can track the dimension of the nonconstant derivative space and require it to return to zero.

8.7.3 Higher-order Möbius coefficients

The ANF coefficients are the Möbius transform of the truth table. Projectors onto degree layers give an exact packet decomposition

$$f = f^{(0)} + f^{(1)} + f^{(2)} + \cdots .$$

A restricted affine circuit remains in degrees zero and one. The ambient circuit can populate higher packets temporarily. A successful lower bound must charge the useful flow through those packets, not merely their instantaneous dimension.

8.8 Four-bit search architecture

A full four-bit ambient group has size 16!, so direct enumeration is impossible. Several exact approaches remain available.

1. Meet in the middle for a fixed candidate target and length.
2. SAT or SMT encoding of a length-L circuit with affine endpoint constraints.
3. Bidirectional search restricted to states within a bounded nonlinear-degree or Toffoli-count budget.
4. Enumeration of two- or four-Toffoli normal forms modulo affine conjugation.

5. Mixed-integer or algebraic solving of ANF coefficient equations.

Protocol 8.1 (First four-bit threshold search)**.**

1. Compute exact affine distances for all 322,560 affine targets.
2. Enumerate ambient words with two Toffoli gates modulo affine gates on the left and right.
3. Retain words whose endpoint is affine.
4. Compare with the exact affine distance.
5. If no defect appears, increase the Toffoli budget and produce an unsatisfiability certificate for every shorter length.

The first step has already been implemented. The simple two-Toffoli sandwich test found no defect, but the full normal-form quotient remains open.

8.9 Resource variants

Temporary nonlinearity may be useful under one metric and useless under another. Natural costs include:

$$\text{Toffoli count}, \quad \text{CNOT count}, \quad T\text{-count after decomposition}, \quad T\text{-depth},$$
$$\text{ancilla count}, \quad \text{total depth}.$$

A Toffoli-assisted affine shortcut in total Boolean gates might disappear after fault-tolerant decomposition. Conversely, a circuit with more Boolean primitives can reduce T-depth.

8.10 Related nonlinear gate families

The same program applies to:

- CCZ-assisted diagonal Boolean phase maps,
- relative-phase Toffoli gates,
- multi-controlled X and Z gates,
- Reed–Muller degree hierarchies,
- reversible arithmetic over $\mathbb{Z}_d$,
- polynomial automorphisms over finite rings.

Conjectural frontier

There exists an affine target on finitely many bits whose total or fault-tolerant resource cost is reduced by a restored nonlinear excursion. The exact three-bit census rules out the smallest possible instance but supplies no evidence for a particular larger threshold.

8.11 Reproducible code

The implementation represents every reversible map as an exact permutation of 2^n basis states. Breadth-first search enumerates both the affine group and the full three-bit reversible group.

Listing 8.1: Exact three-bit affine-versus-Toffoli census.

```
from catalytic_gate_atlas import exact_affine_toffoli_census

print(exact_affine_toffoli_census(3, entangling_cost=False))
print(exact_affine_toffoli_census(3, entangling_cost=True))
```

The expected defect count is zero in both outputs.

Chapter 9

Phase-polynomial and Clifford+T catalysis

9.1 Why the phase setting is different

Fault-tolerant quantum compilation rarely assigns equal cost to every gate. Clifford operations are often treated as inexpensive relative to non-Clifford gates, while connectivity and parity computation can dominate the remaining cost. Consequently there is no single Clifford+T defect. One must specify at least

$$\mathbf{c}(U) = \big(T\text{-count}, T\text{-depth}, \text{two-qubit count}, \text{two-qubit depth}, \text{ancillas}\big).$$

A catalytic excursion may improve one component and worsen another. The natural comparison is therefore Pareto-valued rather than scalar unless a hardware-specific weight vector is fixed.

Let $\omega = e^{i\pi/4}$. A phase-polynomial circuit on n qubits has the form [3, 29]

$$U|x\rangle = \omega^{P(x)}|Ax + b\rangle, \tag{9.1}$$

where $A \in \mathrm{GL}(n, 2)$, $b \in \mathbb{F}_2^n$, and $P : \mathbb{F}_2^n \to \mathbb{Z}_8$ is represented as a sum of binary linear forms with coefficients in $\mathbb{Z}_8$. The diagonal case has $A = I$ and $b = 0$.

The restricted model may be chosen in several ways:

1. diagonal phase gadgets only;
2. CNOT-dihedral circuits generated by X, CNOT, and T;
3. parity networks in which every intermediate transformation remains affine on computational basis labels;
4. a prescribed family of commuting Pauli rotations.

The ambient model may add Hadamards, arbitrary Clifford conjugations, measurements, or ancillas. These choices define different problems.

9.2 Pauli-rotation language

For a Pauli operator P and angle θ, write

$$R_P(\theta) = e^{-i\theta P/2}.$$

Every Clifford C satisfies

$$CR_P(\theta)C^\dagger = R_{CPC^\dagger}(\theta). \tag{9.2}$$

Thus a short ambient basis change can rotate a large family of phase gadgets into a different commuting or nearly commuting family, apply them there, and rotate back.

Proved statement

Equation (9.2) is an exact catalytic transport identity. It does not by itself prove a gate-count defect. A defect additionally requires a lower bound showing that the transported family is more expensive in the restricted representation than the opening and closing Clifford layers plus the transported implementation.

The simplest symbolic sandwich is

$$C\left(\prod_{j=1}^{m} R_{P_j}(\theta_j)\right)C^\dagger = \prod_{j=1}^{m} R_{CP_jC^\dagger}(\theta_j).$$

If the P_j commute, their conjugates commute. A catalyst can therefore change the geometry of the support without changing the algebra of the angles.

9.3 Phase polynomials and parity carriers

A CNOT network computes linear parities

$$\lambda(x) = a \cdot x \in \mathbb{F}_2.$$

Applying T^k to a wire carrying $\lambda(x)$ contributes

$$\omega^{k\lambda(x)}$$

to the phase polynomial. A parity network must create each required parity, expose it to a phase gate, and restore the linear map required at the endpoint.

This makes phase-polynomial synthesis a close relative of the snapshot families in the linear core. A parity row may evolve through

$$\lambda_0, \lambda_1, \ldots, \lambda_m,$$

with selected states consumed by phase gates. The difference is that consumption need not copy the parity to another row; it can deposit phase while leaving the carrier unchanged.

Formal research skeleton

A candidate phase catalyst consists of:

1. a Clifford C that opens a non-diagonal frame;
2. an evolving Pauli or parity carrier in that frame;
3. several phase rotations consuming intermediate carrier states;
4. $C^\dagger$ restoring the original diagonal or CNOT-dihedral model;
5. a certificate that no restricted parity network attains the same resource vector.

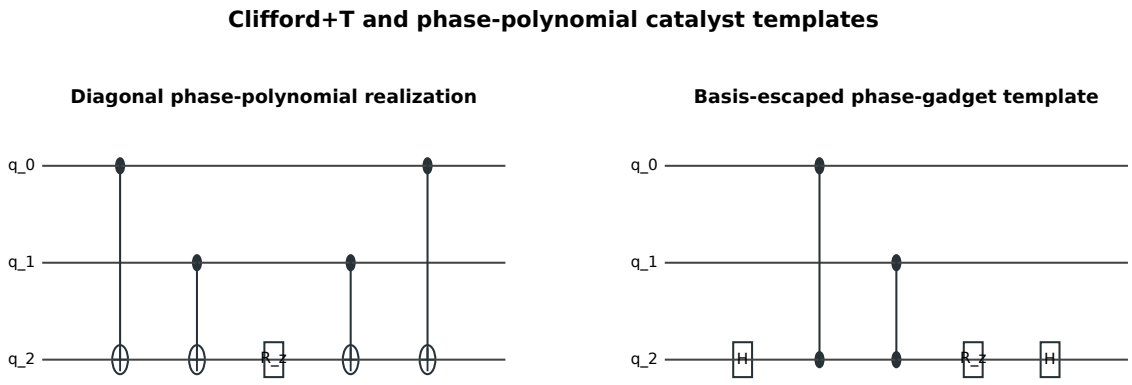


Figure 9.1: Symbolic Clifford+T catalyst: a basis excursion transports a family of phase obligations and is removed at the endpoint.

9.4 T-count as a coding problem

For diagonal Clifford+T synthesis without ancillas, equivalent phase-polynomial descriptions can differ by identities whose coefficients vanish modulo eight on Boolean inputs. Minimizing the number of odd coefficients is naturally expressed as a coset-decoding problem in a punctured Reed–Muller-type code.

Let $a \in \mathbb{Z}_8^{2^n-1}$ index nonzero linear forms and let $\mathrm{odd}(a)$ denote the binary vector recording which coefficients are odd. An abstract restricted T-count is

$$\ell_T(P) = \min_{a:\, P_a = P} \mathrm{wt}(\mathrm{odd}(a)).$$

An ambient Clifford excursion changes not only the coefficient vector but also the Pauli frame in which the rotations are represented. A lower bound that survives ambient Clifford conjugation must therefore be invariant under the relevant symplectic action or must explicitly charge the frame change.

9.5 T-depth and commuting layers

Two phase gadgets may share a T layer when a suitable Clifford network maps their parities to distinct wires or when ancilla resources permit parallel extraction. Define the conflict graph Γ_P whose vertices are odd phase terms and whose edges encode inability to realize two terms in one layer under the chosen ancilla and connectivity rules. Then

$$T\text{-depth}(P) \geq \chi(\Gamma_P)$$

for any conflict graph whose edge relation is sound. Catalysis can alter Γ_P by changing the Pauli frame. The right question is not merely whether C preserves the number of terms, but whether it reduces the chromatic or scheduling obstruction enough to repay its own depth.

9.6 A vector-valued defect

For a target U in a restricted phase model, define

$$\mathcal{C}_{\mathrm{res}}(U) = \{\mathbf{c}(\mathcal{W}) : \mathcal{W} \text{ is a restricted circuit for } U\}$$

and similarly $\mathcal{C}_{\mathrm{amb}}(U)$. Their Pareto frontiers are denoted $\mathrm{Pareto}_{\mathrm{res}}(U)$ and $\mathrm{Pareto}_{\mathrm{amb}}(U)$.

Definition 9.1 (Pareto catalytic defect)**.** A target has a Pareto catalytic defect when the ambient Pareto frontier contains a point strictly dominating every restricted cost vector in at least one declared region of resource space.

Scalarization by positive weights w gives

$$\Delta_w(U) = \min_{c \in \mathcal{C}_{\mathrm{res}}(U)} w \cdot c - \min_{c \in \mathcal{C}_{\mathrm{amb}}(U)} w \cdot c.$$

Different w can change the sign of the defect. This is expected, not pathological.

9.7 Exact finite-search model

A finite exact atlas can be built [20, 4] for small n by representing Clifford+T operators in one of three ways.

1. Exact matrices over the cyclotomic ring $\mathbb{Z}[1/\sqrt{2}, i]$ modulo global phase.
2. Normal forms combining a binary symplectic Clifford with a phase-polynomial record.
3. Quotients by left and right Clifford actions when only T-count is charged.

The third model can be dramatically smaller but requires care: quotienting is sound only when the free Clifford actions in the cost model agree with the quotient relation.

Protocol 9.2 (Small-width Clifford+T threshold atlas)**.**

1. Fix exact versus approximate synthesis, ancilla rules, and the resource vector.
2. Enumerate the restricted target subgroup or a complete set of normal forms.
3. Compute restricted Pareto frontiers by multiobjective shortest path.
4. Enumerate ambient states or use meet in the middle for candidate targets.
5. Store an optimal witness and a lower-bound certificate for every defect.
6. Classify defects under qubit permutation and free Clifford equivalence.

9.8 Lower-bound candidates

9.8.1 Odd-support rank

The span of odd phase terms is invariant under some restricted rewrites but transforms covariantly under linear changes of variables. Its dimension gives a coarse obstruction.

9.8.2 Pauli support geometry

Represent a commuting Pauli family by an isotropic subspace of $\mathbb{F}_2^{2n}$. Restricted diagonal circuits remain in the Z Lagrangian. Ambient Clifford circuits move through the symplectic Grassmannian. Distance from the Z Lagrangian, together with the number of deposited rotations, is the phase analogue of microscopic flag escape.

9.8.3 Moment and derivative packets

For a phase function P, finite differences

$$D_u P(x) = P(x+u) - P(x) \pmod 8$$

separate affine, quadratic, and higher interactions. A hierarchy of derivative tensors can record temporary flow into nonlinear phase packets.

9.8.4 Architecture-aware parity rank

On a coupling graph, each parity has a Steiner cost. A catalyst can be profitable by making several required parities share a Steiner tree. A sound lower bound may combine cut rank with edge congestion and restoration debt.

9.9 Tensor and repeated-module questions

For targets U and V, independent synthesis gives

$$\ell_T(U \otimes V) \le \ell_T(U) + \ell_T(V)$$

when the systems are disjoint and local Clifford cost is ignored. Joint synthesis may be better because phase identities can couple terms across modules. This motivates a stable regularization

$$\tau_T(U) = \liminf_{r\to\infty} \frac{\ell_T(U^{\otimes r})}{r}.$$

A one-copy rigid phase target could still display tensor activation.

Conjectural frontier

There are diagonal or CNOT-dihedral targets whose Pareto frontier is strictly improved by a restored non-diagonal Clifford excursion. The smallest qubit threshold and the resource component in which the first defect appears are open in this monograph.

9.10 What would count as a theorem

A publishable first result in this direction should contain all of the following:

1. an exact target family and gate alphabets;
2. explicit ambient circuits;
3. a restricted lower bound for the same declared cost;

4. exact treatment of global phase and ancillas;
5. classification below the first threshold or a proof of minimality;
6. reproducible symbolic code.

Merely rewriting a phase gadget with Hadamards is not sufficient. The decisive content is the optimality comparison.

Chapter 10

Gaussian, matchgate, and fermionic catalytic models

10.1 Two Gaussian settings

Quantum Gaussianity occurs [34] in two closely related but algebraically different settings.

For n bosonic modes, a Gaussian unitary acts on quadratures

$$\mathbf{r} = (q_1, \ldots, q_n, p_1, \ldots, p_n)^T$$

by an affine symplectic transformation

$$\mathbf{r} \longmapsto S\mathbf{r} + d, \qquad S \in \mathrm{Sp}(2n, \mathbb{R}).$$

For n fermionic modes, a fermionic Gaussian unitary acts on $2n$ Majorana operators by

$$\gamma \longmapsto O\gamma, \qquad O \in SO(2n).$$

Nearest-neighbor matchgates give a circuit presentation of a fermionic Gaussian subgroup on qubits.

In both cases one can choose a restricted subgroup preserving a flag, mode ordering, particle number, or bipartition, and compare it with a larger Gaussian or non-Gaussian gate set.

10.2 An inherited symplectic shear theorem

Every invertible linear transformation $A \in \mathrm{GL}(n, R)$ has a canonical symplectic embedding

$$\iota(A) = \begin{pmatrix} A & 0 \\ 0 & A^{-T} \end{pmatrix}. \tag{10.1}$$

Indeed, with

$$J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix},$$

one checks

$$\iota(A)^T J \iota(A) = J.$$

Thus every row-transvection identity in the established linear theory induces an exact identity between symplectic shear circuits.

Proved statement

Under a Gaussian gate alphabet in which each embedded elementary transvection $\iota(I + aE_{ij})$ is one primitive and direction is the only restriction, the universal six-row catalytic atom embeds into a twelve-quadrature Gaussian atom with restricted length seven and ambient length six.

This is a formal inheritance statement for the declared shear alphabet. It does not assert optimality after arbitrary beam splitters, squeezers, Fourier transforms, or local Gaussian gates are admitted at unit cost.

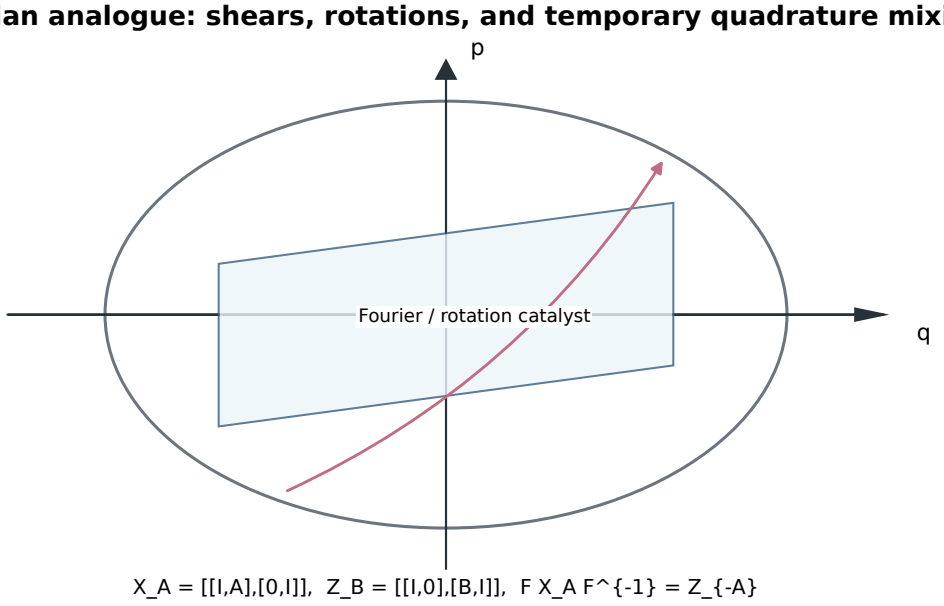


Figure 10.1: The row-transvection atom embedded as a Gaussian symplectic shear circuit.

10.3 Bosonic gate hierarchies

Natural nested models include

triangular shears $\subset$ all Gaussian unitaries $\subset$ Gaussian plus a non-Gaussian primitive $\subset U(\mathcal{H})$.

Typical primitive resources are:

- SUM or controlled displacement gates,
- beam splitters,
- single-mode and two-mode squeezers,
- Fourier transforms,
- cubic phase gates,
- photon counting or non-Gaussian state injection.

Each inclusion supports a different defect. A target Gaussian transformation might be shortened by leaving a triangular decomposition but remaining Gaussian. A more radical question asks whether temporary non-Gaussianity can shorten a final Gaussian target.

10.4 Bloch–Messiah versus catalytic synthesis

A bosonic Gaussian unitary admits a Bloch–Messiah decomposition

$$S = K_1 D K_2,$$

where K_1, K_2 are passive symplectic transformations and D is a diagonal squeezing layer. This normal form optimizes conceptual structure, not necessarily a weighted gate count on a specific architecture.

Catalytic synthesis asks whether a short excursion into a basis where many shear or squeezing obligations align can reduce total cost. The opening and closing passive layers play the role of the catalyst.

Formal research skeleton

A Gaussian catalytic theorem requires:

1. a declared primitive library and cost vector;
2. a target subgroup, such as triangular, number conserving, or local;
3. an explicit sandwich KHK^{-1} returning to the target subgroup;
4. a lower bound based on symplectic rank, singular values, graph cuts, or mode support;
5. treatment of approximation error when continuous parameters are synthesized numerically.

10.5 Fermionic Gaussian and matchgate circuits

Let $c_j, c_j^\dagger$ be fermionic creation and annihilation operators. Quadratic Hamiltonians generate Gaussian unitaries, and their action is determined by an orthogonal transformation on Majoranas. Number-conserving Gaussian operations correspond to a unitary action on modes, while general Gaussian operations allow particle-hole mixing.

This yields the hierarchy

$$\begin{aligned} U(n) \text{ mode transformations} &\subset SO(2n) \text{ fermionic Gaussian transformations} \\ &\subset \text{parity-preserving unitaries.} \end{aligned}$$

A temporary Bogoliubov transformation can violate number conservation while preserving parity. The target may nevertheless return to the number-conserving subgroup.

Definition 10.1 (Bogoliubov catalytic defect)**.** For a number-conserving Gaussian target U, compare the minimum cost using number-conserving primitives with the minimum cost using general Gaussian primitives, requiring the endpoint to conserve number exactly.

10.6 Matchgates and geometry

Nearest-neighbor matchgate circuits [32, 19] have strong classical simulation properties. Their geometry changes when SWAP, fermionic SWAP, long-range connectivity, or a single non-matchgate resource is added. Candidate restricted/ambient pairs include:

1. nearest-neighbor matchgates versus all-to-all matchgates;
2. number-conserving matchgates versus general matchgates;
3. matchgates versus parity-preserving two-qubit gates;
4. Gaussian circuits versus Gaussian circuits with one reusable non-Gaussian ancilla.

A compiler application is immediate: a temporary mode permutation or particle-hole rotation may allow a batch of otherwise expensive interactions to share one routing or basis-change layer.

10.7 Covariance-matrix lower bounds

For a Gaussian state, the covariance matrix Γ transforms as

$$\Gamma \longmapsto S\Gamma S^T$$

in the bosonic case and

$$\Gamma \longmapsto O\Gamma O^T$$

in the fermionic case. Possible restricted lower bounds include:

- rank change of off-diagonal covariance blocks,
- symplectic singular-value obligations,
- growth of mode support,
- graph bandwidth and cut rank,
- number of nontrivial canonical angles between invariant subspaces.

One elementary gate changes only a controlled portion of these data. A useful invariant must also charge preparation of a carrier that makes progress on many cuts simultaneously.

10.8 Temporary non-Gaussianity

Suppose a non-Gaussian primitive g is introduced and later removed:

$$U = gh_m \cdots h_1 g^{-1}, \qquad U \in G_{\text{Gauss}}.$$

If $gh_i g^{-1}$ generates several useful Gaussian effects, the excursion may be catalytic. However, many standard non-Gaussianity monotones vanish only on states rather than unitaries, and cancellation at the endpoint complicates direct lower bounds.

Candidate monotones and certificates include:

- degree in the Weyl characteristic function,
- Wigner negativity generated during the path,
- operator Schmidt rank across mode cuts,

- polynomial degree of Heisenberg-evolved quadratures,
- stabilizer rank for encoded finite-dimensional subspaces.

10.9 Exact and numerical search

Finite exact enumeration is available for binary symplectic or orthogonal quotients, but continuous Gaussian groups require a different strategy.

1. Fix a symbolic gate skeleton and solve polynomial equations for parameters.
2. Use interval arithmetic to certify nonexistence below a length.
3. Exploit canonical decompositions to quotient local or passive layers.
4. For discrete-angle Gaussian libraries, use exact algebraic-number arithmetic.
5. Store both an upper-bound circuit and a dual or algebraic certificate.

10.10 Architecture and routing

On a line or grid, Gaussian and matchgate circuits may be dominated by routing. Let G_{conn} be the hardware graph. A mode permutation π can be treated as a catalyst if it is opened, many interactions are executed in the reordered frame, and π^{-1} restores the logical labeling. The benefit is

$$\text{routing saved across consumers} - 2\,\mathrm{cost}(\pi).$$

This is the same amortization law as the linear snapshot family, now applied to mode geometry.

Conjectural frontier

There exist number-conserving or nearest-neighbor Gaussian targets whose minimum weighted cost is reduced by a restored Bogoliubov, mode-permutation, or non-Gaussian excursion. The first exact threshold depends on the primitive library and is not determined here.

10.11 Research deliverables

A complete Gaussian extension should produce:

1. a finite or semialgebraic model with exact endpoint equality;
2. at least one nontrivial lower-bound invariant;
3. a minimal catalytic atom;
4. a weighted hardware interpretation;
5. a tensor or repeated-mode amplification theorem;
6. code using exact arithmetic or certified intervals.

Chapter 11

Symmetry, ancillas, measurement, and architecture

11.1 Beyond subgroup word metrics

Some of the most important quantum resource enlargements are not inclusions of groups. Measurement introduces branches, reset is irreversible, and ancillas enlarge the Hilbert space. The appropriate abstraction is a nested pair of realization categories

$$\mathcal{C}_{\mathrm{res}} \subseteq \mathcal{C}_{\mathrm{amb}},$$

with a cost assigned to generating morphisms.

For a target channel Φ in the restricted category, define

$$c_{\mathrm{res}}(\Phi) = \inf\{\mathrm{cost}(W) : W \text{ realizes } \Phi \text{ in } \mathcal{C}_{\mathrm{res}}\}$$

and similarly $c_{\mathrm{amb}}(\Phi)$. The value $+\infty$ is allowed. A finite positive difference is a metric defect; an infinite difference is an existence defect.

11.2 Symmetry-preserving circuits

Let Q be a conserved charge and let

$$H_Q = \{U : [U, Q] = 0\}.$$

Restricted circuits use gates commuting with Q. Ambient circuits may temporarily leave H_Q but must return to a charge-preserving endpoint.

Examples include:

- particle-number conservation,
- parity conservation,
- total spin or magnetization,
- lattice gauge constraints,
- stabilizer or CSS sector preservation,
- block-diagonal representation sectors.

The Hilbert space decomposes as

$$\mathcal{H} = \bigoplus_q \mathcal{H}_q.$$

A restricted circuit is block diagonal. An ambient catalyst may couple sectors, use the enlarged route, and decouple them at the endpoint.

Formal research skeleton

A symmetry-breaking catalyst has three obligations:

1. quantify how far the path leaves the symmetry algebra;
2. prove exact return, not merely return on a chosen input state;
3. compare the cost with every symmetry-preserving realization of the full target operator.

11.3 Sector-flow invariants

Let P_q be the projector onto sector q. For an intermediate unitary U_t, define transfer blocks

$$T_{q,r}(t) = P_q U_t P_r.$$

A restricted path has $T_{q,r} = 0$ for $q \neq r$. Ambient paths may populate off-diagonal blocks. Possible excursion measures include

$$\sum_{q \neq r} \operatorname{rank} T_{q,r}, \qquad \max_{q \neq r} \|T_{q,r}\|, \qquad \text{bandwidth in } |q - r|.$$

These are analogues of strict-upper rank and bandwidth in the unitriangular theory.

11.4 Clean and dirty ancillas

A clean ancilla starts in a specified state, usually $|0\rangle$, and must be returned to that state. A dirty ancilla starts in an unknown state and must be restored for every input.

Let U act on the data register. A clean-ancilla realization is a unitary W satisfying

$$W(|\psi\rangle \otimes |0\rangle) = (U|\psi\rangle) \otimes |0\rangle \quad \text{for every } |\psi\rangle.$$

A dirty-ancilla realization requires

$$W(|\psi\rangle \otimes |\phi\rangle) = (U|\psi\rangle) \otimes |\phi\rangle \quad \text{for every } |\psi\rangle, |\phi\rangle.$$

The second condition is strictly stronger.

Ancillas define a family of defects

$$\Delta_{a,\text{clean}}(U), \qquad \Delta_{a,\text{dirty}}(U),$$

where a is the number of ancillary qubits or modes. Stable ancilla capacity asks whether a fixed number of reusable ancillas can yield an unbounded saving as the data size grows.

11.5 Measurement and feedforward

A measurement-assisted realization is described by an instrument with classical outcomes m, conditional channels Φ_m, and feedforward operations. Exact deterministic implementation of a unitary target requires every branch, after correction, to induce the same channel.

The cost vector may include

$$(T\text{-count}, \text{two-qubit count}, \text{measurement rounds}, \text{classical latency}, \text{ancilla qubits}).$$

A measurement can create an existence defect by realizing a channel unavailable unitarily under the same locality or depth constraints, or a metric defect by reducing one resource while increasing another.

Definition 11.1 (Branch-catalytic realization)**.** A measurement-assisted circuit is branch catalytic when the intermediate classical or quantum branch information is absent from the final data channel and all declared work systems are reset or discarded according to the model.

11.6 Teleportation as a canonical ambient excursion

Gate teleportation replaces direct implementation by:

$$\text{prepare resource state} \to \text{entangle and measure} \to \text{classical correction}.$$

The ambient category contains resources not present in the target channel: entanglement, measurement outcomes, and feedforward. The endpoint can nevertheless be a deterministic unitary.

This is an archetypal realization defect, but careful accounting is mandatory. A claimed saving must include resource-state preparation or explicitly treat the state as an external catalyst.

11.7 Architecture-constrained gates

Let $\Gamma = (V, E)$ be a hardware graph. A restricted two-qubit gate may act only on edges of E. The ambient model may permit a small number of nonlocal interactions, temporary logical relabelings, teleportation edges, or bus-mediated gates.

For a target U, define

$$\ell_\Gamma(U) = \text{minimum hardware-respecting cost},$$

and compare it with an enlarged graph Γ'. A matched routing excursion has the form

$$\pi\, W\, \pi^{-1},$$

where π is a logical permutation. The excursion is profitable when many gates in W become local in the permuted frame.

11.8 Depth, count, and space-time volume

Let g_t be the set of gates scheduled in layer t. Common scalar costs are

$$\sum_t |g_t|, \qquad \#\{t : g_t \neq \varnothing\}, \qquad \sum_t \#\{\text{active qubits in layer } t\}.$$

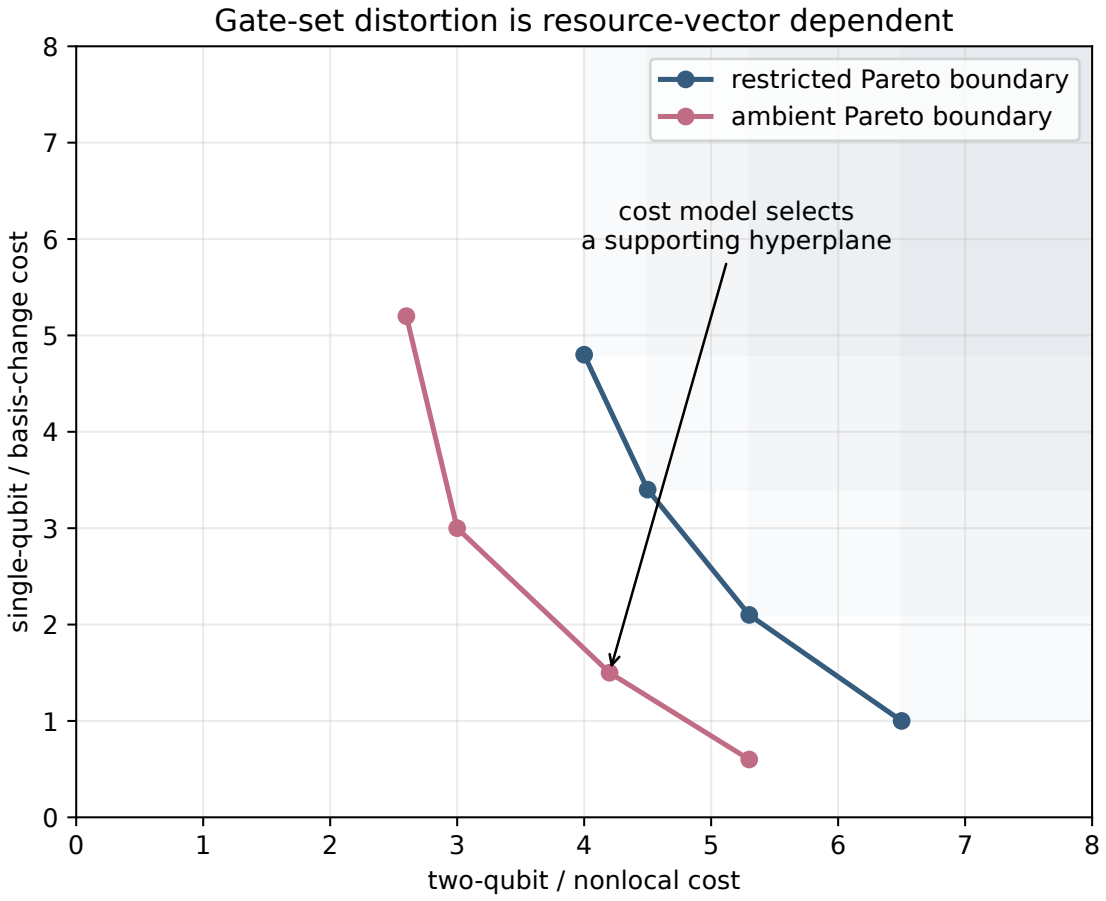


Figure 11.1: Gate-set defects generally live in a multi-resource Pareto geometry.

A catalyst may reduce count but increase depth, or vice versa. Space-time volume is often the more stable hardware quantity.

For stochastic error rates p_g, a first-order weighted cost is

$$C_{\mathrm{err}}(W) = \sum_{g \in W} -\log(1 - p_g).$$

This additive approximation can be inserted directly into Dijkstra search. It does not capture coherent or correlated errors, but it produces a declared and reproducible metric.

11.9 Approximate synthesis

For continuous gate sets, define

$$\ell_{\mathrm{res},\varepsilon}(U) = \min\{\mathrm{cost}(W) : \|W - U\| \le \varepsilon\}.$$

There are two distinct border phenomena:

1. algebraic border length, where a target lies in the closure of a bounded exact word locus;
2. analytic approximation length, where an error tolerance is explicitly permitted.

They should not be conflated.

A catalyst can improve the asymptotic Solovay–Kitaev or number-theoretic approximation constant even when exact synthesis is impossible in both models.

11.10 Resource theories and monotones

A quantum resource theory provides free states, free operations, and monotones. Gate-set distortion adds a path-sensitive question: the endpoint may be free even though the path temporarily creates resource.

A state monotone $M(\rho)$ alone may fail because the catalyst acts at the channel level. Useful channel-level quantities include:

- robustness or magic capacity,
- entangling power,
- operator Schmidt rank,
- Choi-state resource measures,
- symmetry asymmetry of the channel,
- amortized resource generation.

A lower bound must relate the cumulative or peak resource to the declared gate cost and account for cancellation.

11.11 A categorical amplification principle

Suppose a monoidal category admits a tensor product and a cost satisfying

$$c_{\mathrm{amb}}(f \otimes g) \leq c_{\mathrm{amb}}(f) + c_{\mathrm{amb}}(g).$$

If restricted cost is additive on a target family, then any one-copy defect amplifies linearly under independent tensor powers. Superlinear amplification requires a normalization in which dimensions multiply, as in the matrix tensor rule of the core theory, or a joint-synthesis synergy across copies.

11.12 Experimental standards

Every architecture, measurement, or ancilla claim should state:

1. gate alphabet and native connectivity;
2. clean, dirty, catalytic, or consumable ancilla convention;
3. deterministic or postselected implementation;
4. exact or approximate endpoint condition;
5. total, weighted, depth, or Pareto cost;
6. whether resource-state preparation is included;
7. whether qubit relabeling is free;
8. certificate of restoration on all inputs.

Without these declarations, two reported "gate counts" may refer to incomparable problems.

Conjectural frontier

For several common symmetry and architecture restrictions, a constant-size restored excursion should yield an unbounded saving on structured target families. The unitriangular construction proves this phenomenon for directed linear information flow; analogous theorems for charge sectors, measurement branches, and nonlocal routing remain open.

Part IV

Packets, realization, and local-to-global structure

Chapter 12

Packets, types, holonomy, and restricted realization

12.1 Why coarse spectra are not enough

A finite symmetry problem often comes with a natural commuting operator algebra. Its joint spectrum is attractive because it is canonical, compact, and computationally cheap. Yet the joint eigenvalues and their multiplicities can be identical across objects whose realization behavior is different.

This chapter develops the hierarchy

$$\text{spherical packet} \prec \text{embedded type packet} \prec \text{holonomy packet} \prec \text{realization fiber}. \tag{12.1}$$

The relation $\prec$ means "retains strictly more potentially relevant information," not that every refinement is complete.

12.2 Spherical packets

Let $T_1, \dots, T_m$ be commuting normal operators on a finite-dimensional Hilbert space. Their unique coarsest joint spectral resolution is

$$T_j = \sum_{\lambda} t_j(\lambda) B_\lambda, \qquad \sum_{\lambda} B_\lambda = I.$$

The tuple

$$\lambda = (t_1(\lambda), \dots, t_m(\lambda))$$

is the spherical label. The abstract spectral packet records these labels and the ranks of B_λ.

Two different embedded systems can have the same abstract packet. The actual matrices B_λ, endpoint coefficients, and transition operators may still differ.

Principle 12.1 (Spherical collapse)**.** *Passing from embedded operators to eigenvalues and multiplicities can erase exactly the incidence or transition data that controls restricted realization.*

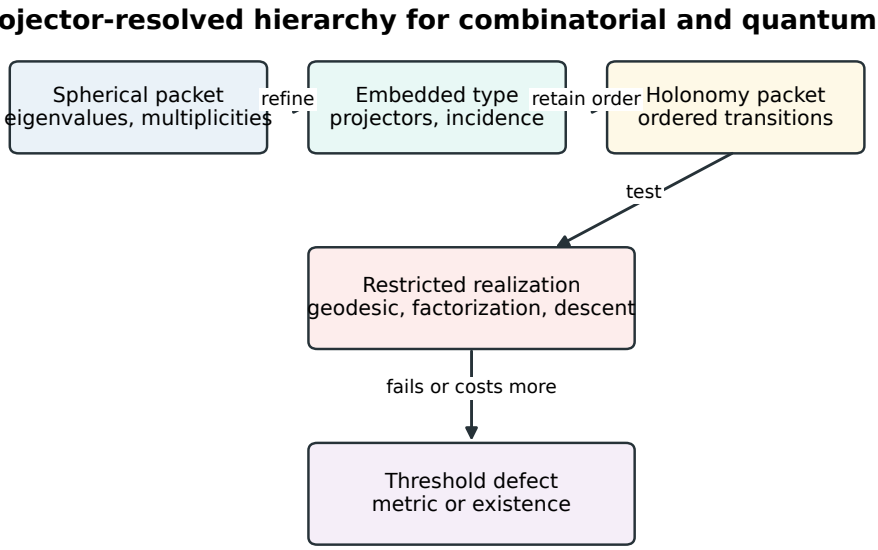


Figure 12.1: The packet hierarchy used to organize finite threshold systems.

12.3 The Latin-square association algebra

A Latin square L of order n has n^2 cells. Let R, C, S be adjacency matrices for equality of row, column, and symbol. The algebra generated by I, J, R, C, S is commutative and its abstract multiplication table depends only on n, not on the paratopy class of L.

Define

$$Q_R = \frac{I+R}{n}, \qquad Q_C = \frac{I+C}{n}, \qquad Q_S = \frac{I+S}{n}, \qquad B_0 = \frac{J}{n^2}.$$

Then the five joint projectors are

$$B_0, \quad Q_R - B_0, \quad Q_C - B_0, \quad Q_S - B_0, \quad I - Q_R - Q_C - Q_S + 2B_0.$$

Their multiplicities are

$$1, \quad n-1, \quad n-1, \quad n-1, \quad (n-1)(n-2).$$

The Latin-square graph $R + C + S$ has universal spectrum

$$3(n-1)^1, \qquad (n-3)^{3(n-1)}, \qquad (-3)^{(n-1)(n-2)}.$$

At $n = 6$ this becomes

$$15^1, \qquad 3^{15}, \qquad (-3)^{20}.$$

Thus the graph spectrum cannot distinguish the twelve main classes of order six.

Exact finite computation

The accompanying atlas transcribes all twelve order-six main classes used in the quantum-Latin-square reduction and verifies that their abstract R, C, S joint eigenvalue packets coincide. This is an exact no-go result for an eigenvalue-only explanation of the order-six obstruction.

12.4 Transition fusion also collapses

Let ρ_i be the row permutations of a Latin square. The symmetrized aggregate of all pair transitions satisfies

$$\sum_{i<j} \left(P_{\rho_i \rho_j^{-1}} + P_{\rho_j \rho_i^{-1}} \right) = n(J - I). \tag{12.2}$$

Hence first-order fusion destroys the class information as well.

The lesson is general: summing typed transitions can be as destructive as discarding the embedded projectors. Order and labels matter.

12.5 Embedded type data

A type packet retains the actual projectors, support partitions, distinguished incidence sets, or local representation types. Examples include:

- the row, column, and symbol partitions of a Latin square;
- a projective-line closure system;
- the flag stabilized by a circuit subgroup;
- support patterns of unitary or quantum-Latin entries;
- ramification or block data in a finite group action.

In the order-six Latin-square computation, the discrete projective algorithm leaves two hard classes with a spread of twelve disjoint projective lines on the 36 cells. The spread adjacency operator commutes with R, C, S and therefore refines the spherical packet. The two hard classes nevertheless retain the same refined abstract spectrum. One type operator is not complete.

12.6 Holonomy packets

Let a family of local charts or rows be represented by permutations ρ_i. Define normalized transitions

$$h_{ij} = \rho_i \rho_j^{-1}$$

and the transition group

$$H_r(L) = \langle h_{ij} : i, j \rangle.$$

A holonomy packet may include:

- the isomorphism type or order of $H_r(L)$,
- the cycle-type inventory of the distinguished h_{ij},
- block systems and orbit partitions,

- characters or matrix coefficients of words in the h_{ij},
- noncommutative moments
$$\mathrm{tr}(h_{i_1 j_1} \cdots h_{i_k j_k}),$$
- the corresponding packets for row, column, and symbol roles.

Isotopy conjugates these transition groups; paratopy permutes the three roles. Therefore the unordered multiset of role signatures is a paratopy invariant.

Exact finite computation

For the twelve order-six main classes, the finite packet consisting of transition-group order, pair-transition cycle inventory, and nontrivial block sizes is invariant under tested isotopies and all six role permutations. The twelve packets are pairwise distinct.

This classification result does not prove orthonormal representability. It demonstrates that noncommutative transition data recover information lost under spherical fusion.

12.7 Imprimitivity as hidden decomposition

A transitive permutation group is imprimitive when it preserves a nontrivial block system. In combinatorial designs, imprimitivity often signals a subsquare, product, congruence, or module decomposition.

For a Latin square of order $2m$, the normalized row-transition group preserves a two-block system of sizes $m + m$ precisely when the square contains an $m \times m$ Latin subsquare, under the normalization used in the atlas. At order six this identifies the four classes containing 3×3 subsquares.

Principle 12.2 (Clifford-type decomposition)**.** *Combinatorial substructure can be detected as imprimitivity of a transition packet. The spherical packet sees aggregate multiplicities; the holonomy packet sees how those multiplicities are assembled into blocks.*

12.8 Realization fibers

Let P denote packet data and let
$$\mathrm{Real}_{\mathrm{res}}(P)$$
be the space of restricted realizations compatible with P. The packet may completely classify the discrete object while the realization fiber remains empty, positive dimensional, disconnected, or singular.

Examples of realization conditions include:

- a flag-preserving geodesic with a declared length;
- a product-state orthonormal representation;
- a positive semidefinite Gram matrix with a zero pattern;
- a tensor factorization;
- a global descent object compatible with local packets;

- a circuit satisfying ancilla restoration on every input.

Thus classification and realization are distinct layers.

12.9 Quantum Latin squares as an existence defect

A quantum Latin square is an array [22] of vectors whose rows and columns are orthonormal bases. Ball and Simoens prove that a pair of non-entangled mutually orthogonal quantum Latin squares of order six does not exist, while earlier work provides entangled order-six solutions and hence AME$(4, 6)$ states [5, 26, 27].

This is naturally expressed as

$$\mathrm{Real}_{\mathrm{product}}(P_6) = \varnothing, \qquad \mathrm{Real}_{\mathrm{entangled}}(P_6) \neq \varnothing.$$

It is an existence defect, not a gate-count defect.

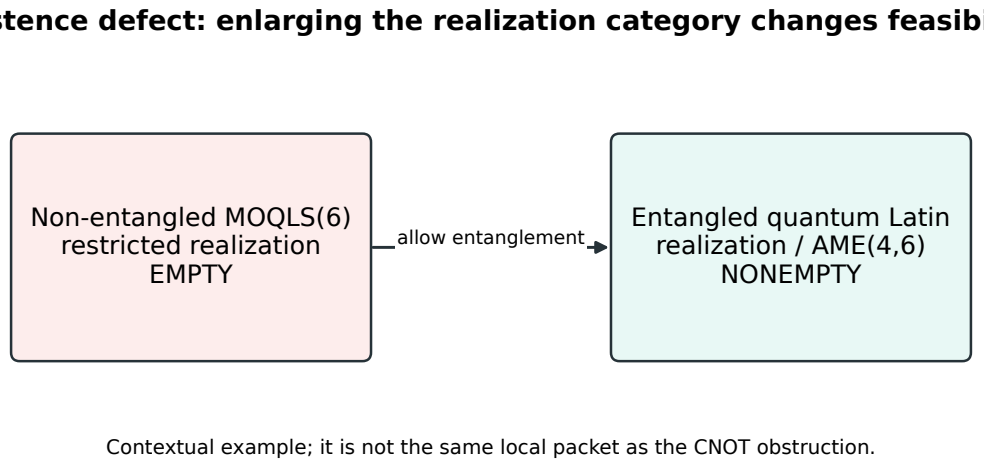


Figure 12.2: The order-six quantum-Latin distinction is a restricted-versus-ambient realization gap.

The source proof uses support patterns, a reduction to one classical square, orthonormal representations of complements of Latin-square graphs, projective-line closure, finite case analysis over twelve main classes, and a continuous argument for the two remaining subsquare cases. No theorem in the unitriangular or Clifford chapters is inferred from this result.

12.10 First-arrival packets

For a finite generator set S, the endpoint Green kernel

$$\mathcal{G}_{S,g}(z) = \langle \delta_g, (I - zA_S)^{-1}\delta_e \rangle$$

contains more information than the spectrum of A_S. Its first nonzero degree is the distance to g.

If

$$A_S = \sum_{\lambda} \lambda B_\lambda,$$

then

$$\mathcal{G}_{S,g}(z) = \sum_{\lambda} \frac{\langle \delta_g, B_\lambda \delta_e \rangle}{1 - \lambda z}.$$

The endpoint matrix coefficients of the embedded projectors determine cancellation between spectral components. This is the metric analogue of retaining a type rather than only an unramified eigencharacter.

12.11 Typed Green kernels

Let $H \leq G$ and let $\mu_k(h)$ count ambient words of length k with holonomy $h \in H$. For an irreducible representation τ of H, define

$$\widehat{\mu}_k(\tau) = \sum_{h \in H} \mu_k(h)\tau(h)$$

and

$$\widehat{\mathcal{G}}_\tau(z) = \sum_{k \geq 0} \widehat{\mu}_k(\tau) z^k.$$

Fourier inversion recovers every $\mu_k(h)$. The trivial representation records only total return to the subgroup; nontrivial types resolve the holonomy.

Proved statement

For finite H, the full family of typed Green kernels over $\widehat{H}$ determines every endpoint Green series and therefore every ambient word distance from the identity to an element of H.

12.12 Operational Langlands analogy

The hierarchy resembles the local organization of representation theory:

$$\text{spherical character} \to \text{type} \to \text{ramification or monodromy} \to \text{realization and descent.}$$

The analogy is useful because it prevents coarse spectral data from being mistaken for a complete correspondence. It does not imply that global Langlands reciprocity predicts either the circuit threshold or the quantum-Latin result.

Scope boundary

The word "Langlands" in this monograph refers to an operational packet architecture: commuting local spectra, type refinement, noncommutative holonomy, tensor assembly, and descent obligations. No general automorphic or Galois correspondence is claimed or used.

12.13 A general finite threshold system

A packet-resolved threshold system consists of:

1. a finite group, groupoid, or association scheme;
2. a commutative spherical algebra with exact projectors;

3. embedded incidence or type data;
4. a transition group or monodromy representation;
5. restricted and ambient realization categories;
6. a metric, multiplicity, or descent obstruction.

The first threshold may then be defined as

$$\tau = \min\{n : \exists P \text{ with } c_{\mathrm{res}}(P) > c_{\mathrm{amb}}(P)\}.$$

The cost can be finite or infinite.

12.14 Open problems

Question 12.3. For which finite association schemes does the full embedded Bose–Mesner algebra determine the restricted realization fiber?

Question 12.4. What bounded degree of noncommutative transition moments separates all Latin squares of a fixed order?

Question 12.5. Can typed Green kernels yield new lower bounds without enumerating the full ambient group?

Question 12.6. When does imprimitivity of a transition packet characterize a tensor or subsquare decomposition in quantum combinatorial designs?

Part V

Computational atlas and theorem templates

Chapter 13

The exact computational atlas

13.1 Purpose and design

The accompanying code is an audit-oriented finite mathematics package rather than a production compiler. Its goals are:

1. exact state representations with no floating-point ambiguity;
2. complete finite search where the state space is tractable;
3. explicit witnesses and reconstructible paths;
4. small, dependency-free core algorithms;
5. frozen JSON results and regression tests;
6. separation between proved, computed, and proposed claims.

The package is named `catalytic_gate_atlas`. Binary matrices are tuples of integer row bitmasks, and reversible Boolean maps are exact permutations of basis-state indices.

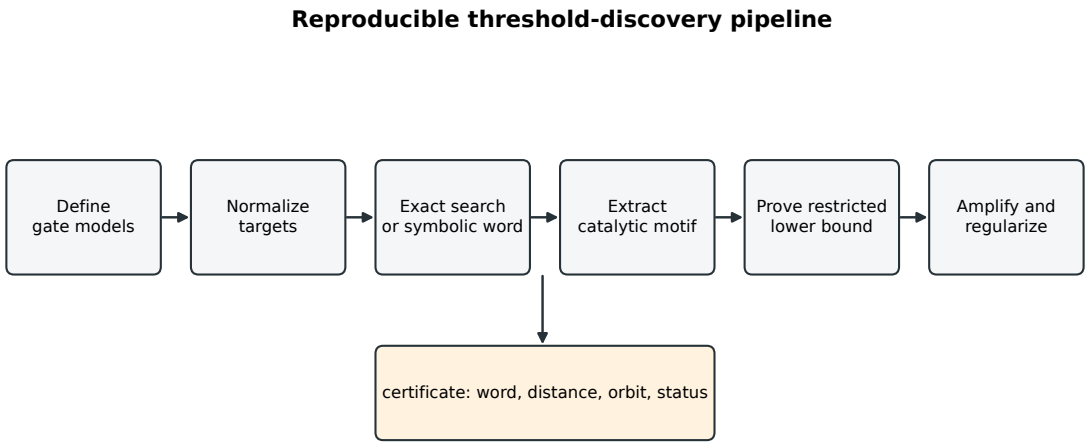


Figure 13.1: The finite certificate pipeline used throughout the computational atlas.

13.2 State models

13.2.1 Binary matrices

An $m \times m$ matrix over $\mathbb{F}_2$ is stored as

$$(r_0, \ldots, r_{m-1}),$$

where bit j of r_i is the (i, j) entry. Matrix multiplication is exact XOR accumulation.

13.2.2 Symplectic Clifford states

An n-qubit Clifford modulo global phase and Pauli translation is represented by its action on Pauli exponent vectors

$$(x \mid z) \in \mathbb{F}_2^{2n}.$$

The state space is $\mathrm{Sp}(2n, 2)$. The restricted parabolic group is generated by S, CZ, and CNOT; the ambient group additionally permits H.

13.2.3 Classical reversible states

A reversible map on n bits is stored as a permutation of $\{0, \ldots, 2^n - 1\}$. Composition is exact. The affine subgroup is generated by X and CNOT, and the ambient three-bit group adds Toffoli.

13.3 Shortest-path engines

The package implements:

- breadth-first search for unit costs;
- 0–1 breadth-first search when gates are free or unit cost;
- Dijkstra search for general nonnegative integer weights;
- parent maps for witness reconstruction;
- defect distributions on a restricted target set.

Algorithm 13.1 (Breadth-first defect census)**.** Given restricted generators S_R, ambient generators S_A, and identity e:

1. compute d_R on $\langle S_R \rangle$;
2. compute d_A on $\langle S_A \rangle$;
3. for every restricted state x, compare $d_R(x)$ and $d_A(x)$;
4. record the pair $(d_R(x), d_A(x))$ and reconstruct witnesses for strict inequalities.

For a finite group this is complete because every vertex reachable from the identity is visited exactly once at its minimum distance.

13.4 Zero-cost gates

When local Clifford gates are free and entangling gates cost one, ordinary BFS is invalid. The package uses a deque-based 0–1 BFS. A zero-cost edge is pushed to the front and a unit-cost edge to the back. The resulting labels are exact shortest entangling costs.

A common error is to quotient by free gates without proving that the quotient preserves the target subset and all left/right actions. The atlas avoids this issue by running 0–1 BFS directly on the finite group for the reported small cases.

13.5 The optimized Clifford verifier

The three-qubit symplectic group has

$$|\operatorname{Sp}(6,2)| = 1{,}451{,}520$$

states. A compact C++ implementation enumerates the full group and the parabolic subgroup exactly. It supports:

- total primitive count;
- total count with unit-cost SWAP;
- entangling count with H and S free;
- witness reconstruction and JSON output.

The C++ program and the independent Python implementation use the same gate matrices but different traversal code.

13.6 Computed Clifford census

n	Metric	SWAP	$\lvert P_n\rvert$	$\lvert\operatorname{Sp}(2n,2)\rvert$	Defects	Max defect
1	total	no	2	6	0	0
1	entangling	no	2	6	0	0
2	total	no	48	720	0	0
2	entangling	no	48	720	4	1
3	total	no	10,752	1,451,520	24	1
3	total	yes	10,752	1,451,520	0	0
3	entangling	no	10,752	1,451,520	1,624	2

The detailed defect distributions are:

model	restricted → ambient pairs
$n = 2$ entangling	$3 \to 2 : 4$
$n = 3$ total	$7 \to 6 : 6, \quad 8 \to 7 : 18$
$n = 3$ entangling	$3 \to 2 : 24,\ 4 \to 3 : 432,\ 5 \to 4 : 1056,\ 6 \to 4 : 48,\ 6 \to 5 : 64.$

Exact finite computation

Every entry in the table is a complete finite enumeration in the declared binary symplectic model. Phases and Pauli translations are not part of the state space. These are exact results for that quotient model, not full hardware Clifford optimizations.

13.7 Computed affine-versus-Toffoli census

For three bits:

$$|\operatorname{AGL}(3,2)| = 1344, \qquad |S_8| = 40320.$$

The complete census finds no affine target with a strict defect under total primitive count or under entangling count with NOT free.

Metric	Restricted size	Ambient size	Restricted diameter	Defects
Total	1,344	40,320	7	0
Entangling	1,344	40,320	6	0

13.8 Linear symbolic checks

The Python module verifies that the displayed six-row ambient and restricted words produce the same canonical target modulo

$$2, 3, 5, 8.$$

The lower bounds and ring universality are proved in the core chapter; these modular checks are regression tests rather than the logical basis of the theorem.

13.9 Witness format

A witness record contains:

- the target matrix or permutation;
- restricted and ambient costs;
- a named restricted gate path;
- a named ambient gate path;
- the metric and gate alphabet;
- the state-model convention.

For example, the two-qubit entangling Clifford witness has restricted path

$$\mathrm{CX}_{0\to1},\ S_1,\ \mathrm{CX}_{1\to0},\ \mathrm{CX}_{0\to1}$$

and ambient path

$$H_1,\ \mathrm{CX}_{1\to0},\ \mathrm{CX}_{0\to1},\ H_0,\ S_1,\ S_0.$$

Only two gates in the ambient path are charged by the entangling metric.

13.10 Certificates beyond full enumeration

For larger systems, the following certificate styles are supported in principle.

1. **Meet in the middle.** Store forward and backward balls whose radii sum to the candidate length.
2. **SAT/SMT.** Encode gate choices and endpoint equations; use an UNSAT certificate below the claimed optimum.
3. **Integer programming.** Useful when a fixed gate skeleton leaves discrete routing choices.
4. **Algebraic elimination.** Eliminate continuous gate parameters from bounded-length word equations.
5. **Dual potentials.** Exhibit a one-Lipschitz potential attaining the claimed lower bound.
6. **Quotient search.** Search cosets only after proving that the quotient is compatible with the cost and target class.

13.11 Multiobjective search

Let each gate have a vector cost in $\mathbb{N}^r$. At each state, retain only nondominated labels. The algorithm terminates for finite state spaces with a bounded search region, but the frontier can grow exponentially.

A practical exact strategy is to scalarize with several positive weight vectors, use the resulting optima to identify candidate Pareto points, and then certify each candidate by constrained search.

13.12 Reproducibility commands

From the `code` directory:

Listing 13.1: Python atlas commands.

```
python -m pip install -e . --no-build-isolation
python scripts/run_atlas.py --output ../results/atlas_small.json
python -m pytest tests
```

The optimized three-qubit verifier can be compiled by

Listing 13.2: C++ Clifford verifier.

```
g++ -O3 -std=c++20 scripts/clifford_census.cpp -o scripts/clifford_census
./scripts/clifford_census --n 3 --metric total --output ../results/clifford_n3_total.
    json
./scripts/clifford_census --n 3 --metric entangling --output ../results/
    clifford_n3_entangling.json
```

13.13 Validation policy

The release uses four layers of validation.

1. Unit tests for matrix, permutation, and path primitives.

2. Cross-checks of group sizes against known finite formulas.
3. Independent Python and C++ computations for selected Clifford cases.
4. Frozen JSON outputs and SHA-256 manifests in the release archive.

13.14 Scaling limits

Full group enumeration grows rapidly:

$$|\operatorname{Sp}(2n, 2)| = 2^{n^2} \prod_{j=1}^{n} (4^j - 1).$$

The four-qubit symplectic group is far beyond direct Python BFS. Future work should exploit:

- double-coset normal forms,
- Bruhat or parabolic decompositions,
- admissible heuristics for A* search,
- SAT encodings,
- orbit reduction,
- tensor-network representations,
- distributed meet in the middle.

13.15 What the atlas does not prove

A finite census establishes only the declared finite model. It does not prove:

- an asymptotic theorem without a separate proof,
- a hardware advantage under a different cost,
- robustness to approximation,
- equivalence after adding ancillas,
- a result for global phases or Pauli translations if they were quotiented out.

The status metadata in every result file is intended to make these boundaries machine readable.

Chapter 14

Reusable theorem templates

14.1 Purpose

This chapter collects proof architectures that recur across gate logics. Each template separates the easy upper-bound identity from the difficult lower-bound obligation. The templates are not claims that every gate hierarchy satisfies the hypotheses.

14.2 Template I: catalytic conjugation

Let $H \le G$ have restricted and ambient costs. Choose $g \in G$ and $h_1, \dots, h_m \in H$ such that

$$U = g h_m \cdots h_1 g^{-1} \in H.$$

Then

$$\ell_{\mathrm{amb}}(U) \le 2\ell_{\mathrm{amb}}(g) + \sum_i \ell_{\mathrm{amb}}(h_i).$$

Theorem 14.1 (Catalytic comparison template)**.** *Assume a restricted lower bound of the form*

$$\ell_{\mathrm{res}}(U) \ge \sum_i r_i - \beta,$$

where r_i is the restricted cost of the ith transported obligation and β is the maximum certified sharing. Then

$$\Delta(U) \ge \sum_i r_i - \beta - 2\ell_{\mathrm{amb}}(g) - \sum_i \ell_{\mathrm{amb}}(h_i).$$

The proof is immediate. The research content lies in bounding β.

14.3 Template II: amortized consumers

Suppose a catalyst costs $c_{\mathrm{open}} + c_{\mathrm{close}}$ and each of m consumers saves at least s_i restricted cost. Then

$$\Delta(U) \ge \sum_{i=1}^{m} s_i - c_{\mathrm{open}} - c_{\mathrm{close}} - r,$$

where r is repair or interference cost. A threshold occurs when the right side first becomes positive.

This template explains the three-cross rule in the linear atom and guides searches in phase, Gaussian, and routing models.

14.4 Template III: first-arrival Green valuation

Let S be a finite symmetric generating set and

$$A_S = \sum_{s \in S} \rho(s).$$

Then

$$\mathcal{G}_{S,g}(z) = \sum_{k \geq 0} \langle \delta_g, A_S^k \delta_e \rangle z^k.$$

Theorem 14.2 (First-arrival template)**.** *If word multiplicities are nonnegative, then*

$$d_S(e, g) = \operatorname{ord}_{z=0} \mathcal{G}_{S,g}(z).$$

For restricted and ambient sets,

$$\Delta(g) = \operatorname{ord}_0 \mathcal{G}_{S_R,g} - \operatorname{ord}_0 \mathcal{G}_{S_A,g}.$$

This template provides an exact spectral restatement. To turn it into a lower bound without enumeration, one needs control of the endpoint projector coefficients.

14.5 Template IV: one-Lipschitz potentials

Let F be a function on circuit states satisfying

$$F(I) = 0, \qquad |F(gs) - F(g)| \leq c(s)$$

for every allowed restricted gate s. Then

$$\ell_{\mathrm{res}}(U) \geq F(U).$$

Potential design steps:

1. choose a decomposition into cuts, packets, or degree layers;
2. prove a gate changes the total charge by at most its cost;
3. ensure cancellations cannot create free progress on many layers;
4. evaluate the target exactly.

Static sums of ranks are often too coarse in ambient models; time-expanded or target-relative potentials may be required.

14.6 Template V: dimension per gate

Let X be an algebraic family of elementary gates whose irreducible components have dimension at most r. Let V be an irreducible target variety of dimension N.

Theorem 14.3 (Dimension-per-gate template)**.** *The closure of the locus reachable in L gates has dimension at most rL. If $rL < N$, a generic point of V is not reachable in L gates.*

If every target in V has a restricted construction of length at most N/r, then generic restricted and ambient lengths coincide at that value, provided the same parameter bound applies to the ambient alphabet.

14.7 Template VI: defect codimension

Assume every target has restricted cost at most M. A defect of size at least d implies ambient cost at most $M - d$. If the $(M - d)$-gate ambient locus has dimension at most $N - d$, then

$$\operatorname{codim} \overline{\mathcal{D}_{\geq d}} \geq d.$$

This is the general mechanism behind algebraic rarity.

14.8 Template VII: finite-field sparsity

Suppose there are at most B gate directions and at most q^r parameter choices per gate over $\mathbb{F}_q$. Then the number of words of length at most L is at most

$$\sum_{j=0}^{L} (Bq^r)^j.$$

If the target family has q^N points and defect d forces $L \leq (N - d)/r$, this yields a density bound of order q^{-d} up to a fixed combinatorial factor.

The estimate counts words, not distinct endpoints, and is therefore an upper bound without injectivity assumptions.

14.9 Template VIII: block retraction

Let a large restricted system contain a smaller one through an embedding ι, and suppose a gate-nonincreasing retraction π maps large restricted generators to small restricted generators or the identity. Then

$$\ell_{\text{large,res}}(\iota U) \geq \ell_{\text{small,res}}(U).$$

If the ambient embedding preserves gate cost,

$$\Delta_{\text{large}}(\iota U) \geq \Delta_{\text{small}}(U).$$

This transfers a catalytic atom to architectures containing it as a retract.

14.10 Template IX: exact block coupling

For block-lower matrices

$$M = \begin{pmatrix} A & 0 \\ XA & D \end{pmatrix},$$

projection to diagonal blocks forces the costs of A and D, while rank of the cross block forces at least $\operatorname{rank} X$ cross transmissions. If X is a partial monomial matrix, the lower bound is attained by direct construction.

The same logic extends to trees of blocks whenever every coupling layer has an exact rank certificate and the projections are homomorphisms.

14.11 Template X: tensor Leibniz rule

Suppose a restricted complexity is tree tight and satisfies

$$\ell_{\mathrm{res}}(A \otimes B) = n\ell_{\mathrm{res}}(A) + m\ell_{\mathrm{res}}(B)$$

for A of size m and B of size n. If ambient complexity satisfies the corresponding upper bound, then

$$\Delta(A \otimes B) \geq n\Delta(A) + m\Delta(B).$$

For tensor powers,

$$\Delta(A^{\otimes r}) \geq rm^{r-1}\Delta(A).$$

This is the standard route from a finite atom to an $N \log N$ defect.

14.12 Template XI: stable capacity

If

$$a_{r+s} \leq m^s a_r + m^r a_s,$$

then

$$b_r = \frac{a_r}{m^{r-1}}$$

is subadditive and

$$\lim_{r\to\infty} \frac{a_r}{rm^{r-1}}$$

exists by Fekete's lemma. This defines a regularized ambient cost and exposes possible tensor activation.

14.13 Template XII: vector costs

Let gates carry costs in $\mathbb{R}^k_+$. A lower bound may be a vector inequality

$$F(U) \preceq \sum_{g\in W} c(g)$$

or a family of scalar inequalities indexed by dual weights $w \in \mathbb{R}^k_+$:

$$w \cdot F(U) \leq w \cdot c(W).$$

A complete Pareto proof may require enough supporting weights to separate the restricted and ambient convex lower hulls.

14.14 Template XIII: ancilla restoration

Let W act on data D and ancilla A. To certify a dirty catalyst, prove the operator identity

$$W = U_D \otimes I_A$$

on the full joint space, not only on one ancilla input. For a clean catalyst, prove

$$W(|\psi\rangle \otimes |0\rangle) = (U|\psi\rangle) \otimes |0\rangle$$

for all $|\psi\rangle$. The distinction changes both the target equations and lower bounds.

14.15 Template XIV: symmetry-sector escape

Given projectors P_q, define off-sector transfer blocks

$$T_{qr} = P_q U P_r.$$

A restricted gate preserves all diagonal sectors. An ambient gate changes a controlled collection of blocks. A candidate potential is a weighted sum of ranks or norms of obligations relative to the final target. The proof must charge both creation and useful consumption of off-sector coherence.

14.16 Template XV: exhaustive threshold classification

A finite first-threshold theorem should contain:

1. no defects below size n_0;
2. at least one defect at n_0;
3. exact classification of every defect at n_0 modulo declared equivalence;
4. analytic generation of the exceptional motifs;
5. equality between the analytic set and computational census;
6. witness and completeness certificates.

This is the standard achieved by the six-wire linear core and is the target standard for future gate logics.

14.17 A proof-obligation ledger

Before promoting a catalytic identity to a theorem, record:

Obligation	Required evidence
Endpoint membership	Symbolic proof that the target lies in the restricted class.
Ambient upper bound	Explicit circuit evaluated exactly.
Restricted lower bound	Potential, rank, algebraic, search, or certificate argument.
Restoration	Identity on all data and work inputs under the chosen convention.
Cost compatibility	Same gate alphabet, weights, and free operations in both metrics.
Minimal threshold	Exhaustive or symbolic exclusion of smaller instances.
Equivalence classification	Orbit or normal-form analysis of all minimal defects.
Amplification	Proof that the lower bound and upper construction compose.
Reproducibility	Source, frozen outputs, and deterministic commands.

Chapter 15

Research program and paper architecture

15.1 The central research problem

For a nested pair of gate models, determine the distortion function

$$D_n = \sup_{U \in \mathcal{T}_n} \big(c_{\mathrm{res}}(U) - c_{\mathrm{amb}}(U)\big),$$

where $\mathcal{T}_n$ is the target class. The basic questions are:

1. Does any defect exist?
2. What is the first size threshold?
3. What is the minimal catalytic motif?
4. Is the effect generic, sparse, or confined to special strata?
5. How does it scale under direct sum, recursion, and tensor products?
6. Which resource vector reveals the first defect?
7. Can the effect survive realistic architecture and fault-tolerant costs?

15.2 Established base camp

The unitriangular theory provides a complete first example:

- a universal six-row atom over nonzero commutative rings;
- exact binary and ternary first-threshold evidence;
- exact classification of twenty binary minimal defects;
- connected families with two reverse gates and linear saving;
- microscopic rank-one, bandwidth-one escape;
- generic rigidity and defect codimension;
- recursive and tensor amplification to $\Omega(n \log n)$.

The small Clifford atlas adds a second gate logic with genuine exact defects and shows that changing the primitive library by adding SWAP can remove them.

15.3 Program A: complete the Clifford threshold theory

The highest-priority quantum project is a paper on parabolic-versus-ambient Clifford synthesis.

15.3.1 Immediate tasks

1. Classify the four two-qubit entangling defects under local Clifford and qubit-swap equivalence.
2. Classify the twenty-four three-qubit total-count defects under parabolic conjugation and qubit permutations.
3. Derive analytic normal forms for the witness circuits.
4. Prove lower bounds independent of full group enumeration.
5. Test depth, weighted, and architecture-aware metrics.
6. Determine whether a finite defective atom tensor-amplifies.

15.3.2 Longer-term questions

Conjecture 15.1. The parabolic subgroup of $\mathrm{Sp}(2n, 2)$ has unbounded additive distortion in the full symplectic group for at least one natural Clifford cost model.

Question 15.2. Does the maximum total-count defect grow as n, $n \log n$, or faster?

Question 15.3. Is the disappearance of three-qubit defects after adding unit-cost SWAP a small-width accident or a structural geodesic-convexity phenomenon?

15.4 Program B: temporary nonlinearity

The three-bit affine-versus-Toffoli census is rigid. The next target is a complete four-bit normal-form search without enumerating 16! permutations.

15.4.1 Algorithmic plan

1. Compute exact affine distances for all $|\mathrm{AGL}(4, 2)| = 322{,}560$ targets.
2. Enumerate Toffoli sandwiches modulo affine left and right actions.
3. Use ANF equations to retain only affine endpoints.
4. Compare total, Toffoli, CNOT, and fault-tolerant costs.
5. Produce SAT certificates below every claimed optimum.

Question 15.4. What is the first bit width at which a restored nonlinear excursion shortens an affine target?

Question 15.5. Can temporary degree two yield a family with unbounded saving while the endpoint degree remains one?

15.5 Program C: phase-polynomial catalysts

The phase setting is likely to produce a defect first in a resource other than total primitive count.

Priority metrics:

T-depth, CNOT count at fixed T-count, two-qubit depth, architecture-weighted cost.

A first paper should choose one metric and avoid a diffuse comparison across all resources.

Question 15.6. Can a restored Hadamard or Clifford frame reduce T-depth of a diagonal target without additional ancillas?

Question 15.7. Does tensor activation occur for phase-polynomial T-count?

15.6 Program D: Gaussian and matchgate systems

Two concrete first searches are recommended.

1. Number-conserving versus general fermionic Gaussian synthesis on a small number of modes.
2. Nearest-neighbor versus all-to-all matchgate circuits with weighted fermionic SWAP cost.

Both admit exact orthogonal-matrix representations and have clear architecture relevance.

15.7 Program E: symmetry and measurement

This program should begin with a narrowly declared resource theory, such as:

- excitation-preserving two-qubit gates versus arbitrary two-qubit gates;
- unitary Clifford+T versus measurement-assisted Clifford+T;
- clean versus dirty ancillas for a fixed arithmetic family;
- local architecture versus one reusable nonlocal bus gate.

The first objective is not maximal generality but one theorem with exact restoration and cost accounting.

15.8 Program F: root-system catalysis

For a split reductive group with allowed roots $\Psi \subset \Phi$, classify configurations in which a forbidden root acts as a reusable catalyst through Chevalley commutator relations.

Suggested sequence:

$$A_2 \to B_2 = C_2 \to G_2 \to A_r.$$

Low-rank nonsimply laced systems are attractive because longer root strings may create more than one useful effect per consumer.

Question 15.8. Is there a root-system analogue of the three-cross, one-phase threshold?

Question 15.9. Can a G_2 root string produce a multiplicative distortion exceeding the clean type-A value $3/2$?

15.9 Program G: algebraic defect loci

For each finite-dimensional gate hierarchy, define bounded word loci and defect strata. Determine:

- irreducible components,
- exact dimension and codimension,
- singularities and degenerations,
- border length,
- finite-field point counts,
- behavior under tensor products.

Conjecture 15.10. In a broad class of one-parameter elementary gate varieties, defect at least d is generically a codimension-at-least-d condition, with equality on explicit families of components.

15.10 Program H: packet-resolved realization

The packet hierarchy should be developed as a separate mathematical framework rather than appended to every circuit paper.

Core objectives:

1. define spherical, type, and holonomy functors for finite transition groupoids;
2. prove information-loss theorems for commutativization;
3. classify when typed Green kernels are sufficient for distance;
4. relate imprimitivity to combinatorial decomposition;
5. formalize realization fibers and descent obstructions.

15.11 Recommended paper sequence

Paper	Working title	Core result
I	Catalytic escape from unitriangularity	Completed linear threshold, rarity, and asymptotic distortion.
II	Catalytic basis escape in Clifford synthesis	Exact parabolic-versus-Clifford thresholds and analytic motifs.
III	Temporary nonlinearity in reversible circuits	Affine targets with or without Toffoli-assisted defects.
IV	Gate-set distortion and catalytic holonomy	Abstract subgroup, groupoid, and vector-cost theory.
V	Spherical collapse and transition holonomy	Packet refinement, Latin-square holonomy, and realization fibers.
VI	Algebraic defect loci	Word varieties, codimension, border catalysis, and point counts.
VII	Root-system catalytic complexity	Chevalley commutators and thresholds beyond type A.

15.12 Software roadmap

Version 0.1 of the atlas supports exact small finite groups. A mature package should add:

1. generic multiobjective Dijkstra search;
2. disk-backed and parallel BFS;
3. SAT and SMT export;
4. group action and orbit classification;
5. symbolic word-map elimination;
6. circuit diagrams in Quantikz and SVG;
7. architecture graphs and routing costs;
8. certificate verification independent of the search engine;
9. a machine-readable theorem-status ledger.

15.13 Standards for claims

A result should be labeled *proved* only when every lower bound and restoration claim has a proof. A complete finite census should identify the state quotient, enumerate its full size, and make the traversal reproducible. A symbolic skeleton should be valued as a research object but not described as an established shortcut.

15.14 Potential applications

If developed, the theory may contribute to:

- exact and heuristic quantum circuit compilers,
- architecture-aware routing,
- parity and phase-network synthesis,
- stabilizer and Clifford optimization,
- reversible arithmetic,
- Gaussian and fermionic simulation,
- lower-bound methodology,
- geometric group theory of gate sets,
- algebraic geometry of circuit word maps.

The near-term value is methodological and algorithmic. Hardware benefit requires a device-specific cost and noise model.

15.15 Twenty central open problems

1. Prove or refute a universal six-dimensional linear threshold over all coefficient rings.
2. Determine the true asymptotic order of the maximum unitriangular defect.
3. Decide whether multiplicative distortion is bounded.
4. Determine the lower border length of the canonical six-row atom.
5. Classify tree-tight catalytic matroids.
6. Find a one-copy rigid target with tensor activation.
7. Analytically classify the two-qubit Clifford entangling defects.
8. Analytically classify the three-qubit total-count Clifford defects.
9. Determine whether SWAP restores rigidity for all n in the chosen primitive model.
10. Find the first affine target shortened by temporary nonlinearity, or prove none exists.
11. Find the first diagonal target improved by non-diagonal Clifford catalysis.
12. Determine a phase-polynomial Pareto defect.
13. Find a number-conserving Gaussian target shortened by Bogoliubov escape.
14. Find a matchgate routing catalyst with a certified architecture advantage.
15. Establish a symmetry-sector microscopic-escape theorem.
16. Classify when a fixed number of dirty ancillas gives unbounded saving.
17. Develop typed Green-kernel lower bounds that avoid full enumeration.
18. Determine finite moment bounds sufficient for transition-holonomy classification.
19. Classify catalytic motifs in B_2, C_2, and G_2.
20. Prove a general theorem connecting algebraic codimension and catalytic defect.

15.16 Closing perspective

The common phenomenon is not that every restriction can be beaten. It is that the path space matters. A target may obey a symmetry, flag, degree bound, locality rule, or factorization constraint while the cheapest route to that target does not.

The disciplined research question is therefore:

Which temporary resources can be opened, reused, and exactly removed at a net computational gain?

The established linear theory answers this question completely for one important model. The monograph supplies the symbolic and computational machinery to ask it across quantum gate logic without conflating identities, experiments, and theorems.

Appendix A

Gate matrices and symbolic conventions

A.1 Row-transvection convention

Matrices act on column coordinate vectors, while row operations left multiply the current matrix. A row transvection

$$\tau_{t,s}(a) = I + aE_{t,s}$$

performs

$$R_t \leftarrow R_t + aR_s.$$

In the binary CNOT interpretation, s is the control and t is the target.

The canonical six-row target is

$$C = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{pmatrix}.$$

One ambient word is

$$\tau_{1,2}(1), \tau_{3,1}(1), \tau_{1,0}(1), \tau_{4,1}(1), \tau_{5,1}(1), \tau_{1,2}(-1).$$

A restricted word of length seven is

$$\tau_{1,0}(1), \tau_{2,1}(1), \tau_{3,0}(-1), \tau_{3,2}(1), \tau_{4,2}(1), \tau_{5,2}(1), \tau_{2,1}(-1).$$

A.2 Qudit SUM gate

On computational basis labels in $\mathbb{Z}_d$,

$$\mathrm{SUM}_{s\to t}^{(a)}|x_s, x_t\rangle = |x_s, x_t + ax_s \bmod d\rangle.$$

Its classical linear action is the transvection $I + aE_{t,s}$ over $\mathbb{Z}_d$.

A.3 Binary symplectic form

Pauli exponents are ordered as

$$(x_0, \ldots, x_{n-1}, z_0, \ldots, z_{n-1}).$$

The symplectic form is

$$J = \begin{pmatrix} 0 & I_n \\ I_n & 0 \end{pmatrix}$$

over $\mathbb{F}_2$. A matrix M is symplectic when

$$M^T J M = J.$$

A.4 Clifford generator matrices

For qubit i, the Hadamard swaps x_i and z_i:

$$H_i : e_i \leftrightarrow e_{n+i}.$$

The phase gate has

$$z_i \leftarrow z_i + x_i.$$

For CNOT with control c and target t,

$$x_t \leftarrow x_t + x_c, \qquad z_c \leftarrow z_c + z_t.$$

For CZ on i, j,

$$z_i \leftarrow z_i + x_j, \qquad z_j \leftarrow z_j + x_i.$$

SWAP exchanges both x and z coordinates of the two qubits.

The parabolic subgroup used in the census is generated by S_i, CZ_{ij}, and $\mathrm{CX}_{i\to j}$. It stabilizes the Z Lagrangian

$$L_Z = \mathrm{span}(e_n, \ldots, e_{2n-1}).$$

A.5 Two-qubit Clifford witness

The target symplectic matrix is

$$M_2 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}.$$

A restricted path is

$$\mathrm{CX}_{0\to1},\ S_1,\ \mathrm{CX}_{1\to0},\ \mathrm{CX}_{0\to1},$$

with three charged entangling gates. An ambient path is

$$H_1,\ \mathrm{CX}_{1\to0},\ \mathrm{CX}_{0\to1},\ H_0,\ S_1,\ S_0,$$

with two charged entangling gates when H, S are free.

A.6 Three-qubit Clifford witness

The target symplectic matrix is

$$M_3 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \end{pmatrix}.$$

A restricted total-count path is

$$\mathrm{CX}_{0\to2},\ \mathrm{CZ}_{1,2},\ \mathrm{CX}_{2\to0},\ \mathrm{CX}_{0\to2},\ \mathrm{CX}_{0\to1},\ \mathrm{CX}_{1\to0},\ \mathrm{CX}_{0\to1},$$

while an ambient path is

$$H_1,\ \mathrm{CX}_{1\to2},\ \mathrm{CX}_{2\to1},\ \mathrm{CX}_{2\to0},\ \mathrm{CX}_{0\to2},\ H_0.$$

A.7 Classical reversible gates

For a bit string $x \in \mathbb{F}_2^n$:

$$X_i : x_i \leftarrow x_i + 1,$$
$$\mathrm{CX}_{c\to t} : x_t \leftarrow x_t + x_c,$$
$$\mathrm{TOF}_{a,b\to t} : x_t \leftarrow x_t + x_a x_b.$$

All other coordinates are unchanged. The first two generate $\mathrm{AGL}(n, 2)$; all three generate the full reversible group for $n \geq 3$.

A.8 Bosonic symplectic embedding

For $A \in \mathrm{GL}(n, \mathbb{R})$,

$$\iota(A) = \mathrm{diag}(A, A^{-T}) \in \mathrm{Sp}(2n, \mathbb{R}).$$

This convention orders quadratures as $(q_1, \ldots, q_n, p_1, \ldots, p_n)$.

A.9 Pauli rotations

For a Hermitian Pauli P,

$$R_P(\theta) = e^{-i\theta P/2}.$$

Clifford conjugation gives

$$CR_P(\theta)C^\dagger = R_{CPC^\dagger}(\theta).$$

The sign of $CPC^\dagger$ can be absorbed by changing θ to $-\theta$.

A.10 Tensor and direct-sum conventions

For linear maps, $A \oplus B$ denotes a block-diagonal direct sum. The Kronecker product $A \otimes B$ uses lexicographic ordering. In quantum chapters, $U \otimes V$ denotes the physical tensor product on disjoint systems. These operations have different dimension and cost normalizations and are not interchanged silently.

Appendix B

Finite census tables and witnesses

B.1 Status key

Every table in this appendix is generated from exact finite output in the accompanying JSON files. "Restricted size" and "ambient size" refer to the enumerated quotient groups, not necessarily the full unitary groups including phases and translations.

B.2 Clifford census summary

Qubits	Metric	SWAP	Restricted size	Ambient size	diam_R	diam_A	Defects
1	total	no	2	6	1	3	0
1	entangling	no	2	6	0	0	0
2	total	no	48	720	5	8	0
2	entangling	no	48	720	3	3	4
3	total	no	10,752	1,451,520	8	12	24
3	total	yes	10,752	1,451,520	7	11	0
3	entangling	no	10,752	1,451,520	6	6	1,624

B.3 Clifford defect distributions

Model	Distance pair	Count
Two qubits, entangling	$3 \to 2$	4
Three qubits, total	$7 \to 6$	6
Three qubits, total	$8 \to 7$	18
Three qubits, entangling	$3 \to 2$	24
Three qubits, entangling	$4 \to 3$	432
Three qubits, entangling	$5 \to 4$	1,056
Three qubits, entangling	$6 \to 4$	48
Three qubits, entangling	$6 \to 5$	64

B.4 Affine-versus-Toffoli census

Bits	Metric	Restricted size	Ambient size	Restricted diameter	Defects
3	total	1,344	40,320	7	0
3	entangling	1,344	40,320	6	0

B.5 Linear atom modular checks

Modulus	Restricted word length	Ambient word length	Both match target
2	7	6	yes
3	7	6	yes
5	7	6	yes
8	7	6	yes

These checks confirm the symbolic words. Exact optimality over general rings is proved separately in the core chapter.

B.6 Core binary six-dimensional census

For $U_6(\mathbb{F}_2)$ there are $2^{15} = 32{,}768$ targets. The complete comparison is:

Restricted distance	Ambient distance	Number of targets
0	0	1
1	1	15
2	2	125
3	3	720
4	4	2,914
5	5	7,851
6	6	12,656
7	6	20
7	7	7,926
8	8	540

The twenty strict defects are exactly the analytic three-cross, one-phase motifs.

B.7 Endpoint Green coefficients for the canonical atom

Restricted word counts for degrees zero through ten are

$$0, 0, 0, 0, 0, 0, 0, 543, 25080, 510703, 10824940.$$

Ambient word counts for degrees zero through eight are

$$0, 0, 0, 0, 0, 0, 2, 776, 39454.$$

Thus first arrival occurs at seven and six, respectively.

B.8 Machine-readable locations

The release contains:

- `results/atlas_results.json`,
- `results/clifford_n3_total.json`,
- `results/clifford_n3_total_swap.json`,
- `results/clifford_n3_entangling.json`,
- the complete unitriangular supplement under `core_extract/supplement/`.

The JSON schema records model, metric, group sizes, diameters, defect distributions, and selected witnesses.

Appendix C

Python and C++ interface

C.1 Package overview

The Python package `catalytic_gate_atlas` exports exact primitives for the linear, Clifford, and classical reversible experiments. It requires Python 3.10 or later and has no runtime dependencies.

C.2 Installation

```
cd code
python -m venv .venv
. .venv/bin/activate
python -m pip install -e . --no-build-isolation
python -m pytest tests
```

On Windows PowerShell, activation is

```
.\.venv\Scripts\Activate.ps1
```

C.3 Top-level API

```
from catalytic_gate_atlas import (
    RowGate,
    canonical_ambient_word,
    canonical_restricted_word,
    canonical_six_target,
    snapshot_target,
    verify_canonical_atom,
    exact_clifford_census,
    exact_affine_toffoli_census,
    gate_word_to_openqasm,
)
```

C.4 Linear atom

```
from catalytic_gate_atlas import verify_canonical_atom

for modulus in (2, 3, 5, 8):
    print(verify_canonical_atom(modulus))
```

The function evaluates the displayed restricted and ambient words over $\mathbb{Z}/m\mathbb{Z}$ and compares both endpoints with the canonical target.

C.5 Snapshot targets

```
from catalytic_gate_atlas import snapshot_target

# u=4 source updates; snapshots at phases 0, 2, and 4.
A = snapshot_target(4, phases=(0, 2, 4), modulus=2)
for row in A:
    print(row)
```

The generator checks phase order and produces the exact lower-unitriangular target matrix used in the symbolic family.

C.6 Clifford census

```
from catalytic_gate_atlas import exact_clifford_census

print(exact_clifford_census(2, entangling_cost=True))
print(exact_clifford_census(2, entangling_cost=False))
```

The optional three-qubit Python census is slower. The release includes an optimized C++ program for the full 1,451,520-state group.

C.7 Affine-versus-Toffoli census

```
from catalytic_gate_atlas import exact_affine_toffoli_census

print(exact_affine_toffoli_census(3, entangling_cost=False))
print(exact_affine_toffoli_census(3, entangling_cost=True))
```

The complete ambient enumeration is intentionally limited to three bits.

C.8 OpenQASM witness export

```
from catalytic_gate_atlas import gate_word_to_openqasm

word = ["H1", "CX12", "CX21", "CX20", "CX02", "H0"]
print(gate_word_to_openqasm(word, 3, title="ambient witness"))
```

The exporter supports the named H, S, X, CX, CZ, and SWAP gates used by the finite atlas and writes OpenQASM 3. The supplied files in `examples/` encode the two- and three-qubit restricted and ambient witness words. The Clifford census is performed modulo global phase and Pauli translations, so equality of an exported pair refers to its binary symplectic action unless the ignored Pauli and phase data are separately reconstructed.

```
PYTHONPATH=. python scripts/export_qasm_examples.py
```

C.9 Search primitives

```
from catalytic_gate_atlas.search import (
    NamedGenerator,
    breadth_first_distances,
    breadth_first_with_parents,
    zero_one_distances,
    zero_one_with_parents,
    dijkstra_distances,
    reconstruct_path,
    defect_distribution,
)
```

Every state must be immutable and hashable. The caller supplies a multiplication function with the convention

`multiply(generator, state)`.

C.10 Binary matrix primitives

The module `algebra.py` provides:

- `gf2_identity`,
- `gf2_mul`,
- `gf2_transpose`,
- `gf2_rank`,
- dense modular matrix multiplication,
- exact permutation composition and inversion.

C.11 C++ verifier

```
g++ -O3 -std=c++20 scripts/clifford_census.cpp -o scripts/clifford_census
./scripts/clifford_census --n 3 --metric total
./scripts/clifford_census --n 3 --metric total --swap
./scripts/clifford_census --n 3 --metric entangling
```

The executable can write JSON using the `--output` option.

C.12 Figure generation

```
python scripts/make_monograph_figures.py
```

All new monograph figures are generated by Matplotlib without external data. The established unitriangular figures are preserved from the Paper I source archive.

C.13 Adding a new finite gate model

A new module should implement:

1. a canonical immutable state representation;
2. exact multiplication;
3. named restricted and ambient generators with nonnegative costs;
4. an exact target-membership test;
5. a census function returning the standard JSON fields;
6. smoke tests for group size and witness evaluation.

C.14 Standard result schema

A census result should include:

```
{
    "model": "descriptive-name",
    "size": n,
    "metric": "total-or-weighted",
    "restricted_group_size": ...,
    "ambient_group_size": ...,
    "restricted_diameter": ...,
    "ambient_diameter": ...,
    "defect_count": ...,
    "defect_distribution": {...},
    "maximum_defect": ...,
    "witness": {...},
    "status": "complete finite enumeration"
}
```

A partial search must not use the status string “complete finite enumeration.”

C.15 Error handling

The package raises `ValueError` for invalid dimensions, repeated controls, non-square matrices, incompatible costs, and requests that exceed intentionally supported complete censuses. Search code does not silently truncate the state space.

Appendix D

Conjecture and obligation ledger

D.1 How to read the ledger

The ledger separates established statements, complete computations, partial checks, and open conjectures. "Open" means no proof is claimed in this monograph. "Computed" refers only to the stated finite model.

ID	Statement	Status	Decisive next step
L1	Canonical six-row transvection target has lengths 7 and 6 over nonzero commutative rings.	Proved	Independent external proof audit.
L2	Dimensions below six are rigid over every coefficient field or ring.	Open	Symbolic bounded-skeleton elimination for $n \leq 5$.
L3	Maximum unitriangular defect is $\Theta(n \log n)$.	Open	Matching upper bound or larger interacting-catalyst family.
L4	Multiplicative unitriangular distortion is bounded by 3/2.	Open	Normal form beyond one clean carrier.
L5	Canonical atom has lower border length seven.	Open	Eliminate six-gate word-map components algebraically.
L6	A one-copy rigid target exhibits tensor activation.	Open	Exact tensor-power search or asymptotic construction.
C1	Two-qubit parabolic Clifford entangling model has four $3 \to 2$ defects.	Computed	Analytic orbit classification and lower bound.
C2	Three-qubit swap-free total Clifford model has twenty-four defects.	Computed	Analytic motif classification.
C3	Unit-cost SWAP eliminates all three-qubit total defects.	Computed	Determine whether rigidity persists for larger n.
C4	Parabolic-versus-full Clifford defect is unbounded.	Open	Construct and certify an amplifiable atom.
N1	Three-bit affine targets receive no Toffoli-assisted saving in the two tested metrics.	Computed	Independent implementation and formal certificate.

N2	Some finite affine target is shortened by temporary nonlinearity.	Open	Complete four-bit normal-form search.
P1	A diagonal target has a Pareto improvement from non-diagonal Clifford escape.	Open	Exact small-width multiobjective atlas.
P2	Phase-polynomial T-count displays tensor activation.	Open	Joint synthesis of repeated phase targets.
G1	A number-conserving Gaussian target is shortened by Bogoliubov escape.	Open	Exact low-mode orthogonal/symplectic search.
G2	A nearest-neighbor matchgate target is shortened by restored mode permutation.	Open	Architecture-weighted routing theorem.
S1	A constant-size symmetry-breaking excursion can save unbounded cost.	Open	Explicit sector-coupled family and lower bound.
A1	A fixed number of dirty ancillas can yield unbounded saving in a non-linear quantum gate model.	Open	Family with all-input restoration proof.
R1	A nonsimply laced root system has catalytic ratio exceeding 3/2.	Open	Low-rank Chevalley word search and restricted lower bound.
H1	Typed Green kernels yield non-enumerative distance lower bounds.	Open	Representation-theoretic coefficient inequalities.
H2	Finite-degree transition moments suffice to distinguish all Latin squares of fixed order.	Open	Moment hierarchy and counterexample search.
Q1	The order-six quantum-Latin product realization is empty while the entangled realization is nonempty.	Established externally	See Ball–Simoens and cited entangled constructions.

D.2 Negative results worth preserving

The following failed or negative conclusions are part of the research record.

- The circuit Pasch configuration and the hard Latin line spread are not the same local projective motif.
- Abstract row-column-symbol eigenvalues do not distinguish order-six Latin main classes.
- One projective spread operator does not distinguish the two hard classes.
- Three-bit Toffoli assistance does not shorten affine targets in the tested costs.
- Adding unit-cost SWAP removes the three-qubit total-count Clifford defects in the declared quotient model.

These results prevent overgeneralization and sharpen the next search spaces.

D.3 Publication obligations

Before any skeleton chapter is separated into a theorem paper, complete:

1. targeted literature review,
2. independent verification of all exact computations,
3. a formal statement of equivalence conventions,
4. proof or certificate for every lower bound,
5. archive of source, expected outputs, and manifests,
6. explicit discussion of what the model omits.

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